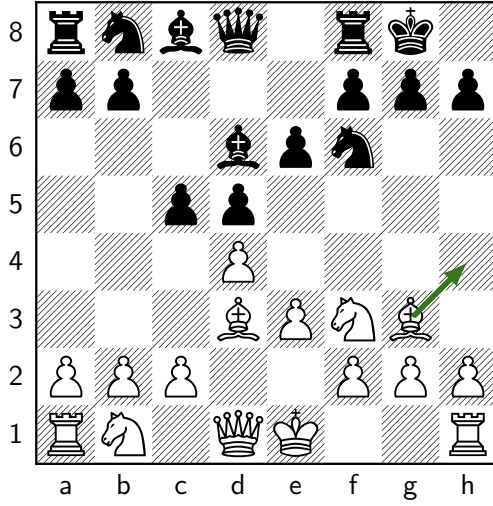


Melanie-BOTS (CCRL engines) — London System

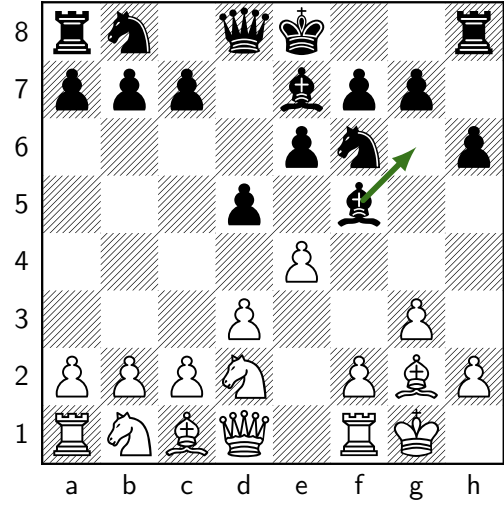
Melanie (Deep_Vision)

Abstract

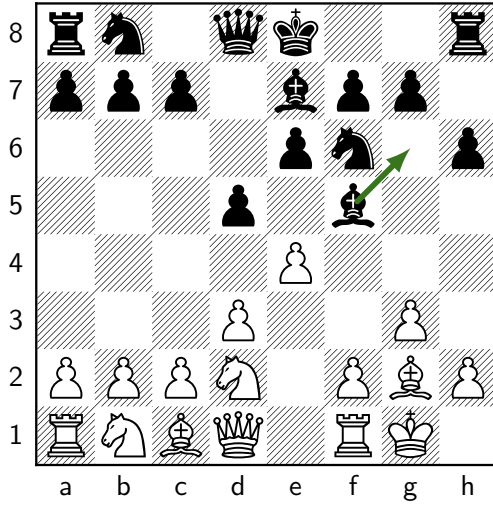
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



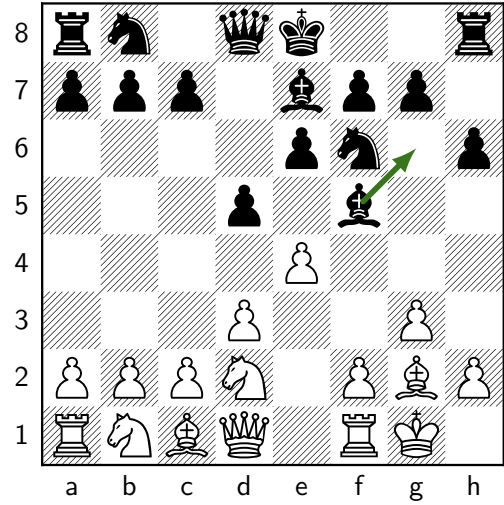
(a)



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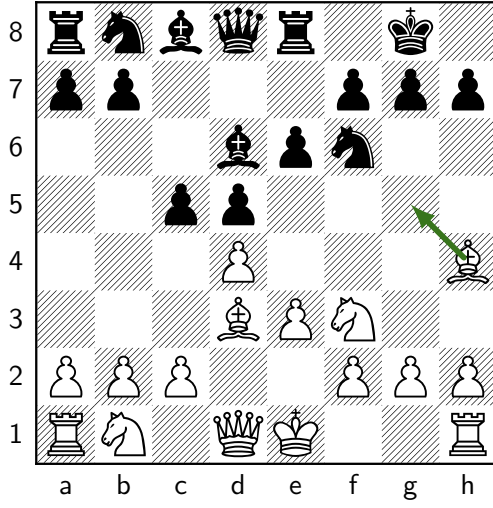


(c)

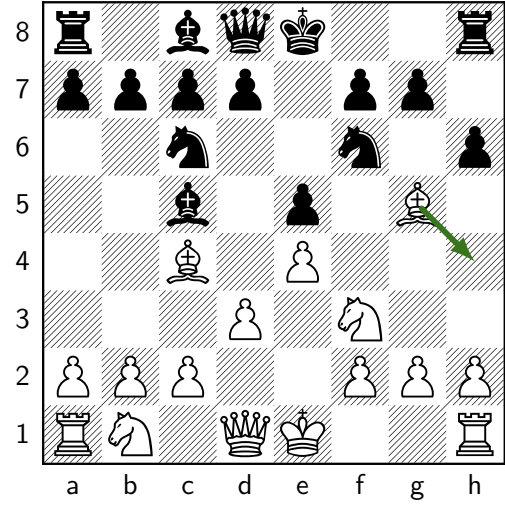


(d)

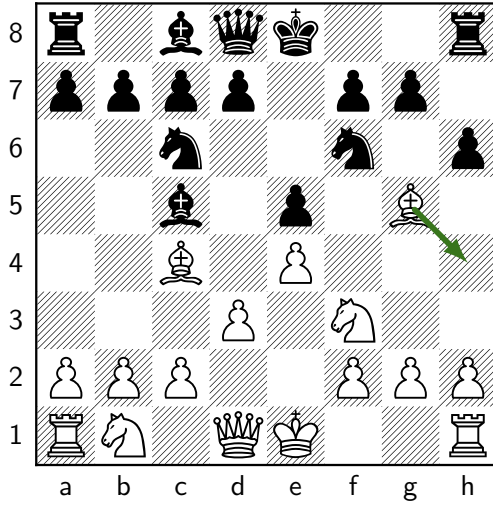
Figure 1: **Move 7, White: Bh4**. (a) Our position; Melanie plays Bh4 (g3h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#226` (ply 13) [structural distance $d=56$, QAP obj=31]: the master played Bg6; under μ our Bg3 plays the role of that game’s Bf4; cross-type roles: our Pc2 as their Nb8, our Pe3 as their Nf6; transferred move g3h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#227` (ply 13) [structural distance $d=56$, QAP obj=31]: the master played Bg6; under μ our Bg3 plays the role of that game’s Bf4; cross-type roles: our Pc2 as their Nb8, our Pe3 as their Nf6; transferred move g3h4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3520` (ply 13) [structural distance $d=56$, QAP obj=31]: the master played Bg6; under μ our Bg3 plays the role of that game’s Bf4; cross-type roles: our Pc2 as their Nb8, our Pe3 as their Nf6; transferred move g3h4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Bh4 (summed QAP objective 186, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



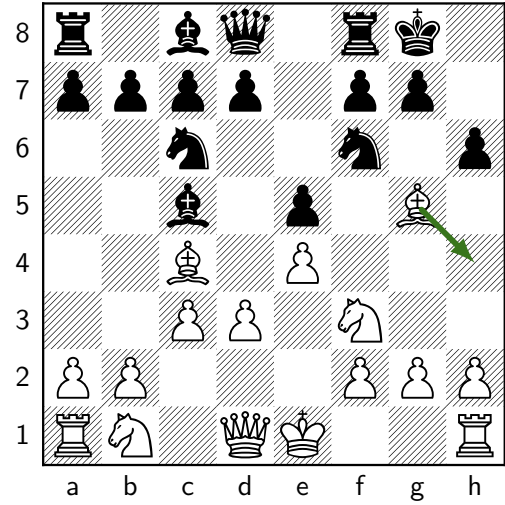
(a)



(b)

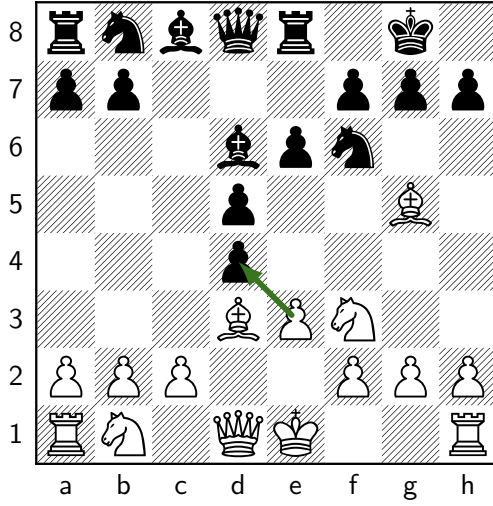


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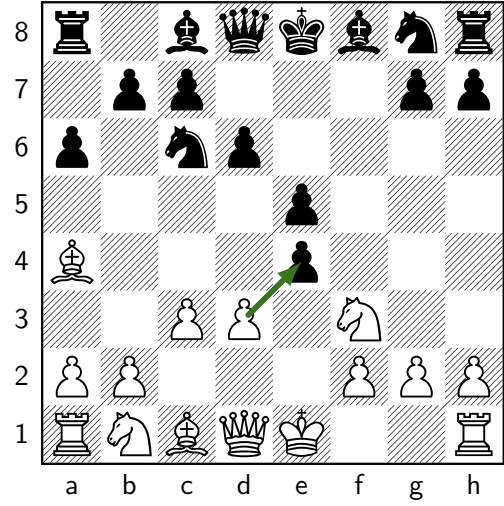


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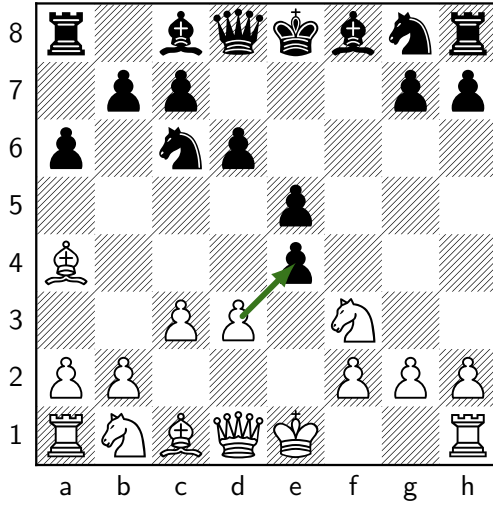
Figure 2: **Move 8, White: Bg5.** (a) Our position; Melanie plays Bg5 (h4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3565` (ply 10) [structural distance $d=54$, QAP obj=38]: the master played Bh4; under μ our Bh4 plays the role of that game’s Bg5; cross-type roles: our Pc2 as their Nb1, our Nb1 as their Pb2; transferred move h4g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3566` (ply 10) [structural distance $d=54$, QAP obj=38]: the master played Bh4; under μ our Bh4 plays the role of that game’s Bg5; cross-type roles: our Pc2 as their Nb1, our Nb1 as their Pb2; transferred move h4g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1743` (ply 12) [structural distance $d=45$, QAP obj=38]: the master played Bh4; under μ our Bh4 plays the role of that game’s Bg5; cross-type roles: our pb7 as their Ra1, our pc5 as their Nb1; transferred move h4g5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bg5 (summed QAP objective 152, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



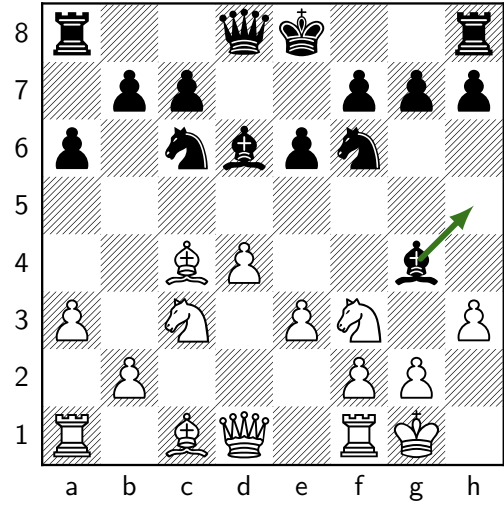
(a)



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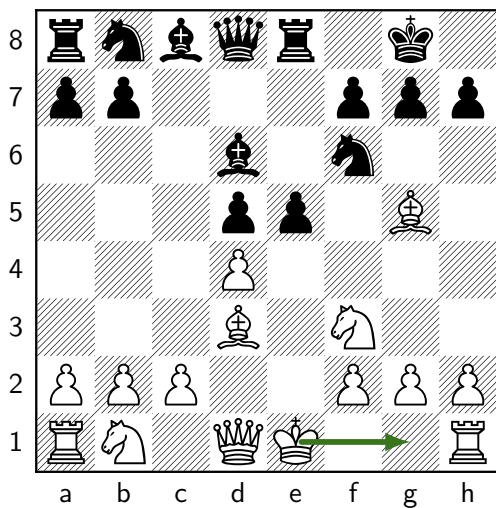


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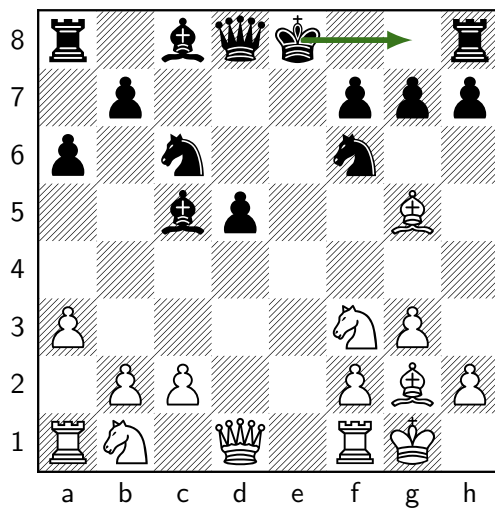


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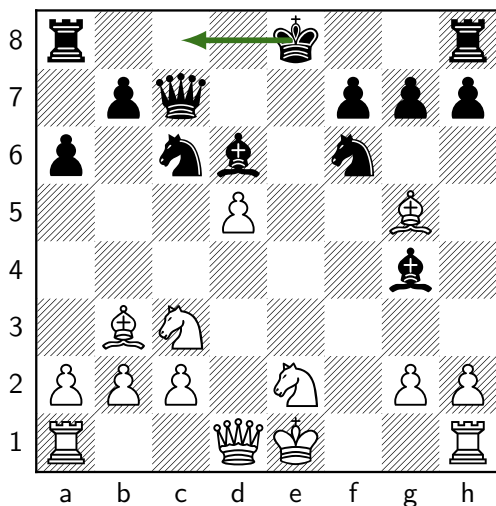
Figure 3: **Move 9, White: exd4.** (a) Our position; Melanie plays exd4 (e3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2790 (ply 12) [structural distance $d=72$, QAP obj=45]: the master played dxe4; under μ our Pe3 plays the role of that game’s Pd3; cross-type roles: our pa7 as their Ra1, our pe6 as their Nb1; transferred move e3d4 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2791 (ply 12) [structural distance $d=72$, QAP obj=45]: the master played dxe4; under μ our Pe3 plays the role of that game’s Pd3; cross-type roles: our pa7 as their Ra1, our pe6 as their Nb1; transferred move e3d4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3776 (ply 17) [structural distance $d=65$, QAP obj=67]: the master played Bh5; under μ our Bg5 plays the role of that game’s Bg5; cross-type roles: our Pb2 as their Ra8, our Rh1 as their Qd8; transferred move g5h4. Synthesis: 2 retrieved analogs transferred to exd4 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



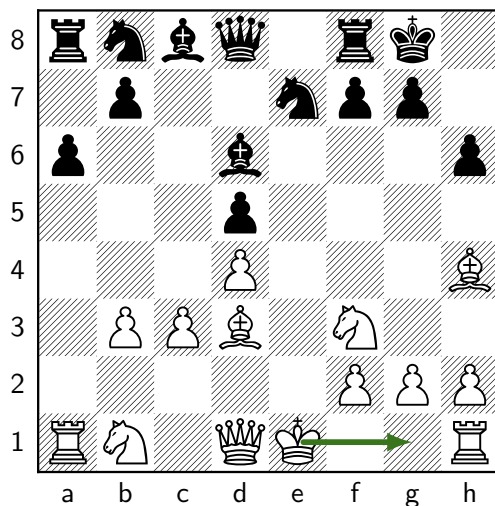
(a)



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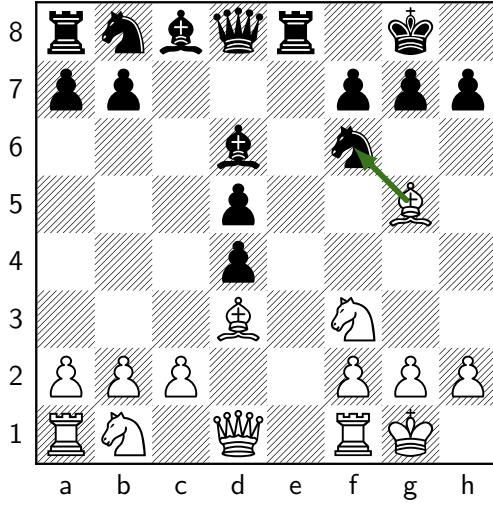


(c)

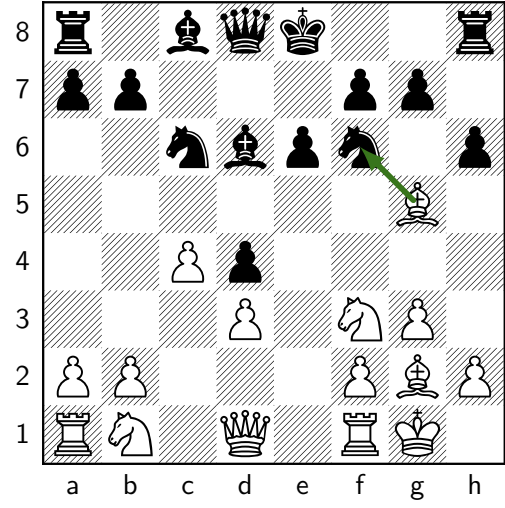


(d)

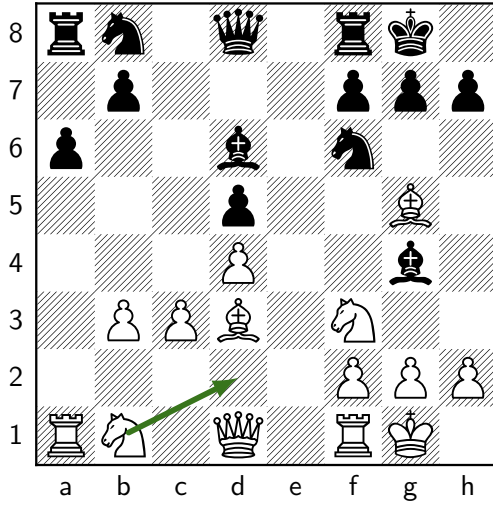
Figure 4: **Move 10, White: 0-0.** (a) Our position; Melanie plays 0-0 (**e1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2506` (ply 19) [structural distance $d=71$, QAP obj=49]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bd3* as their *Ra8*, our *Pg2* as their *Bc8*; transferred move **e1g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#136` (ply 25) [structural distance $d=70$, QAP obj=47]: the master played 0-0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pa2* as their *Rh8*, our *Bd3* as their *Pb7*; transferred move **e1g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#1943` (ply 22) [structural distance $d=45$, QAP obj=31]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bd3* as their *Ra1*, our *Ph2* as their *Nb1*; transferred move **e1g1 (endorses this move)**. Synthesis: 5 retrieved analogs transferred to 0-0 (summed QAP objective 155, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



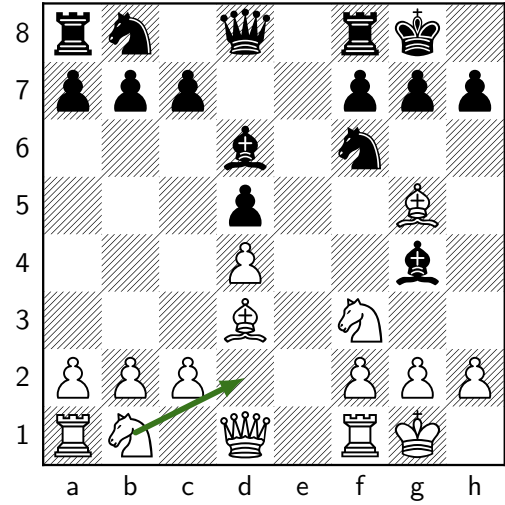
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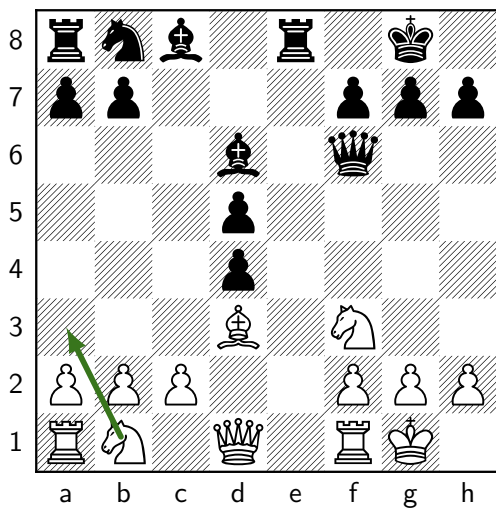


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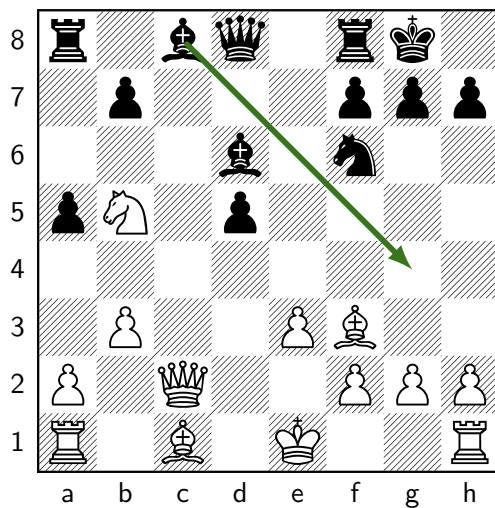


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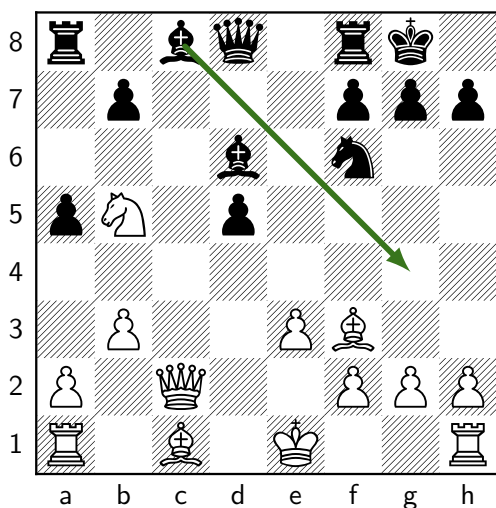
Figure 5: **Move 11, White: Bxf6.** (a) Our position; Melanie plays Bxf6 (g5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#989` (ply 18) [structural distance $d=79$, QAP obj=66]: the master played Bxf6; under μ our Bg5 plays the role of that game’s Bg5; cross-type roles: our Ph2 as their Ra1, our Ra1 as their Nb1; transferred move g5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1942` (ply 22) [structural distance $d=57$, QAP obj=90]: the master played Nbd2; under μ our Pc2 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our bd6 as their Ra1, our re8 as their Bd3; transferred move c2c3. (d) Retrieved from `ccr1_top_p0003#2906` (ply 14) [structural distance $d=41$, QAP obj=89]: the master played Nbd2; under μ our Rf1 plays the role of that game’s Nb1 (a **cross-type** correspondence); cross-type roles: our Nb1 as their Ra1, our Nf3 as their Rf1; transferred move f1e1. Synthesis: 1 retrieved analog transferred to Bxf6 (summed QAP objective 66, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



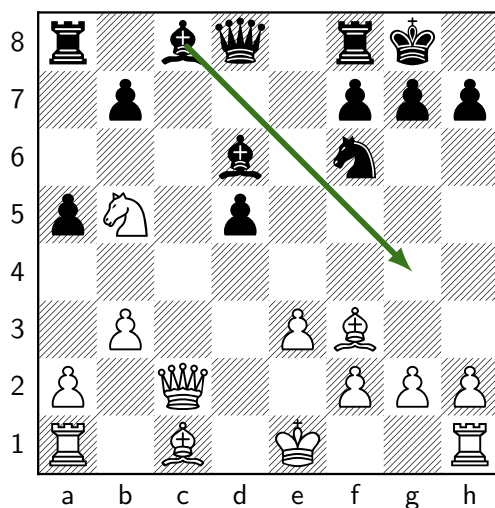
(a)



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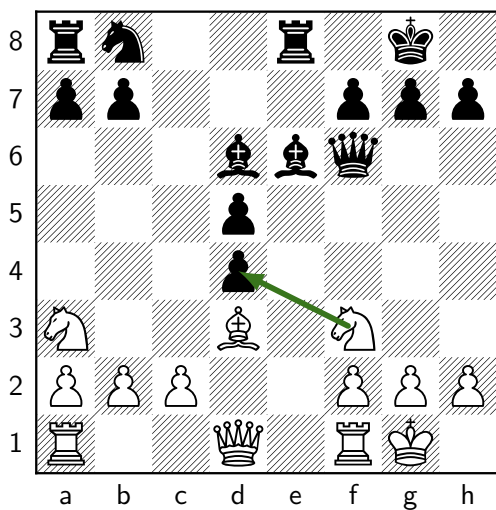


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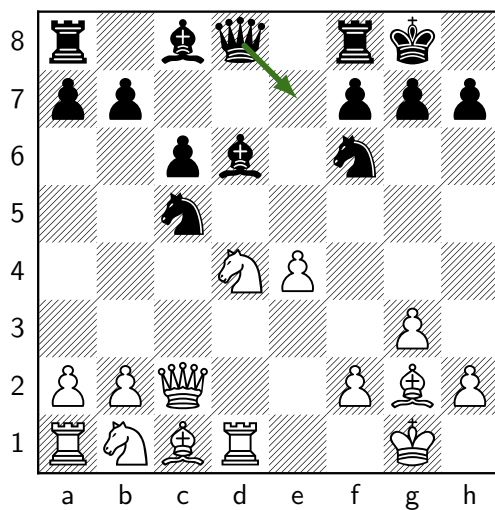


(d)

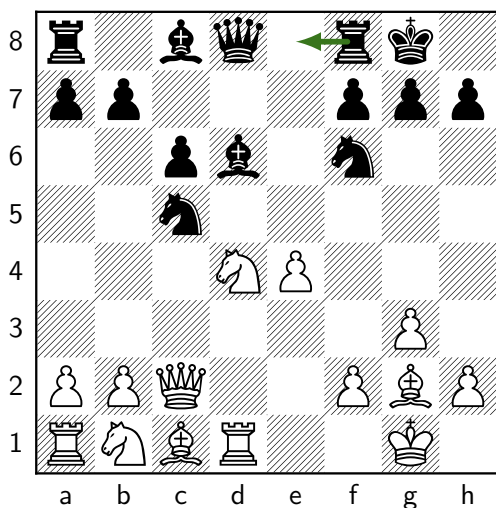
Figure 6: **Move 12, White: Na3.** (a) Our position; Melanie plays Na3 (b1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#258` (ply 25) [structural distance $d=69$, QAP obj=24]: the master played Bg4; under μ our Nb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Ra8, our bd6 as their Pb7; transferred move b1a3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1473` (ply 25) [structural distance $d=69$, QAP obj=24]: the master played Bg4; under μ our Nb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Ra8, our bd6 as their Pb7; transferred move b1a3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#401` (ply 25) [structural distance $d=69$, QAP obj=24]: the master played Bg4; under μ our Nb1 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our bc8 as their Ra8, our bd6 as their Pb7; transferred move b1a3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Na3 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



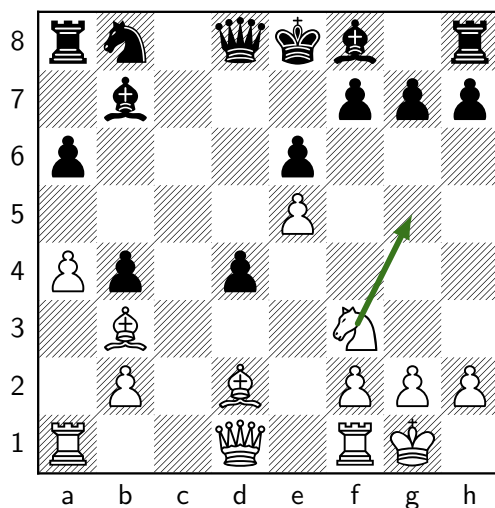
(a)



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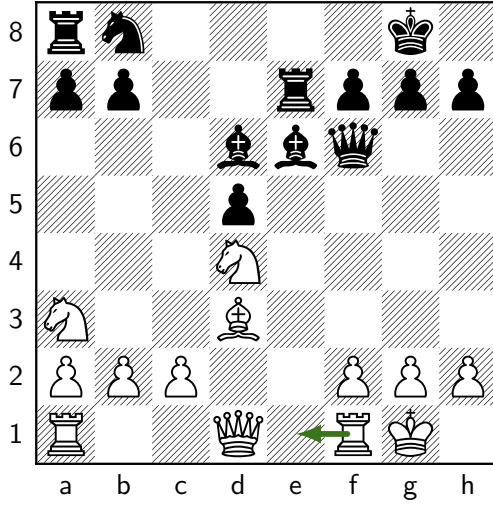


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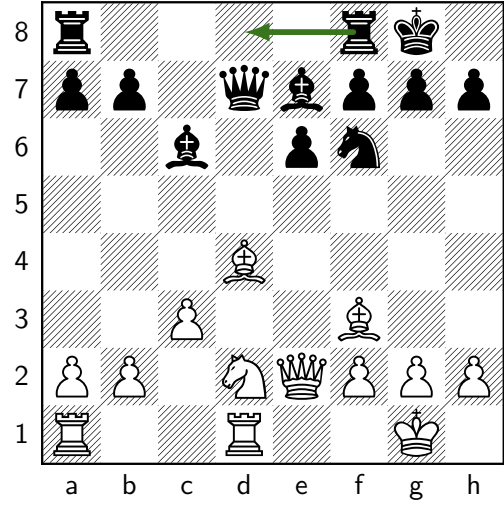


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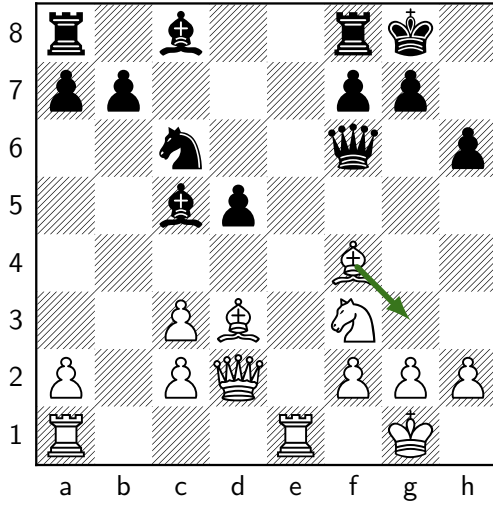
Figure 7: **Move 13, White: Nxd4.** (a) Our position; Melanie plays Nxd4 (f3d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2822` (ply 23) [structural distance $d=70$, QAP obj=73]: the master played Qe7; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pb7 as their Bc8, our be6 as their Pb7; transferred move d1e2. (c) Retrieved from `ccr1_top_p0002#2823` (ply 23) [structural distance $d=70$, QAP obj=73]: the master played Re8; under μ our Rf1 plays the role of that game’s Rf1; cross-type roles: our pb7 as their Bc8, our be6 as their Pb7; transferred move f1e1. (d) Retrieved from `ccr1_top_p0000#1717` (ply 24) [structural distance $d=71$, QAP obj=60]: the master played Ng5; under μ our Nf3 plays the role of that game’s Nf3; cross-type roles: our pb7 as their Ra1, our ra8 as their Qd1; transferred move f3g5. Synthesis: no retrieved analog transferred to Nxd4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



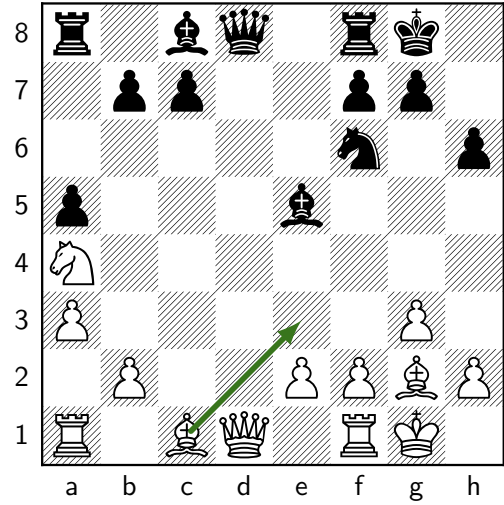
(a)



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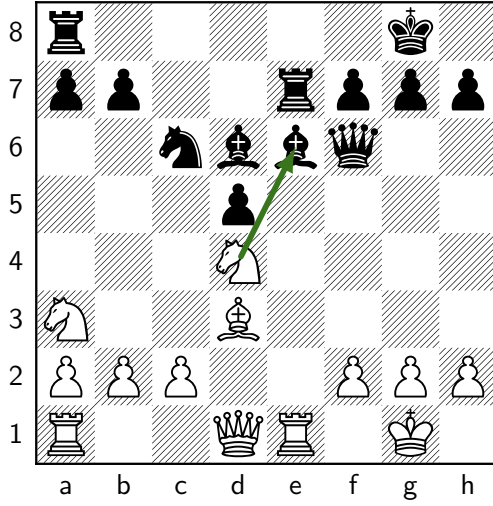


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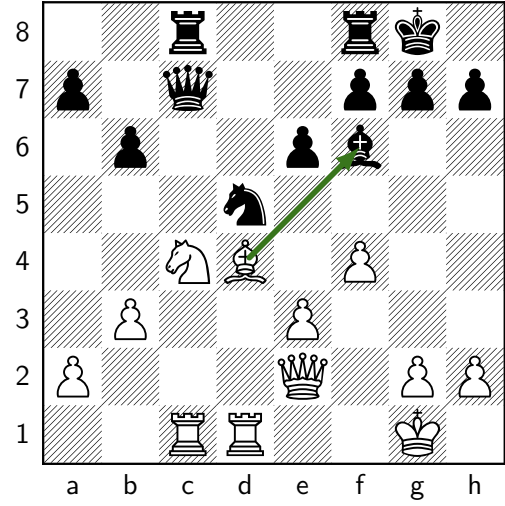


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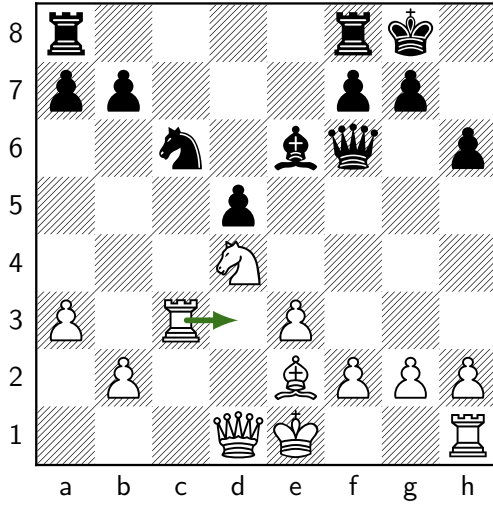
Figure 8: **Move 14, White: Re1.** (a) Our position; Melanie plays **Re1** (**f1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1450` (ply 27) [structural distance $d=58$, QAP obj=75]: the master played **Rfd8**; under μ our **Rf1** plays the role of that game’s **Rf1**; cross-type roles: our **nb8** as their **Pa7**, our **Qd1** as their **Ph7**; transferred move **f1e1** (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2734` (ply 30) [structural distance $d=75$, QAP obj=75]: the master played **Bg3**; under μ our **Nd4** plays the role of that game’s **Bf4** (a **cross-type** correspondence); cross-type roles: our **pd5** as their **Ra1**, our **nb8** as their **Pc2**; transferred move **d4b3**. (d) Retrieved from `ccrl_top_p0003#285` (ply 26) [structural distance $d=74$, QAP obj=46]: the master played **Be3**; under μ our **Bd3** plays the role of that game’s **Bc1**; cross-type roles: our **Qd1** as their **Ra1**, our **Na3** as their **Qd1**; transferred move **d3c4**. Synthesis: 1 retrieved analog transferred to **Re1** (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



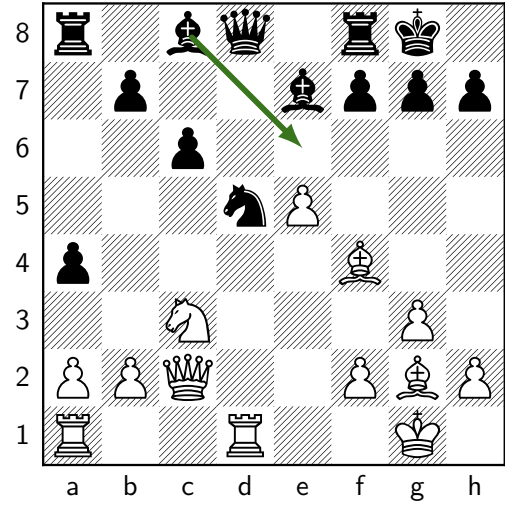
(a)



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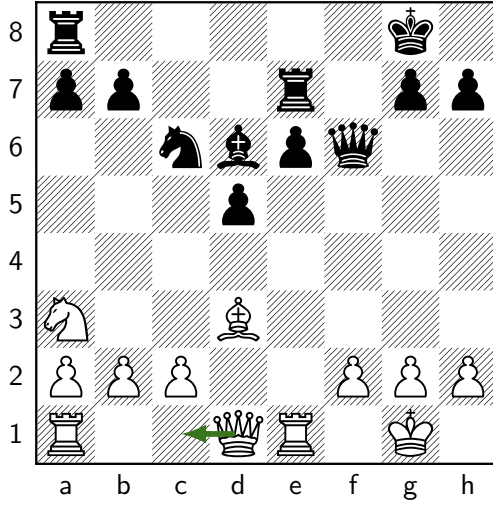


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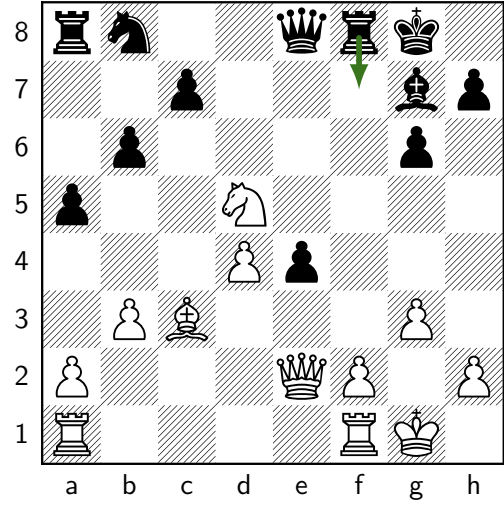


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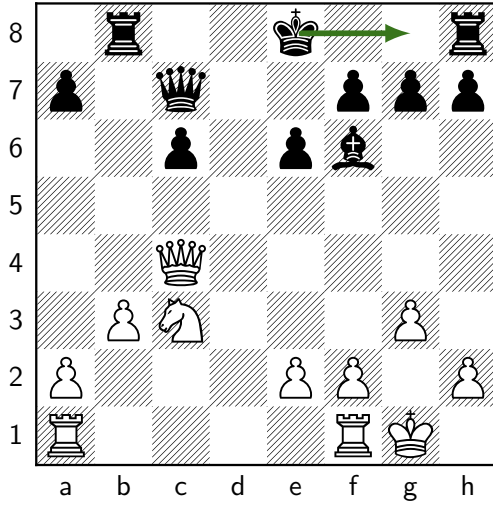
Figure 9: **Move 15, White: Nxe6.** (a) Our position; Melanie plays Nxe6 (d4e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2744` (ply 38) [structural distance $d=88$, QAP obj=80]: the master played Bxf6; under μ our Nd4 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our Qd1 as their Pa2, our Pa2 as their Qe2; transferred move d4e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2688` (ply 26) [structural distance $d=86$, QAP obj=142]: the master played Rd3; under μ our Nd4 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Re1 as their Qd1, our Pg2 as their Rh1; transferred move d4e2. (d) Retrieved from `ccr1_top_p0001#1488` (ply 31) [structural distance $d=89$, QAP obj=125]: the master played Be6; under μ our Na3 plays the role of that game’s Bc1 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Ra8, our Ra1 as their Pb7; transferred move a3c4. Synthesis: 1 retrieved analog transferred to Nxe6 (summed QAP objective 80, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



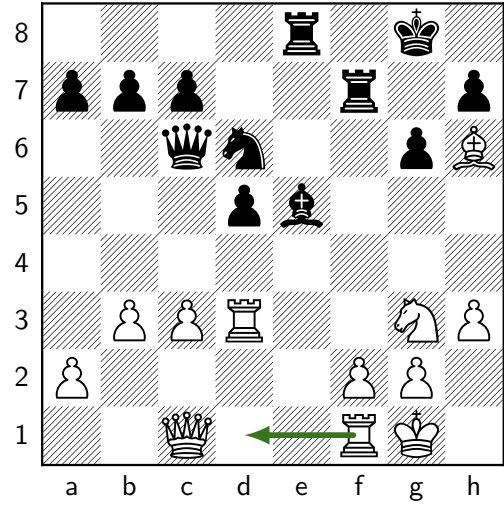
(a)



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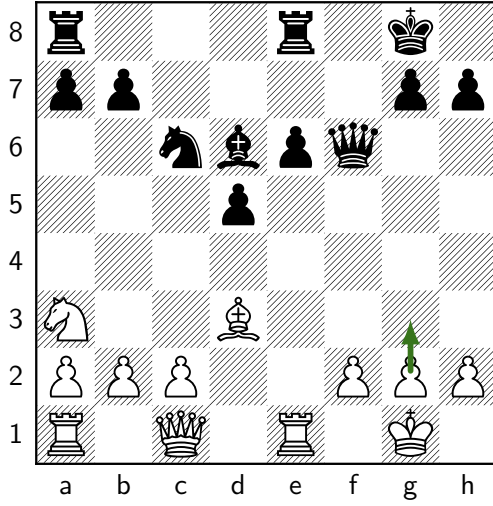


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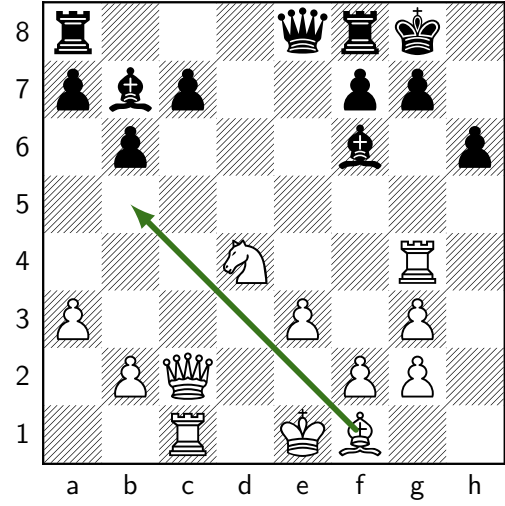


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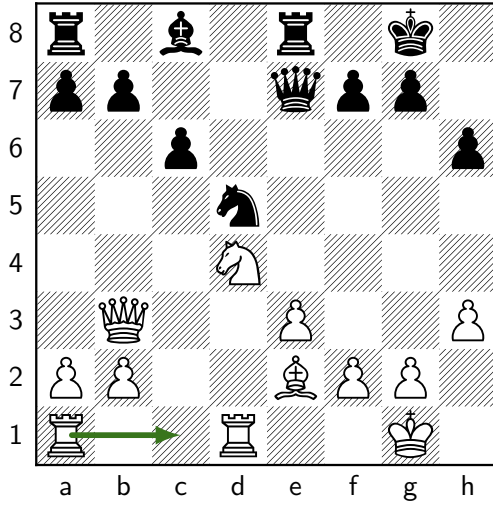
Figure 10: **Move 16, White: Qc1.** (a) Our position; Melanie plays Qc1 (d1c1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2962` (ply 35) [structural distance $d=71$, QAP obj=51]: the master played Rf7; under μ our Re1 plays the role of that game’s Rf1; cross-type roles: our Pg2 as their Nb8, our Na3 as their Pc7; transferred move e1f1. (c) Retrieved from `ccr1_top_p0003#821` (ply 31) [structural distance $d=73$, QAP obj=44]: the master played 0-0; under μ our Kg1 plays the role of that game’s Ke1; cross-type roles: our Qd1 as their Rh8, our Bd3 as their Qc7; transferred move g1h1. (d) Retrieved from `ccr1_top_p0000#774` (ply 42) [structural distance $d=68$, QAP obj=37]: the master played Rfd1; under μ our Pg2 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Qc1, our Qd1 as their Pb3; transferred move g2g3. Synthesis: no retrieved analog transferred to Qc1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



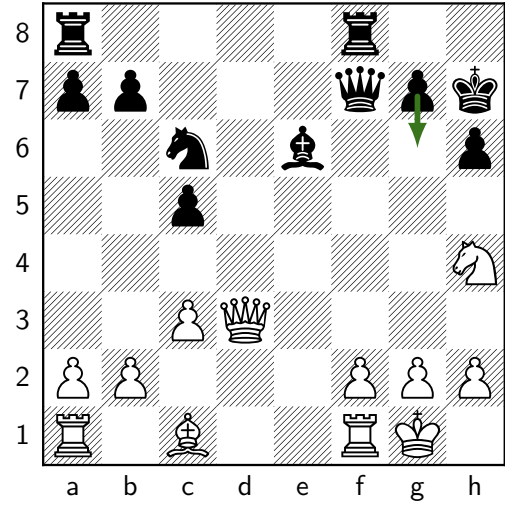
(a)



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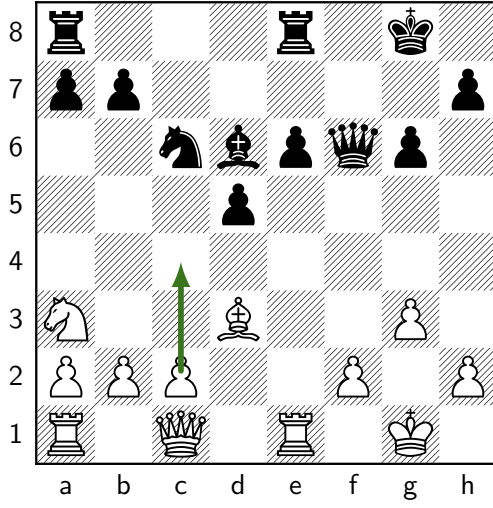


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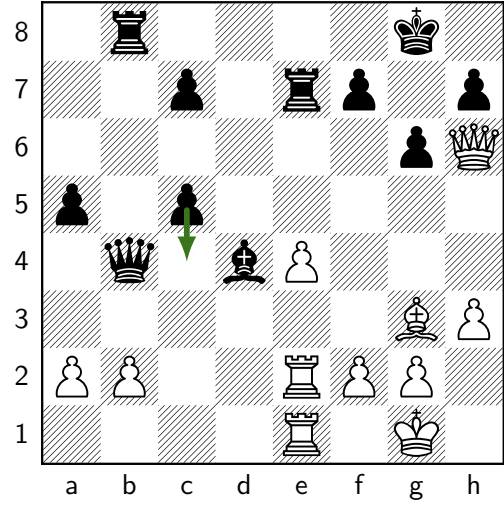


(d)

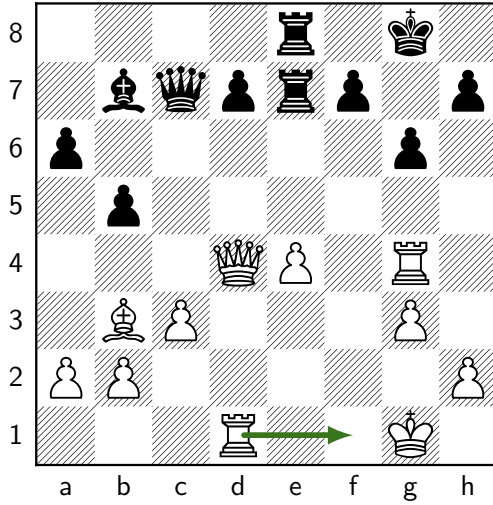
Figure 11: **Move 17, White: g3.** (a) Our position; Melanie plays g3 (g2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1885` (ply 38) [structural distance $d=66$, QAP obj=43]: the master played Bb5; under μ our *Pg2* plays the role of that game’s *Bf1* (a **cross-type** correspondence); cross-type roles: our *Pf2* as their *Rc1*, our *Ra1* as their *Pb2*; transferred move g2g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3339` (ply 34) [structural distance $d=69$, QAP obj=38]: the master played Rac1; under μ our *Pg2* plays the role of that game’s *Ra1* (a **cross-type** correspondence); cross-type roles: our *ra8* as their *Pa2*, our *Qc1* as their *Pb2*; transferred move g2g3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#1230` (ply 39) [structural distance $d=69$, QAP obj=25]: the master played g6; under μ our *Pg2* plays the role of that game’s *Pg2*; cross-type roles: our *pd5* as their *Rf8*, our *Bd3* as their *Pb7*; transferred move g2g3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g3 (summed QAP objective 106, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



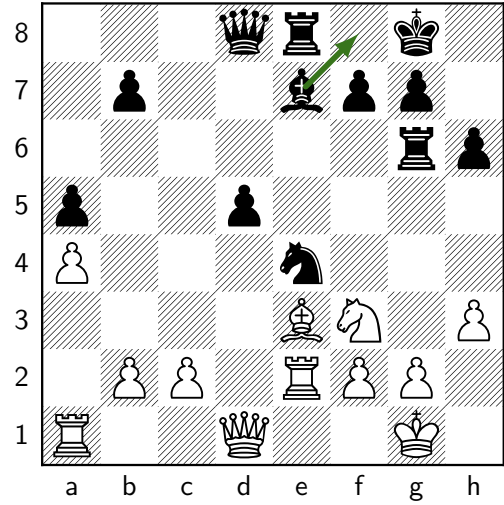
(a)



(b)

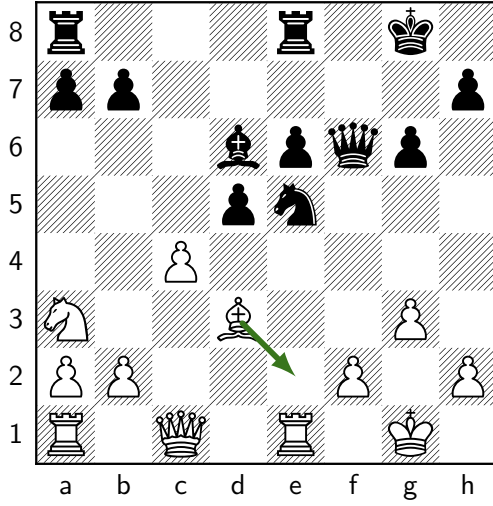


(c)

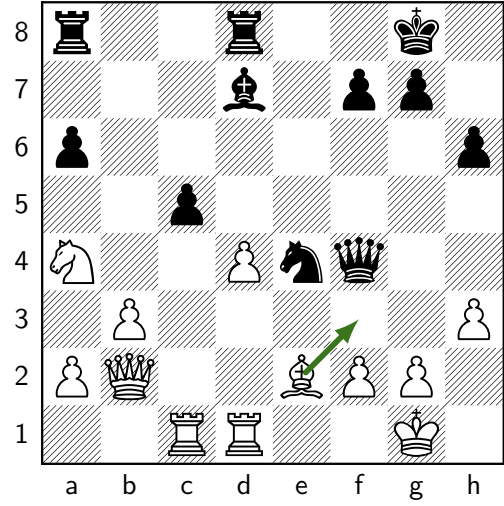


(d)

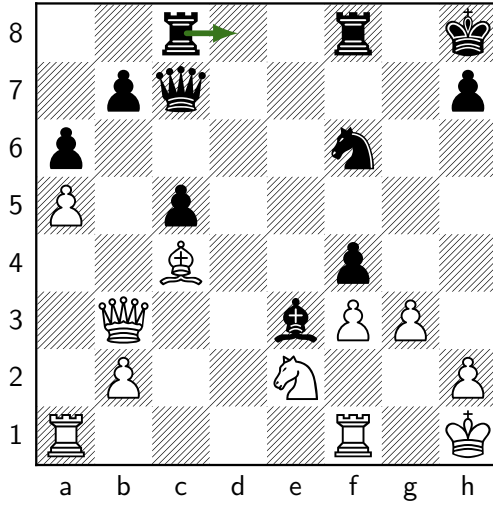
Figure 12: **Move 18, White: c4.** (a) Our position; Melanie plays c4 (c2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2528` (ply 51) [structural distance $d=65$, QAP obj=50]: the master played c4; under μ our $Pc2$ plays the role of that game’s $Pc4$; cross-type roles: our $nc6$ as their $Rb8$, our $gf6$ as their $Re7$; transferred move c2c4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#421` (ply 48) [structural distance $d=56$, QAP obj=47]: the master played Rf1; under μ our $Pc2$ plays the role of that game’s $Rd1$ (a **cross-type** correspondence); cross-type roles: our $ra8$ as their $Pa2$, our $Re1$ as their $Bb3$; transferred move c2c4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1170` (ply 37) [structural distance $d=66$, QAP obj=61]: the master played Bf8; under μ our $Pg3$ plays the role of that game’s $Be2$ (a **cross-type** correspondence); cross-type roles: our $pa7$ as their $Qd8$, our $pb7$ as their $Re8$; transferred move g3g4. Synthesis: 2 retrieved analogs transferred to c4 (summed QAP objective 97, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



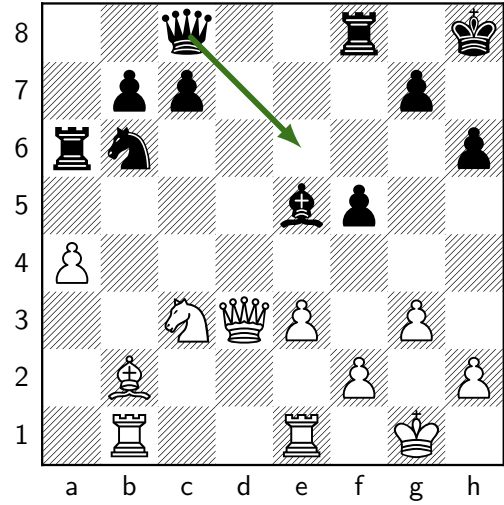
(a)



(b)

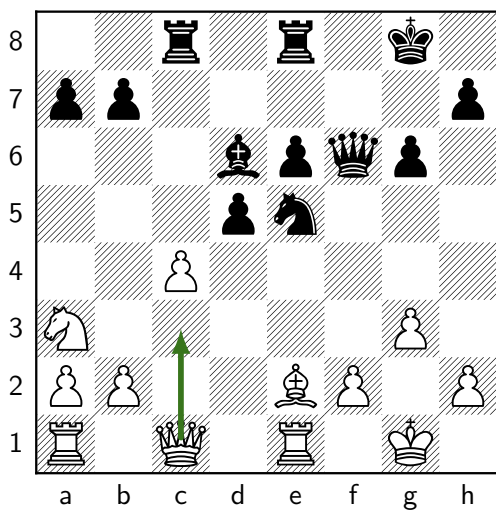


(c)

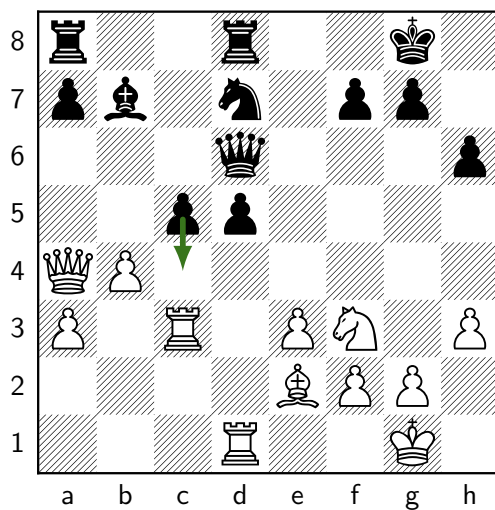


(d)

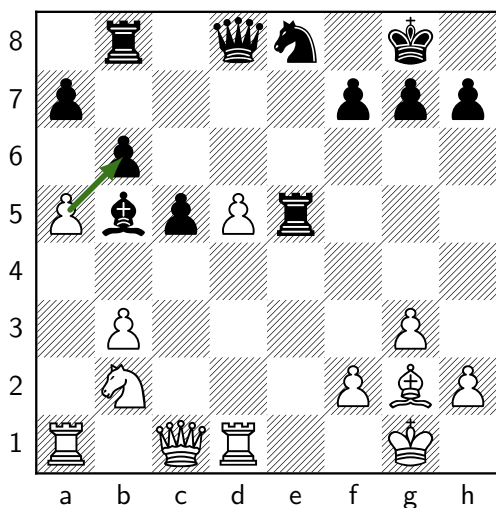
Figure 13: **Move 19, White: Be2.** (a) Our position; Melanie plays Be2 (d3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1697` (ply 44) [structural distance $d=68$, QAP obj=66]: the master played Bf3; under μ our Na3 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Bd3 as their Rc1, our pe6 as their Rd1; transferred move a3b5. (c) Retrieved from `ccr1_top_p0001#3765` (ply 53) [structural distance $d=65$, QAP obj=61]: the master played Rcd8; under μ our Pg3 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Ra1 as their Pb7, our Bd3 as their Pa6; transferred move g3g4. (d) Retrieved from `ccr1_top_p0000#575` (ply 45) [structural distance $d=66$, QAP obj=49]: the master played Qe6; under μ our Qc1 plays the role of that game’s Qc1; cross-type roles: our Bd3 as their Ra6, our Pc4 as their Nb6; transferred move c1c3. Synthesis: no retrieved analog transferred to Be2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



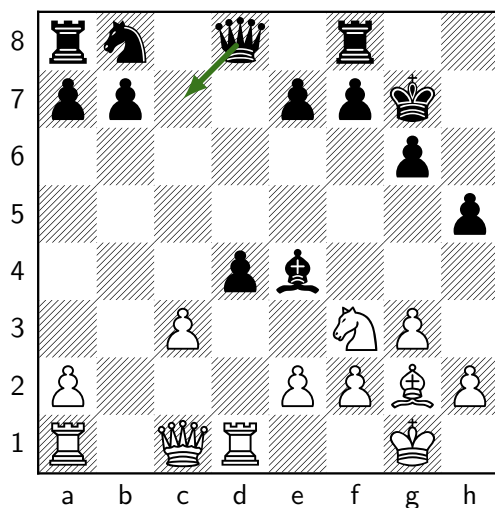
(a)



(b)

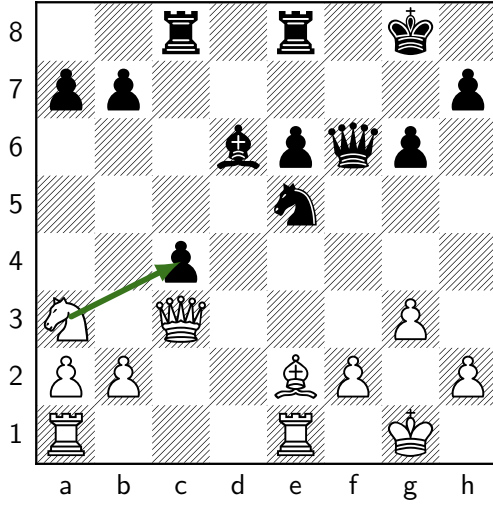


(c)

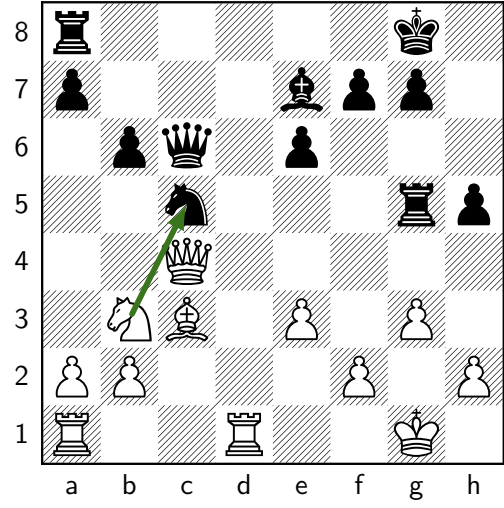


(d)

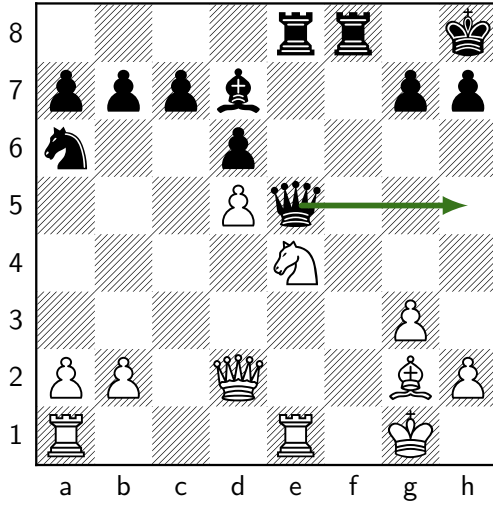
Figure 14: **Move 20, White: Qc3.** (a) Our position; Melanie plays Qc3 (c1c3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2023` (ply 35) [structural distance $d=61$, QAP obj=117]: the master played c4; under μ our $Pc4$ plays the role of that game’s $Pc4$; cross-type roles: our $Pb2$ as their $Ra8$, our $Na3$ as their $Pa7$; transferred move c4c5. (c) Retrieved from `ccr1_top_p0000#3616` (ply 46) [structural distance $d=61$, QAP obj=82]: the master played axb6; under μ our $Pc4$ plays the role of that game’s $Pa5$; cross-type roles: our $Qc1$ as their $Ra1$, our $bd6$ as their $Qc1$; transferred move c4d5. (d) Retrieved from `ccr1_top_p0003#257` (ply 31) [structural distance $d=66$, QAP obj=64]: the master played Qc7; under μ our $Qc1$ plays the role of that game’s $Qd1$; cross-type roles: our $Pb2$ as their $Ra8$, our $Ra1$ as their $Nb8$; transferred move c1h6. Synthesis: no retrieved analog transferred to Qc3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



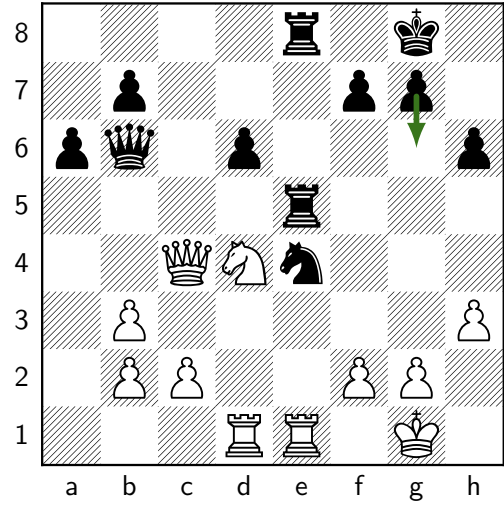
(a)



(b)

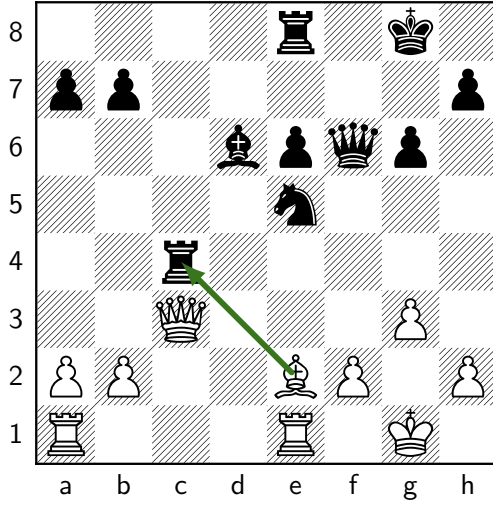


(c)

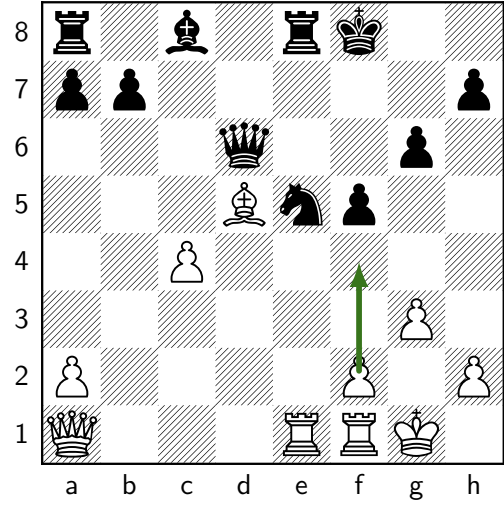


(d)

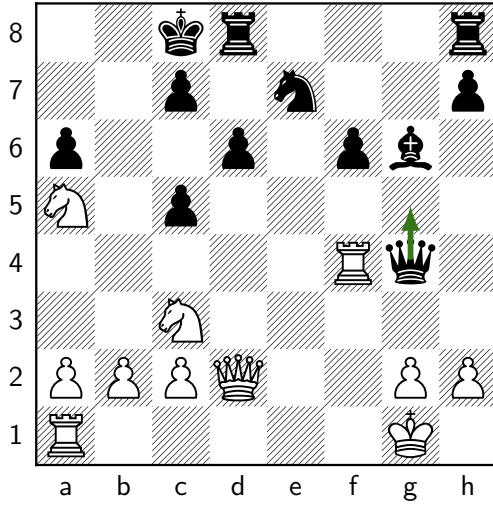
Figure 15: **Move 21, White: Nxc4.** (a) Our position; Melanie plays Nxc4 (a3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#468` (ply 44) [structural distance $d=70$, QAP obj=48]: the master played Nxc5; under μ our Na3 plays the role of that game’s Nb3; cross-type roles: our Be2 as their Rd1, our Pg3 as their Bc3; transferred move a3c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#816` (ply 33) [structural distance $d=70$, QAP obj=62]: the master played Qh5; under μ our Qc3 plays the role of that game’s Qe4; cross-type roles: our Pf2 as their Rf8, our Ph2 as their Bd7; transferred move c3a5. (d) Retrieved from `ccr1_top_p0002#240` (ply 47) [structural distance $d=67$, QAP obj=60]: the master played g6; under μ our Be2 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our Na3 as their Pa6, our Ra1 as their Ph6; transferred move e2g4. Synthesis: 1 retrieved analog transferred to Nxc4 (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



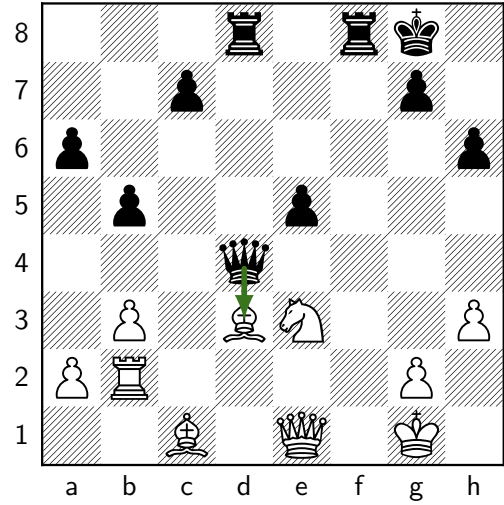
(a)



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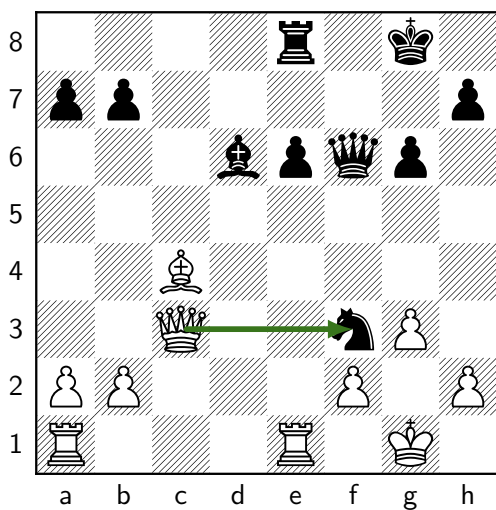


(c)

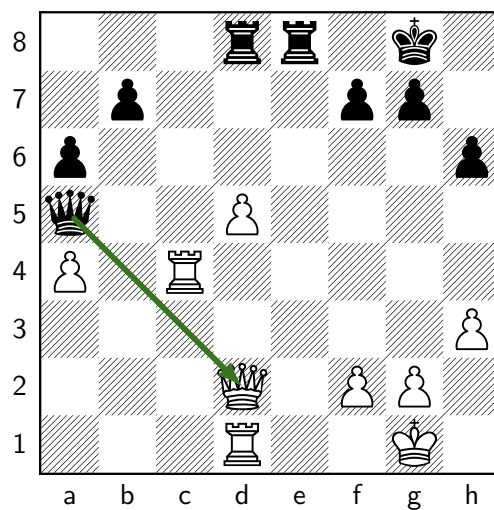


(d)

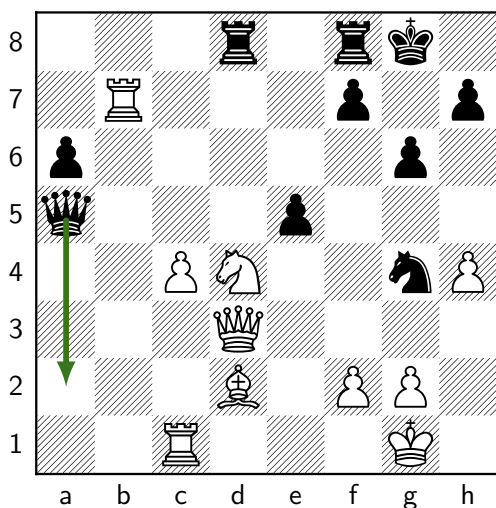
Figure 16: **Move 22, White: Bxc4.** (a) Our position; Melanie plays Bxc4 (e2c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2662 (ply 44) [structural distance $d=80$, QAP obj=26]: the master played f4; under μ our *Be2* plays the role of that game’s *Pf2* (a **cross-type** correspondence); cross-type roles: our *pe6* as their *Qa1*, our *Qc3* as their *Re1*; transferred move e2c4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1237 (ply 37) [structural distance $d=72$, QAP obj=56]: the master played Qg5; under μ our *Qc3* plays the role of that game’s *Qg5*; cross-type roles: our *Pa2* as their *Rh8*, our *Ra1* as their *Pa6*; transferred move c3b3. (d) Retrieved from ccr1_top_p0003#1194 (ply 47) [structural distance $d=73$, QAP obj=52]: the master played Qxd3; under μ our *Qc3* plays the role of that game’s *Qd5*; cross-type roles: our *Be2* as their *Rf8*, our *Ra1* as their *Pc7*; transferred move c3a5. Synthesis: 1 retrieved analog transferred to Bxc4 (summed QAP objective 26, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



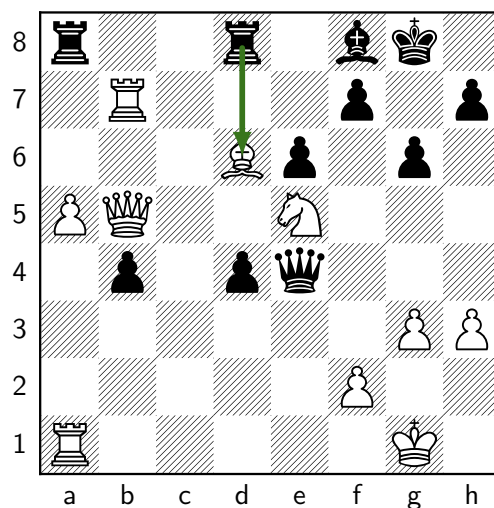
(a)



(b)

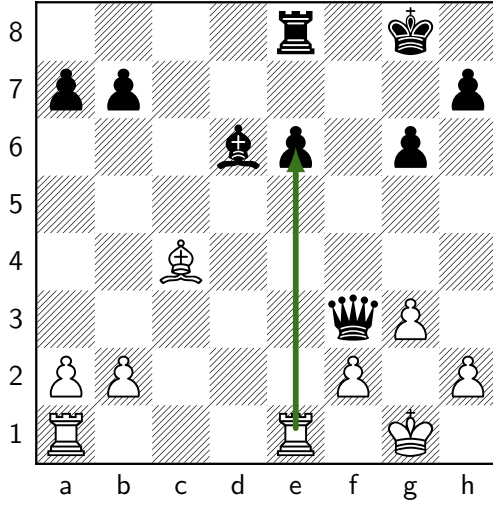


(c)

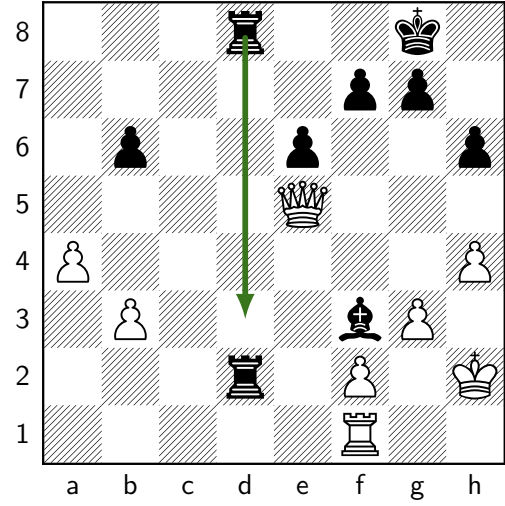


(d)

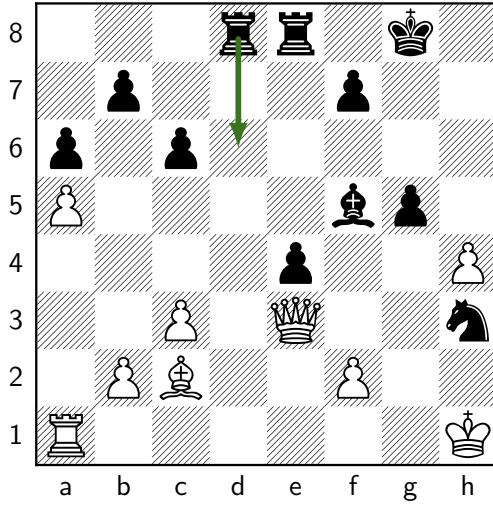
Figure 17: **Move 23, White: Qxf3**. (a) Our position; Melanie plays Qxf3 (c3f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1039` (ply 55) [structural distance $d=87$, QAP obj=24]: the master played Qxd2; under μ our $Qc3$ plays the role of that game's $Qa4$; cross-type roles: our $Bc4$ as their $Rd8$, our $Re1$ as their $Pb7$; transferred move c3f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#753` (ply 55) [structural distance $d=78$, QAP obj=20]: the master played Qa2; under μ our $Qc3$ plays the role of that game's $Qa4$; cross-type roles: our $Bc4$ as their $Rd8$, our $nf3$ as their $rb7$; transferred move c3f3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#314` (ply 59) [structural distance $d=76$, QAP obj=20]: the master played Rxd6; under μ our $Qc3$ plays the role of that game's $Rd1$ (a **cross-type** correspondence); cross-type roles: our $Bc4$ as their $Ra8$, our $Re1$ as their $Bf8$; transferred move c3f3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qxf3 (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



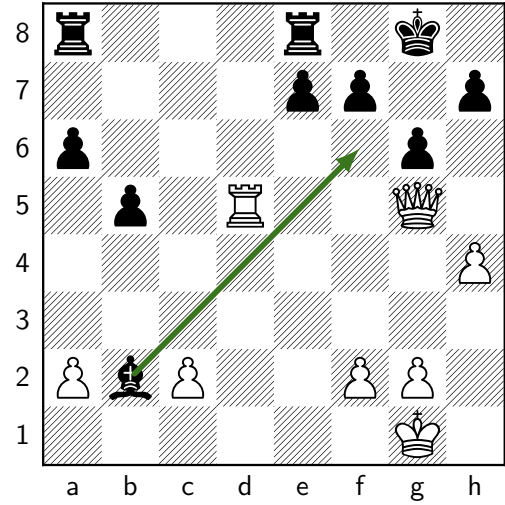
(a)



(b)

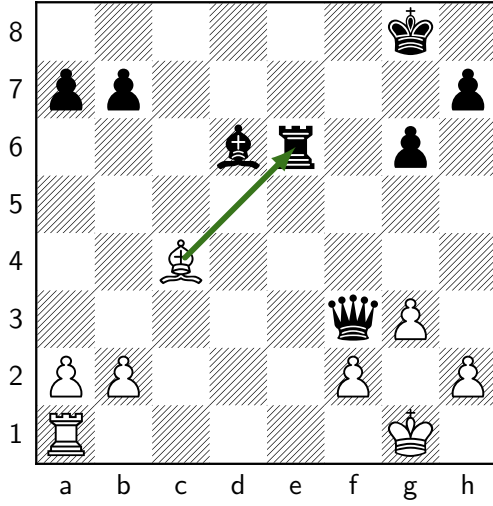


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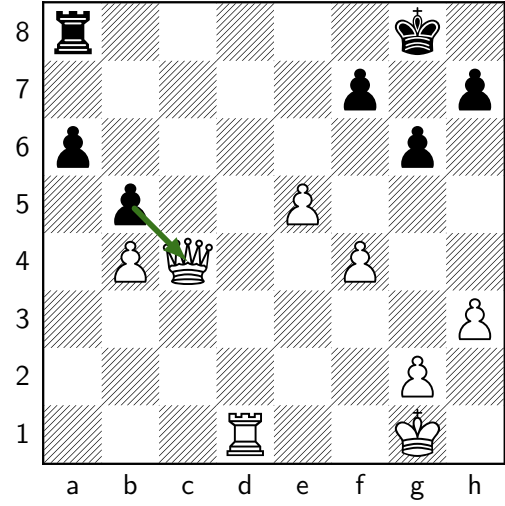


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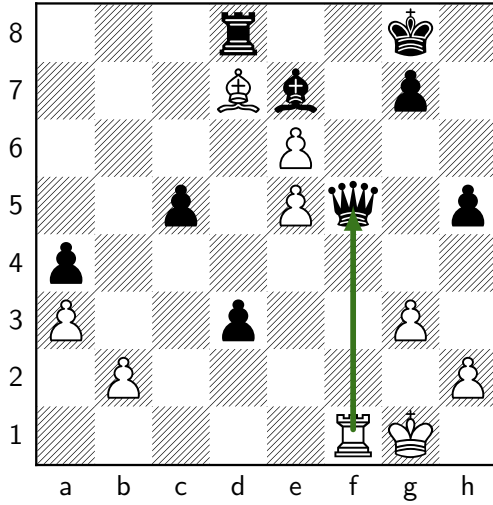
Figure 18: **Move 24, White: Rxe6.** (a) Our position; Melanie plays **Rxe6** (**e1e6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#997` (ply 57) [structural distance $d=47$, QAP obj=75]: the master played **R8d3**; under μ our *Ra1* plays the role of that game’s *Rd1*; cross-type roles: our *bd6* as their *pa4*, our *Re1* as their *Bf3*; transferred move **a1c1**. (c) Retrieved from `ccrl_top_p0001#1206` (ply 59) [structural distance $d=69$, QAP obj=73]: the master played **Rd6**; under μ our *Pa2* plays the role of that game’s *Rd1* (a **cross-type** correspondence); cross-type roles: our *Bc4* as their *Pb7*, our *Ra1* as their *Pa6*; transferred move **a2a4**. (d) Retrieved from `ccrl_top_p0001#2109` (ply 47) [structural distance $d=66$, QAP obj=50]: the master played **Bf6**; under μ our *Pa2* plays the role of that game’s *Bb7* (a **cross-type** correspondence); cross-type roles: our *Bc4* as their *rd5*, our *re8* as their *pc2*; transferred move **a2a3**. Synthesis: no retrieved analog transferred to **Rxe6** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



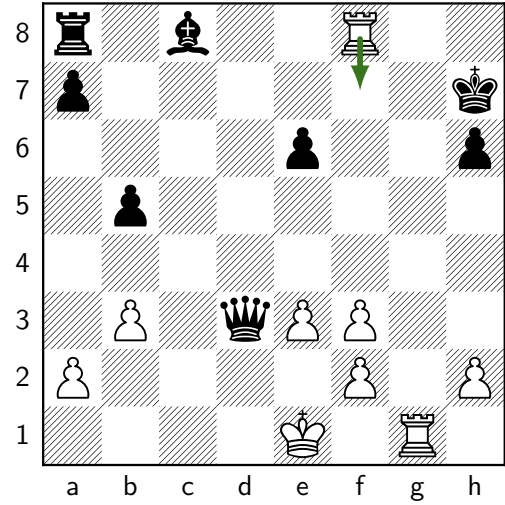
(a)



(b)

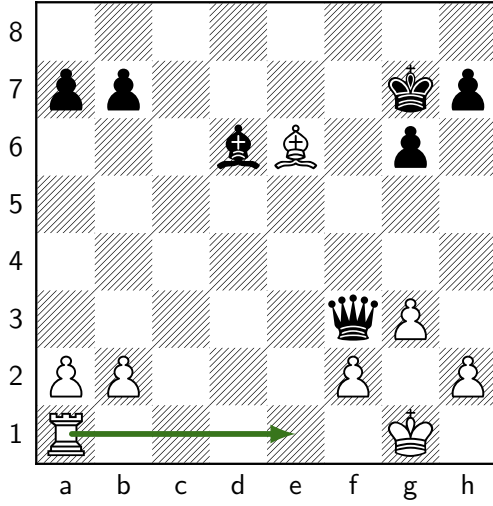


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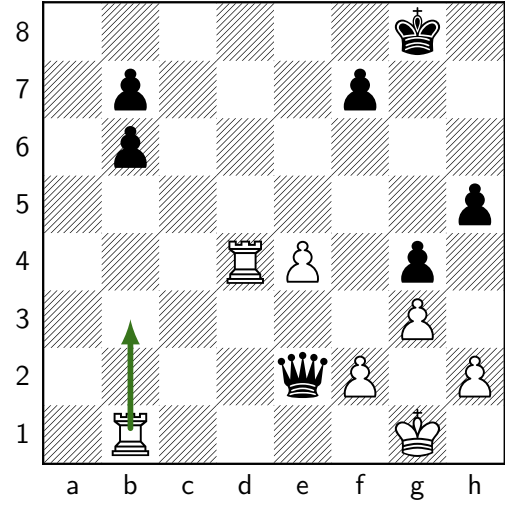


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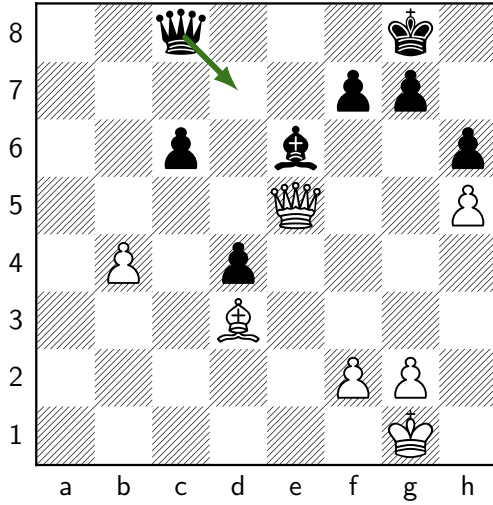
Figure 19: **Move 25, White: Bxe6+.** (a) Our position; Melanie plays Bxe6+ (c4e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#4000 (ply 55) [structural distance $d=63$, QAP obj=9]: the master played **bxc4**; under μ our *Bc4* plays the role of that game’s *Pb4* (a **cross-type** correspondence); cross-type roles: our *qf3* as their *Pg6*, our *bd6* as their *pb4*; transferred move **c4e6** (**endorses this move**). (c) Retrieved from ccr1_top_p0000#845 (ply 60) [structural distance $d=69$, QAP obj=32]: the master played **Rxf5**; under μ our *Pg3* plays the role of that game’s *Rf1* (a **cross-type** correspondence); cross-type roles: our *re6* as their *Bd7*, our *Ra1* as their *pg7*; transferred move **g3g4**. (d) Retrieved from ccr1_top_p0003#2399 (ply 52) [structural distance $d=62$, QAP obj=28]: the master played **Rf7+**; under μ our *Bc4* plays the role of that game’s *Rf8* (a **cross-type** correspondence); cross-type roles: our *bd6* as their *pb5*, our *ph7* as their *ra8*; transferred move **c4b5**. Synthesis: 1 retrieved analog transferred to Bxe6+ (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



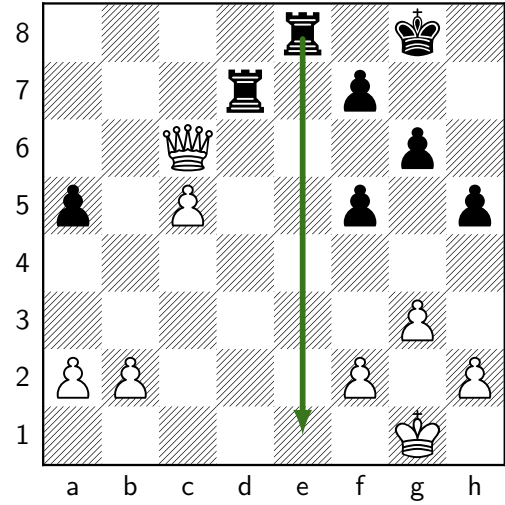
(a)



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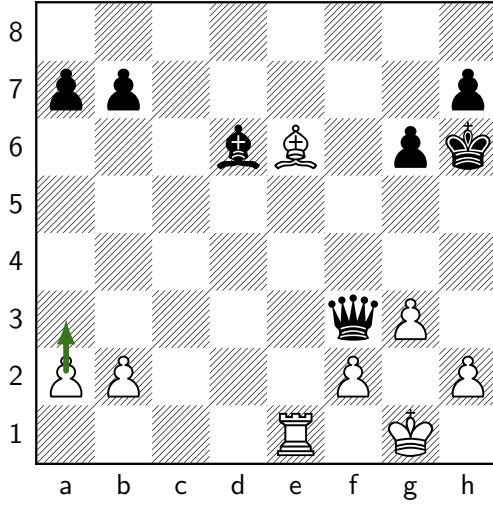


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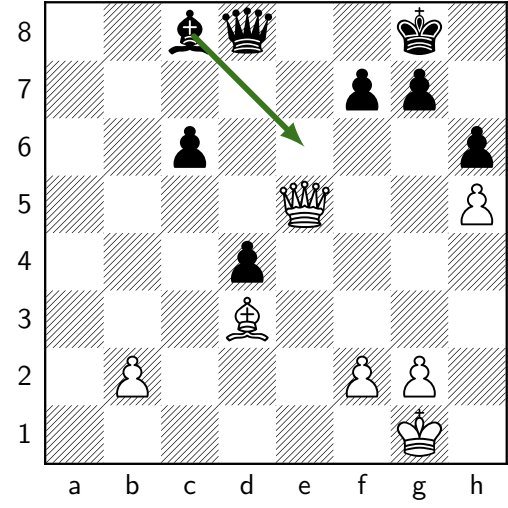


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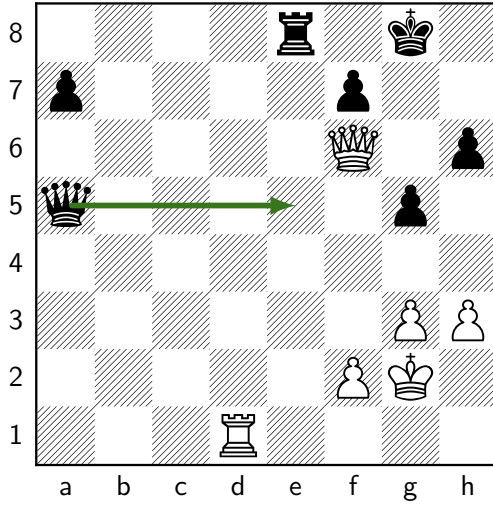
Figure 20: **Move 26, White: Re1.** (a) Our position; Melanie plays Re1 (a1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#97` (ply 66) [structural distance $d=58$, QAP obj=41]: the master played Rb3; under μ our *Ra1* plays the role of that game’s *Rb1*; cross-type roles: our *bd6* as their *Pg3*, our *Ph2* as their *Rd4*; transferred move a1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3697` (ply 59) [structural distance $d=54$, QAP obj=20]: the master played Qd7; under μ our *Ra1* plays the role of that game’s *Qc1* (a **cross-type** correspondence); cross-type roles: our *Be6* as their *Pg7*, our *Pg3* as their *Be6*; transferred move a1e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1317` (ply 57) [structural distance $d=56$, QAP obj=3]: the master played Re1+; under μ our *Ra1* plays the role of that game’s *Re1*; cross-type roles: our *Pf2* as their *Rd7*, our *Be6* as their *Pa5*; transferred move a1e1 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Re1 (summed QAP objective 103, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



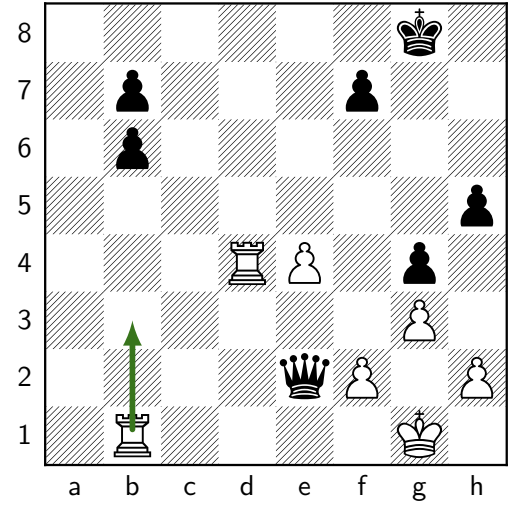
(a)



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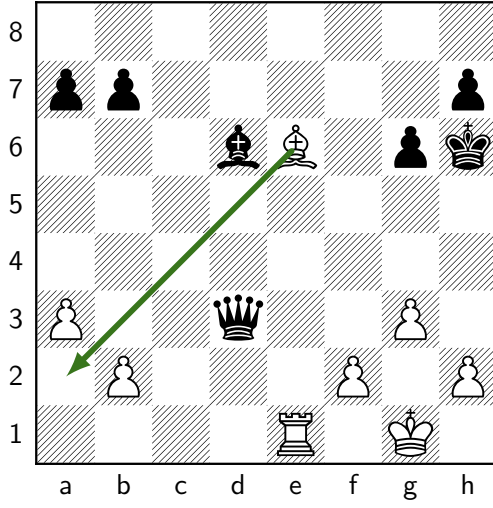


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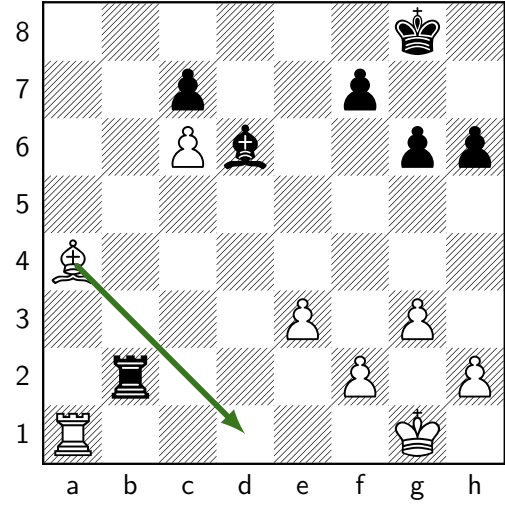


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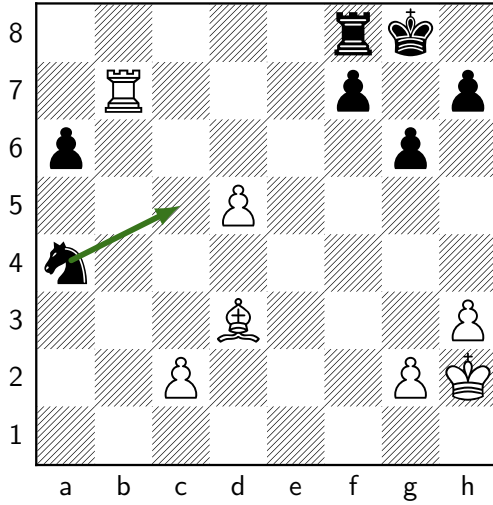
Figure 21: **Move 27, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3697` (ply 53) [structural distance $d=47$, QAP obj=38]: the master played **Be6**; under μ our *Pa2* plays the role of that game’s *Bc1* (a **cross-type** correspondence); cross-type roles: our *Re1* as their *Qd8*, our *Be6* as their *pb2*; transferred move **a2a3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#970` (ply 63) [structural distance $d=56$, QAP obj=76]: the master played **Qe5**; under μ our *Be6* plays the role of that game’s *Qa4* (a **cross-type** correspondence); cross-type roles: our *bd6* as their *ph3*, our *pg6* as their *rd1*; transferred move **e6f5**. (d) Retrieved from `ccr1_top_p0003#97` (ply 66) [structural distance $d=57$, QAP obj=40]: the master played **Rb3**; under μ our *Re1* plays the role of that game’s *Rb1*; cross-type roles: our *bd6* as their *Pg3*, our *Ph2* as their *Rd4*; transferred move **e1e2**. Synthesis: 1 retrieved analog transferred to **a3** (summed QAP objective 38, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



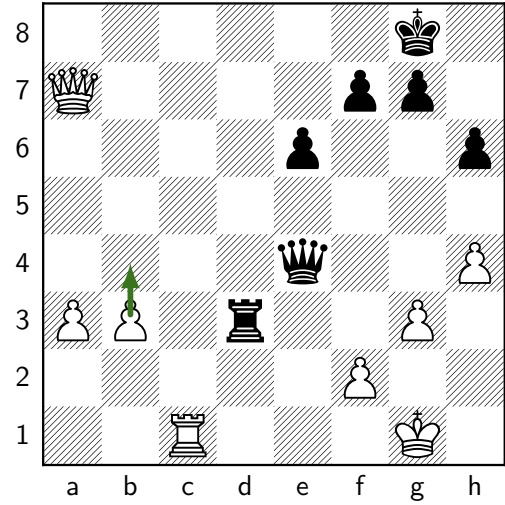
(a)



(b)

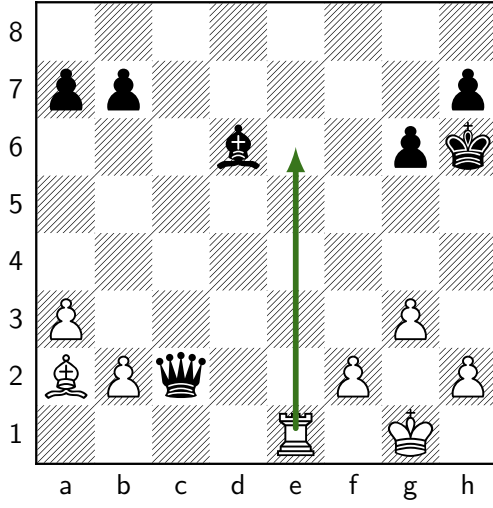


(c)

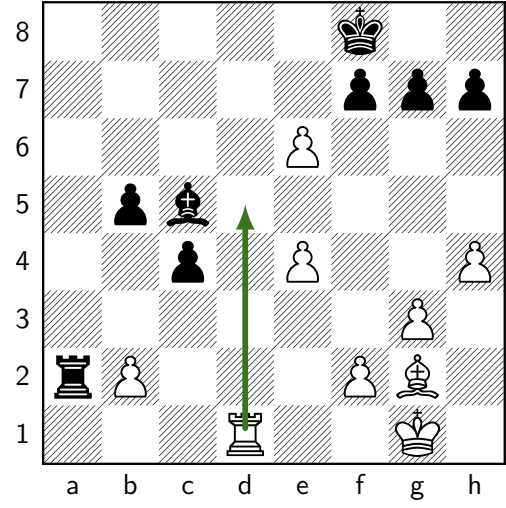


(d)

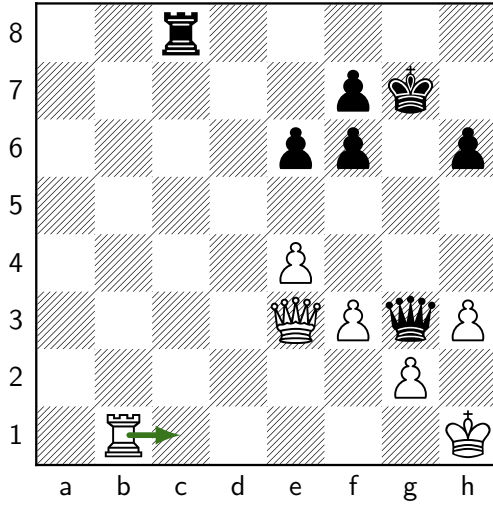
Figure 22: **Move 28, White: Ba2.** (a) Our position; Melanie plays Ba2 (e6a2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2791` (ply 62) [structural distance $d=57$, QAP obj=34]: the master played Bd1; under μ our *Be6* plays the role of that game’s *Ba4*; cross-type roles: our *qd3* as their *rb2*; transferred move **e6a2 (endorses this move)**. (c) Retrieved from `ccrl_top_p0000#2358` (ply 57) [structural distance $d=55$, QAP obj=49]: the master played Nc5; under μ our *Be6* plays the role of that game’s *Na5* (a **cross-type** correspondence); cross-type roles: our *qd3* as their *rb7*; transferred move **e6d5**. (d) Retrieved from `ccrl_top_p0003#314` (ply 70) [structural distance $d=54$, QAP obj=39]: the master played b4; under μ our *Pg3* plays the role of that game’s *Pb3*; cross-type roles: our *bd6* as their *rd3*, our *Be6* as their *Qa7*; transferred move **g3g4**. Synthesis: 1 retrieved analog transferred to Ba2 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



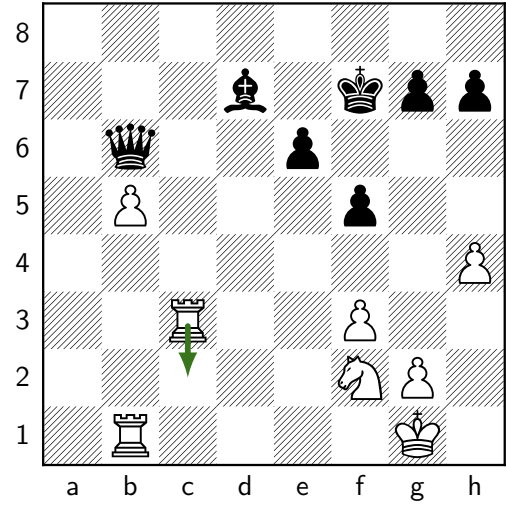
(a)



(b)

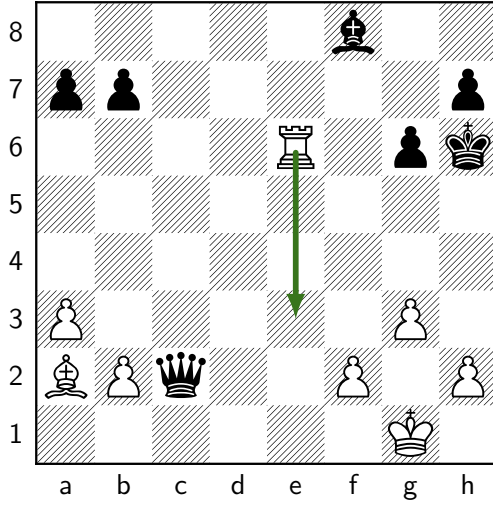


(c)

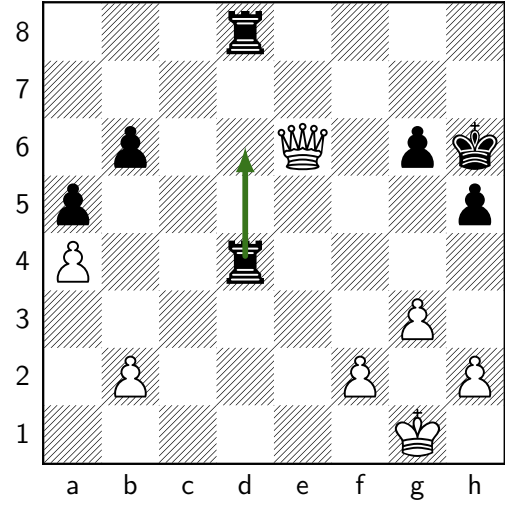


(d)

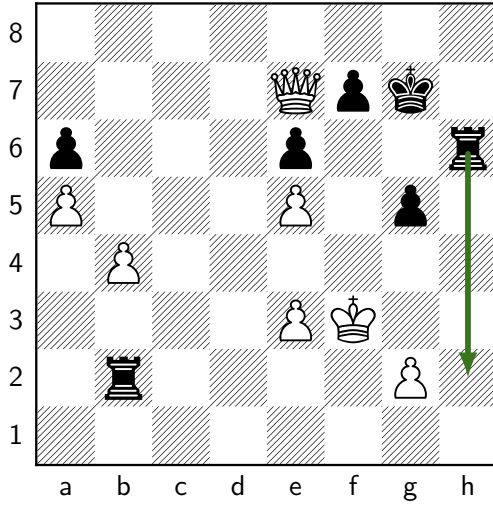
Figure 23: **Move 29, White: Re6.** (a) Our position; Melanie plays Re6 (e1e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#472` (ply 54) [structural distance $d=46$, QAP obj=35]: the master played Rd5; under μ our Re1 plays the role of that game’s Rd1; cross-type roles: our qc2 as their ra2, our Pa3 as their Bg2; transferred move e1e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2836` (ply 68) [structural distance $d=46$, QAP obj=79]: the master played Rc1; under μ our Re1 plays the role of that game’s Rb1; cross-type roles: our bd6 as their Qe3, our Ba2 as their rc8; transferred move e1d1. (d) Retrieved from `ccr1_top_p0002#3969` (ply 64) [structural distance $d=47$, QAP obj=72]: the master played Rc2; under μ our Ba2 plays the role of that game’s Rc3 (a **cross-type** correspondence); cross-type roles: our Pb2 as their Nf2, our qc2 as their bd7; transferred move a2d5. Synthesis: 1 retrieved analog transferred to Re6 (summed QAP objective 35, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



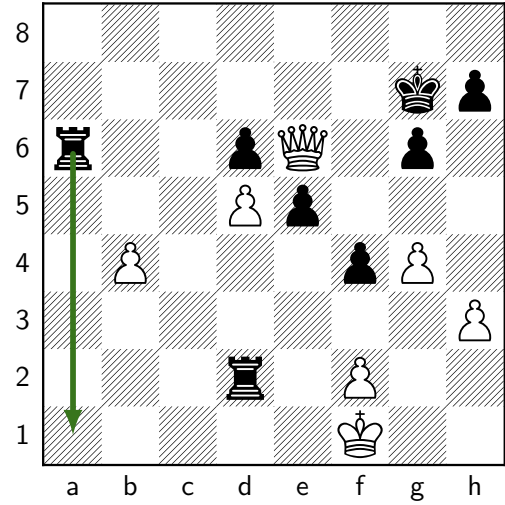
(a)



(b)

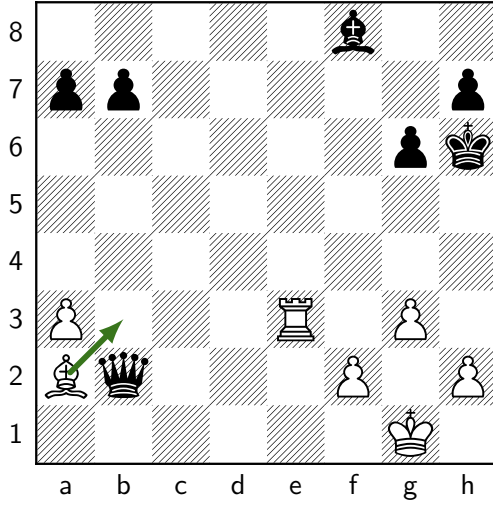


(c)

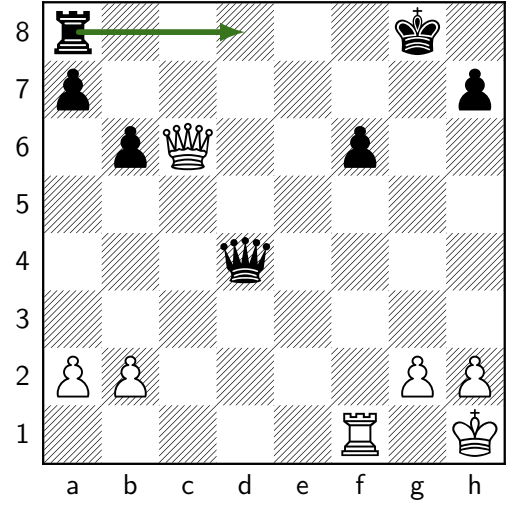


(d)

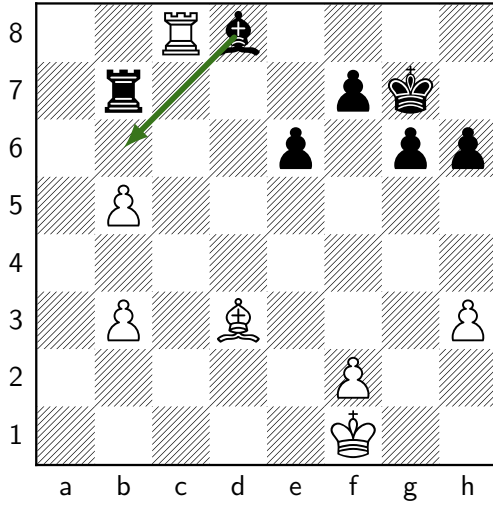
Figure 24: **Move 30, White: Re3.** (a) Our position; Melanie plays **Re3** (e6e3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1135` (ply 57) [structural distance $d=45$, QAP obj=37]: the master played **R4d6**; under μ our *Re6* plays the role of that game’s *Rd5*; cross-type roles: our *Ba2* as their *Rd8*, our *bf8* as their *pg3*; transferred move **e6e3** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2116` (ply 75) [structural distance $d=43$, QAP obj=84]: the master played **Rh2**; under μ our *Ba2* plays the role of that game’s *Rh3* (a **cross-type** correspondence); cross-type roles: our *bf8* as their *pg2*; transferred move **a2d5**. (d) Retrieved from `ccr1_top_p0002#2930` (ply 69) [structural distance $d=47$, QAP obj=84]: the master played **Ra1+**; under μ our *Ba2* plays the role of that game’s *Ra3* (a **cross-type** correspondence); cross-type roles: our *bf8* as their *pb4*; transferred move **a2d5**. Synthesis: 1 retrieved analog transferred to **Re3** (summed QAP objective 37, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



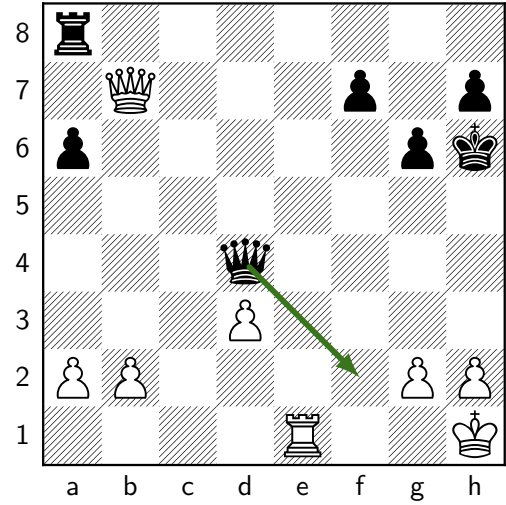
(a)



(b)

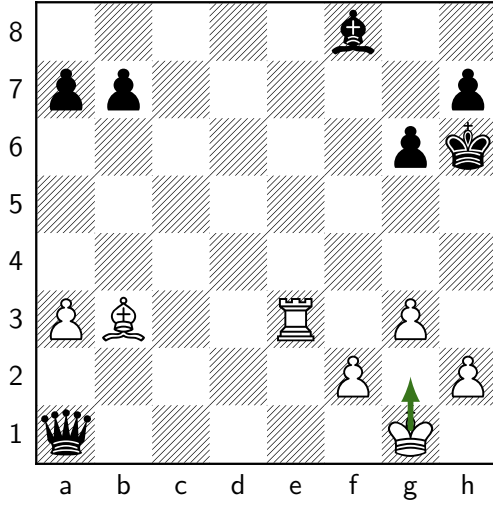


(c)

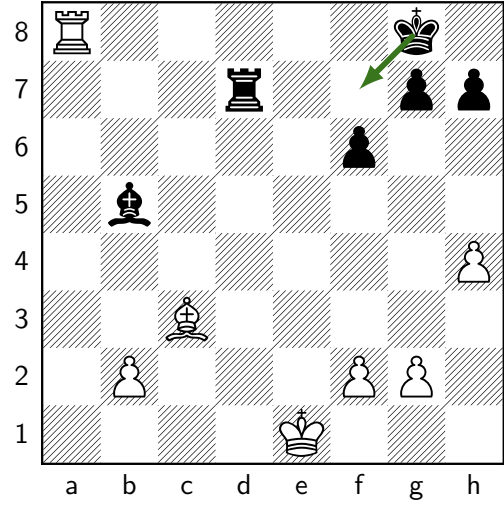


(d)

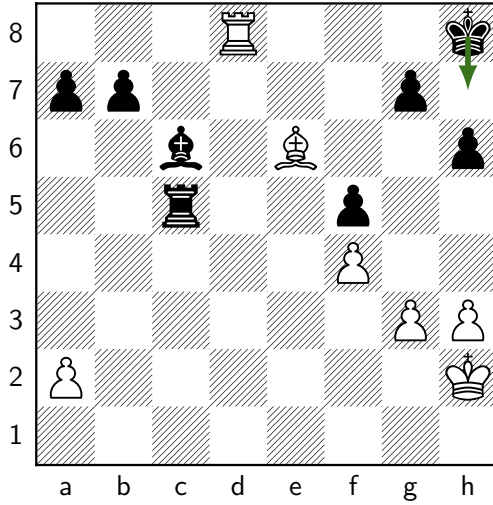
Figure 25: **Move 31, White: Bb3.** (a) Our position; Melanie plays Bb3 (a2b3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#1841` (ply 47) [structural distance $d=44$, QAP obj=30]: the master played Rd8; under μ our Ba2 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Re3 as their Qd4, our bf8 as their rf1; transferred move a2b3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1494` (ply 63) [structural distance $d=44$, QAP obj=21]: the master played Bb6; under μ our Ba2 plays the role of that game’s Bd1; cross-type roles: our qb2 as their rc8; transferred move a2f7. (d) Retrieved from `ccrl_top_p0002#2903` (ply 49) [structural distance $d=46$, QAP obj=18]: the master played Qf2; under μ our Ph2 plays the role of that game’s Qd5 (a **cross-type** correspondence); cross-type roles: our Ba2 as their Ra8, our Re3 as their Ph7; transferred move h2h4. Synthesis: 1 retrieved analog transferred to Bb3 (summed QAP objective 30, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



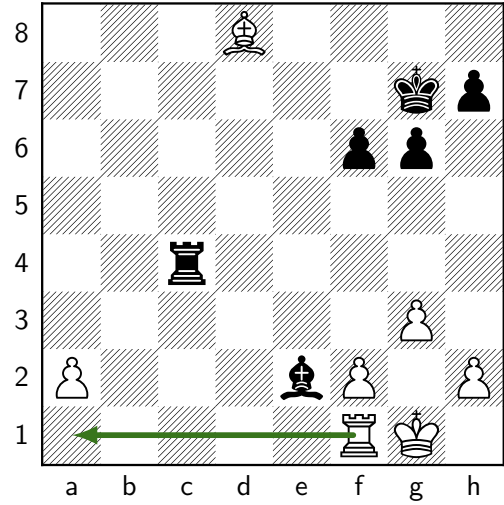
(a)



(b)

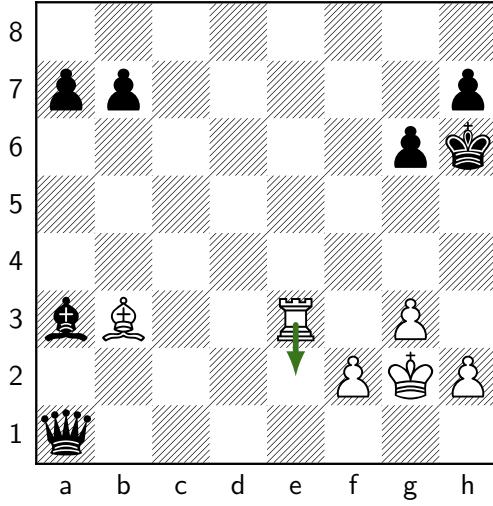


(c)

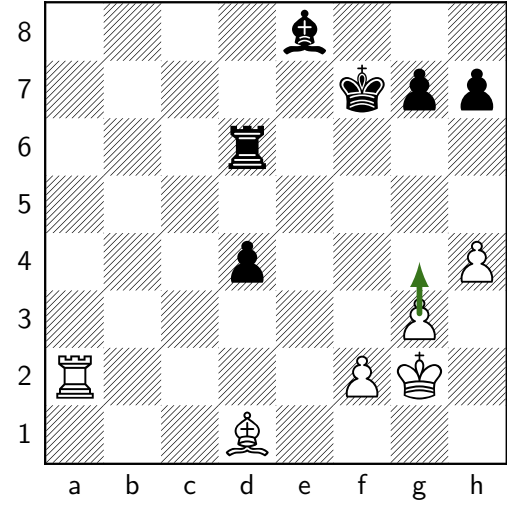


(d)

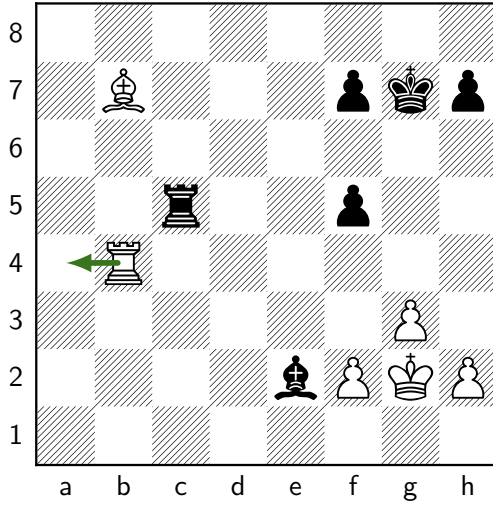
Figure 26: **Move 32, White: Kg2.** (a) Our position; Melanie plays Kg2 (**g1g2**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2991` (ply 71) [structural distance $d=34$, QAP obj=74]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg1; cross-type roles: our qa1 as their ra8; transferred move g1g2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#920` (ply 79) [structural distance $d=37$, QAP obj=46]: the master played Kh7; under μ our Kg1 plays the role of that game’s Kh1; cross-type roles: our qa1 as their rd8, our pg6 as their Bc6; transferred move g1g2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#805` (ply 56) [structural distance $d=38$, QAP obj=1]: the master played Ra1; under μ our Kg1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our qa1 as their be2, our Re3 as their Pf2; transferred move g1g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg2 (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



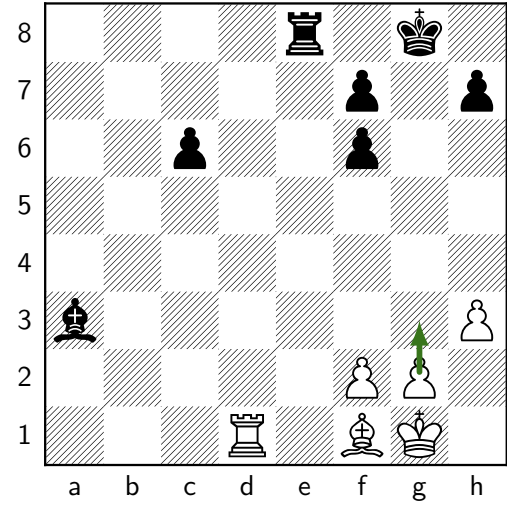
(a)



(b)

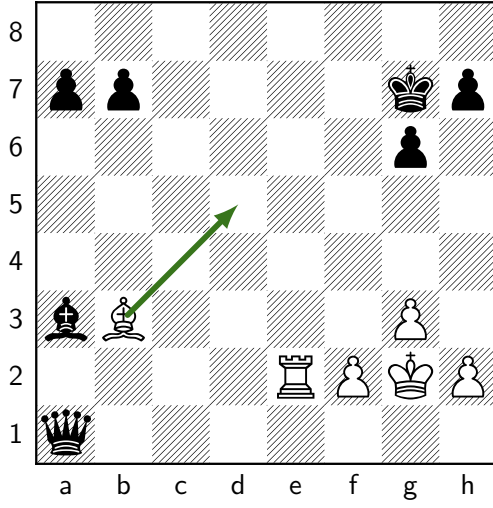


(c)

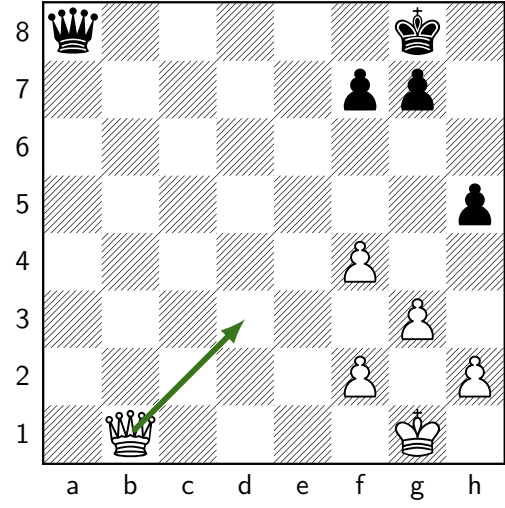


(d)

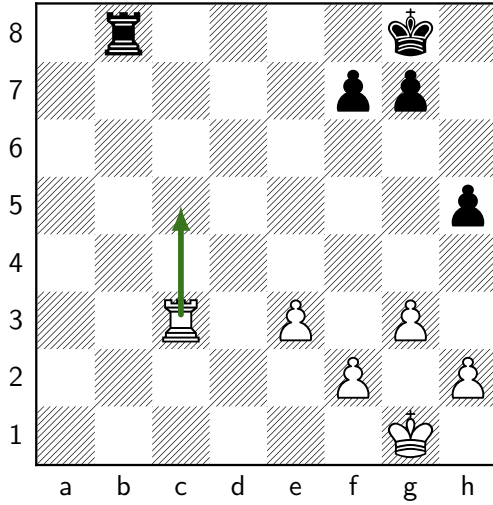
Figure 27: **Move 33, White: Re2 (e3e2).** (a) Our position; Melanie plays Re2 (e3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3349` (ply 78) [structural distance $d=27$, QAP obj=3]: the master played g4; under μ our *Re3* plays the role of that game’s *Pg3* (a **cross-type** correspondence); cross-type roles: our *pg6* as their *Bd1*, our *qa1* as their *Ra2*; transferred move e3e2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#771` (ply 66) [structural distance $d=32$, QAP obj=1]: the master played Ra4; under μ our *Re3* plays the role of that game’s *Rb4*; cross-type roles: our *qa1* as their *be2*, our *Bb3* as their *Pf2*; transferred move e3e2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3321` (ply 54) [structural distance $d=30$, QAP obj=2]: the master played g3; under μ our *Pg3* plays the role of that game’s *Pg2*; cross-type roles: our *qa1* as their *Rd1*, our *Pf2* as their *Bf1*; transferred move g3g4. Synthesis: 2 retrieved analogs transferred to Re2 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



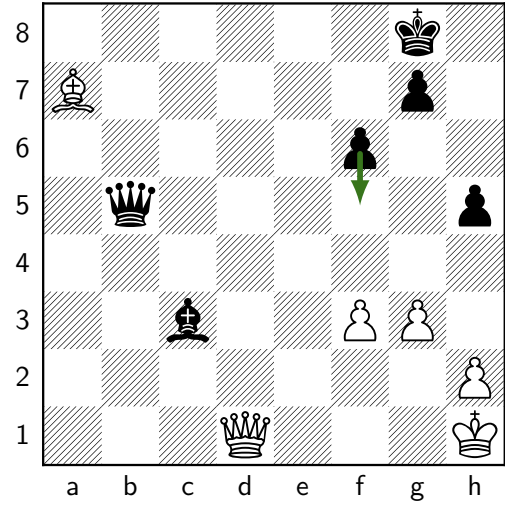
(a)



(b)

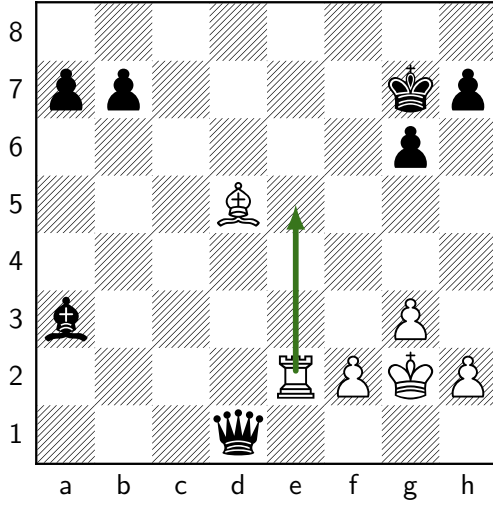


(c)

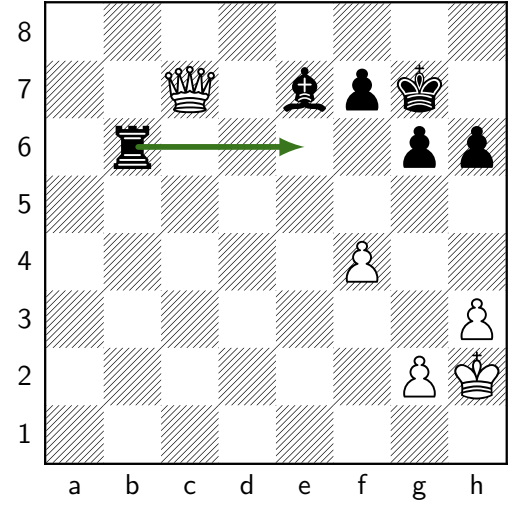


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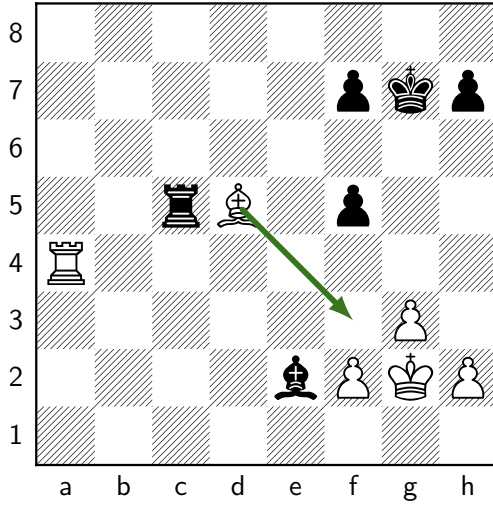
Figure 28: **Move 34, White: Bd5 (b3d5).** (a) Our position; Melanie plays Bd5 (b3d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#2118` (ply 62) [structural distance $d=33$, QAP obj=4]: the master played Qd3; under μ our *Bb3* plays the role of that game’s *Qb1* (a **cross-type** correspondence); cross-type roles: our *Re2* as their *Pf2*, our *ba3* as their *Pf4*; transferred move b3d5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3376` (ply 68) [structural distance $d=31$, QAP obj=3]: the master played Rc5; under μ our *Bb3* plays the role of that game’s *Rc3* (a **cross-type** correspondence); cross-type roles: our *Re2* as their *Pf2*, our *ba3* as their *Pe3*; transferred move b3d5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#2181` (ply 81) [structural distance $d=32$, QAP obj=3]: the master played f5; under μ our *Pg3* plays the role of that game’s *Pf3*; cross-type roles: our *Re2* as their *Qb5*; transferred move g3g4. Synthesis: 2 retrieved analogs transferred to Bd5 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



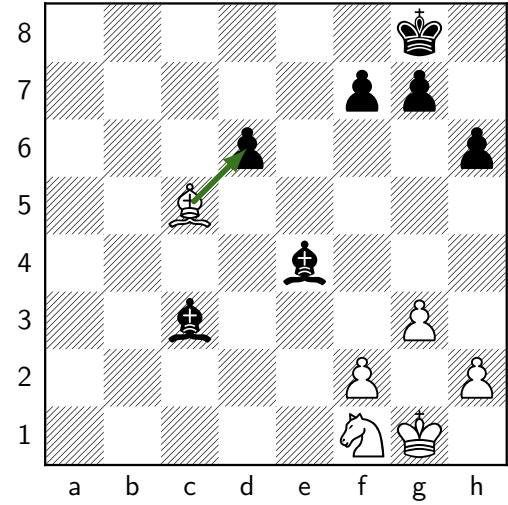
(a)



(b)

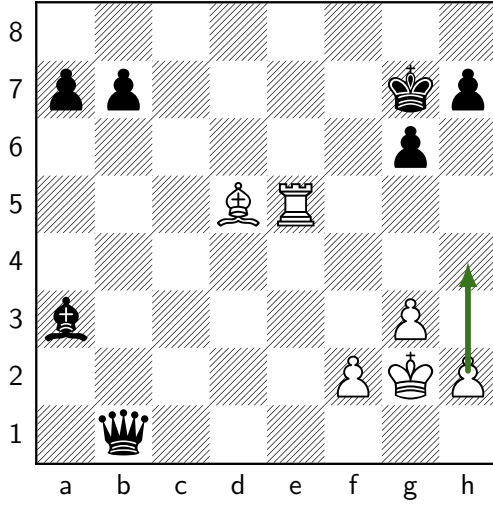


(c)

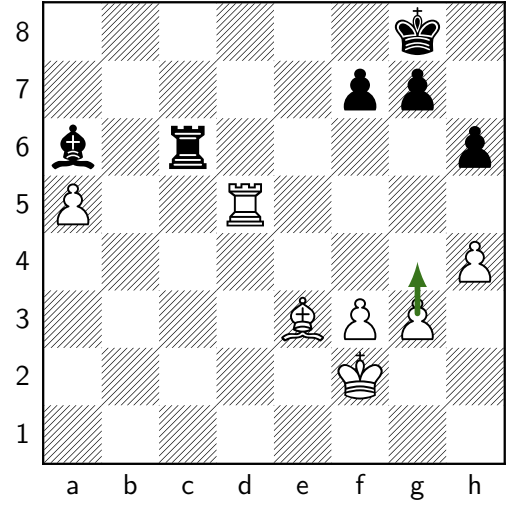


(d)

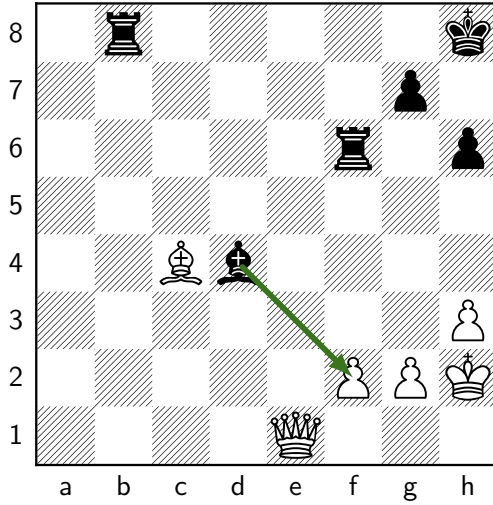
Figure 29: **Move 35, White: Re5.** (a) Our position; Melanie plays Re5 (e2e5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3707` (ply 85) [structural distance $d=21$, QAP obj=24]: the master played Re6; under μ our Re2 plays the role of that game’s Rb3; transferred move e2e5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#771` (ply 70) [structural distance $d=25$, QAP obj=23]: the master played Bf3; under μ our Re2 plays the role of that game’s Bd5 (a **cross-type** correspondence); cross-type roles: our pg6 as their be2, our ph7 as their Ra4; transferred move e2e5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1252` (ply 64) [structural distance $d=35$, QAP obj=13]: the master played Bxd6; under μ our Re2 plays the role of that game’s Bc5 (a **cross-type** correspondence); cross-type roles: our Pf2 as their Nf1, our pb7 as their be4; transferred move e2e5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Re5 (summed QAP objective 68, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



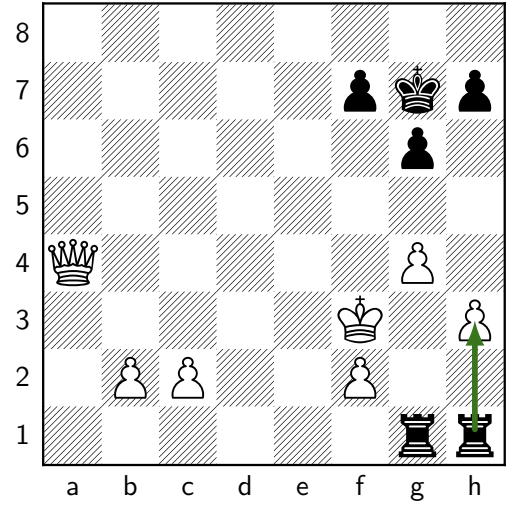
(a)



(b)



(c)



(d)

Figure 30: **Move 36, White: h4.** (a) Our position; Melanie plays **h4** (**h2h4**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1522` (ply 74) [structural distance $d=34$, QAP obj=16]: the master played **g4**; under μ our *Ph2* plays the role of that game’s *Pg3*; cross-type roles: our *ba3* as their *Pa5*, our *pg6* as their *Rd5*; transferred move **h2h4** (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3088` (ply 73) [structural distance $d=35$, QAP obj=26]: the master played **Bxf2**; under μ our *Bd5* plays the role of that game’s *Bd5*; cross-type roles: our *Pg3* as their *Rb8*, our *Re5* as their *Pg7*; transferred move **d5b7**. (d) Retrieved from `ccrl_top_p0002#365` (ply 65) [structural distance $d=36$, QAP obj=22]: the master played **Rxh3+**; under μ our *Bd5* plays the role of that game’s *Rh8* (a **cross-type** correspondence); cross-type roles: our *ba3* as their *pb2*, our *Re5* as their *pc2*; transferred move **d5b7**. Synthesis: 1 retrieved analog transferred to **h4** (summed QAP objective 16, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

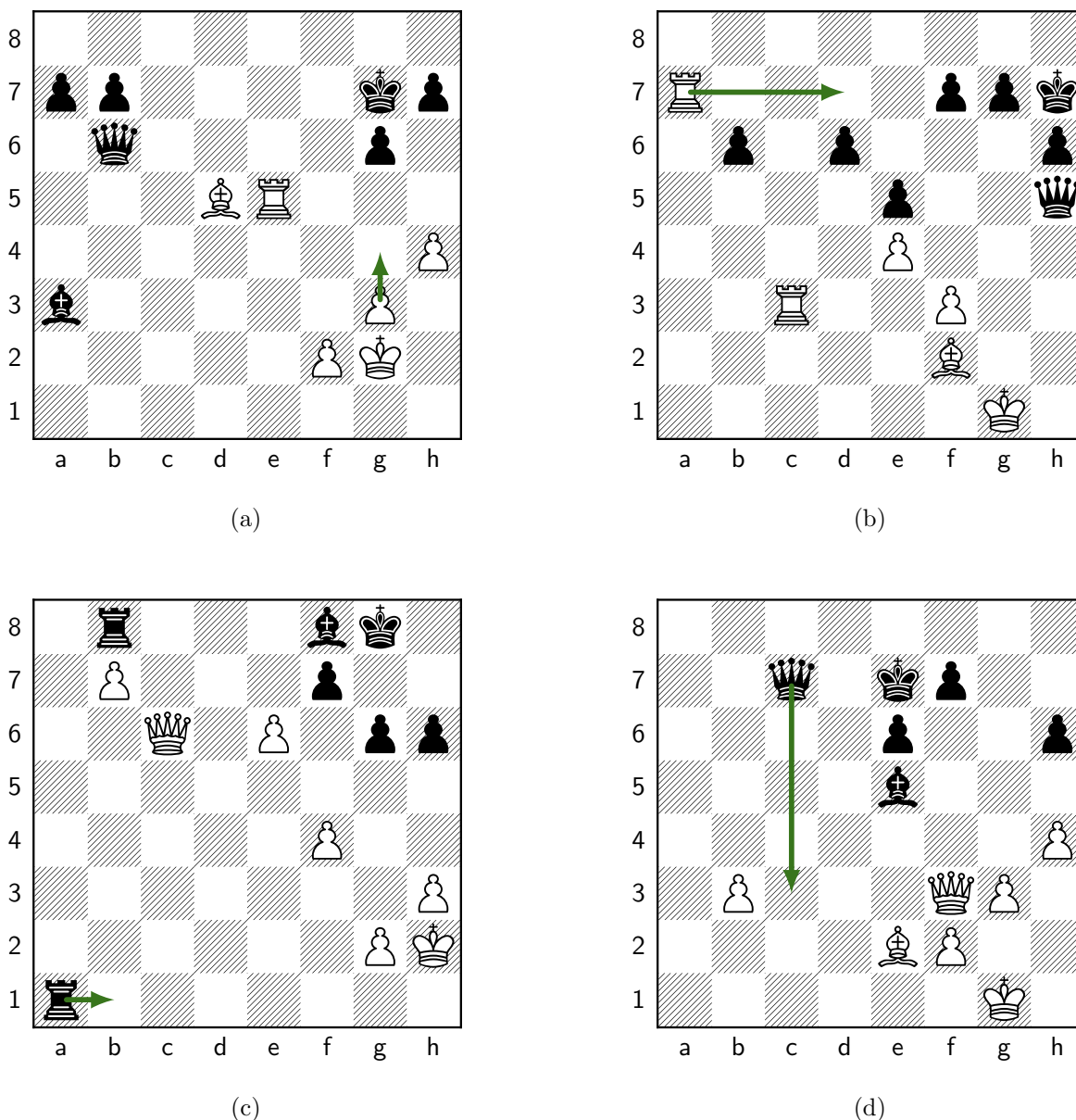


Figure 31: **Move 37, White: g4.** (a) Our position; Melanie plays g4 (g3g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3074` (ply 76) [structural distance $d=23$, QAP obj=29]: the master played Rd7; under μ our Pg3 plays the role of that game’s Ra7 (a **cross-type** correspondence); cross-type roles: our qb6 as their Bf2, our Bd5 as their qh5; transferred move g3g4 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#3707` (ply 75) [structural distance $d=24$, QAP obj=15]: the master played Rb1; under μ our Pg3 plays the role of that game’s Ra8 (a **cross-type** correspondence); cross-type roles: our qb6 as their Rb8, our Re5 as their qc6; transferred move g3g4 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#606` (ply 67) [structural distance $d=33$, QAP obj=35]: the master played Qc3; under μ our Re5 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our pa7 as their be2, our ba3 as their pf2; transferred move e5e6. Synthesis: 2 retrieved analogs transferred to g4 (summed QAP objective 44, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

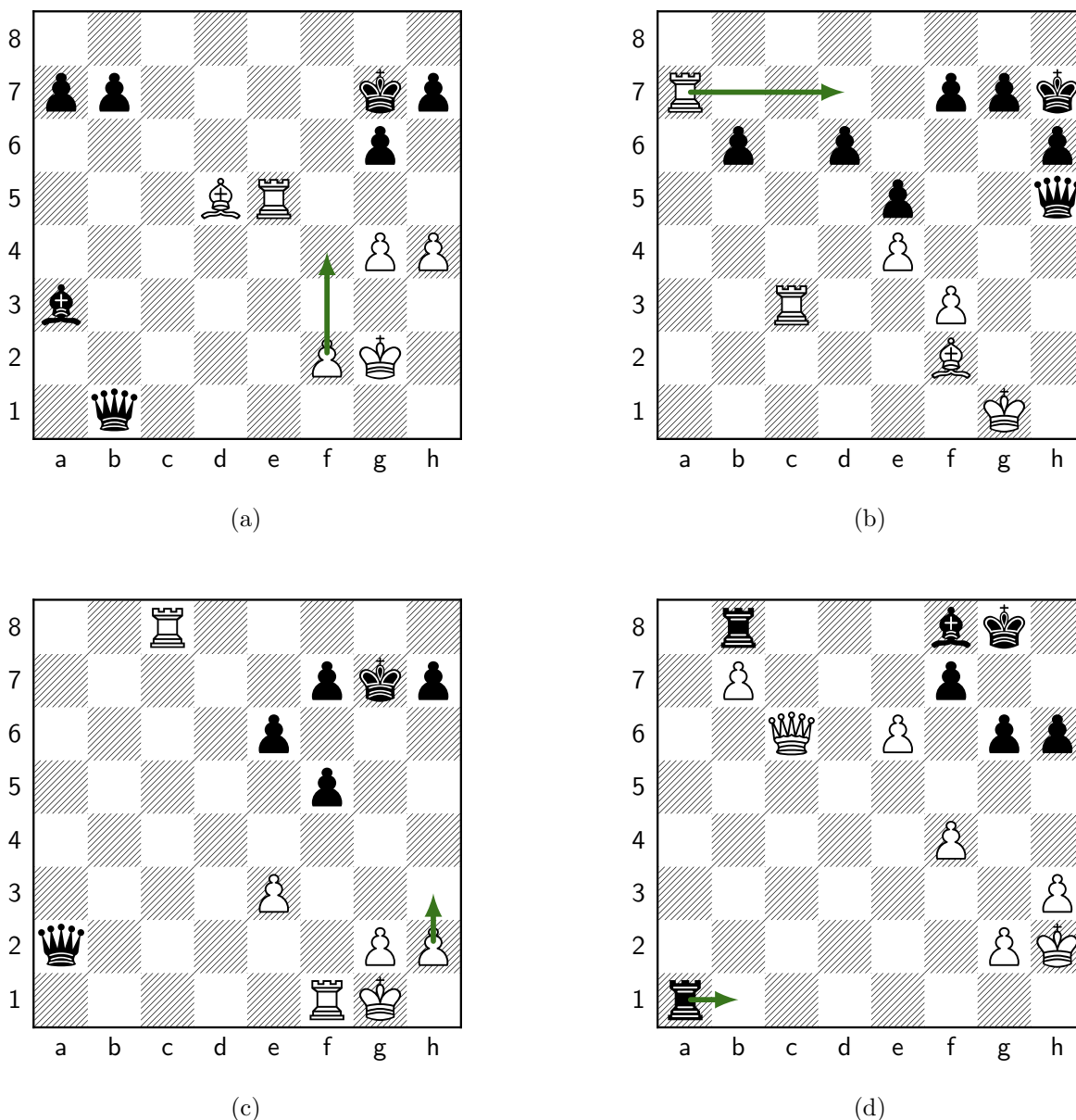
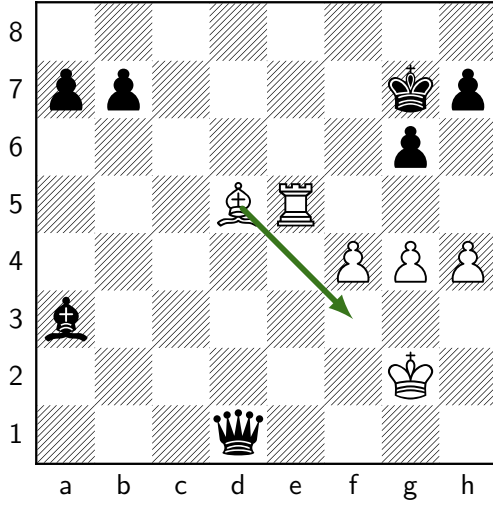
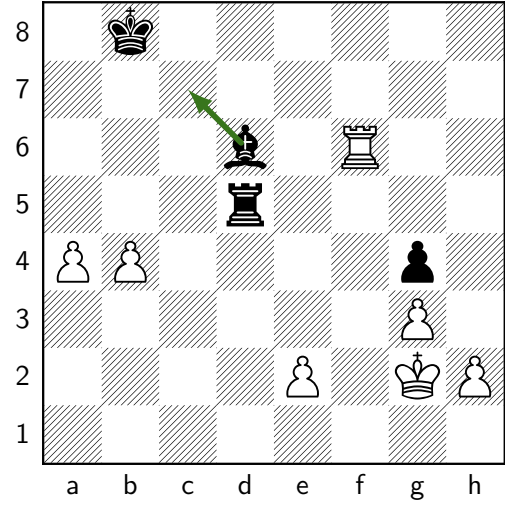


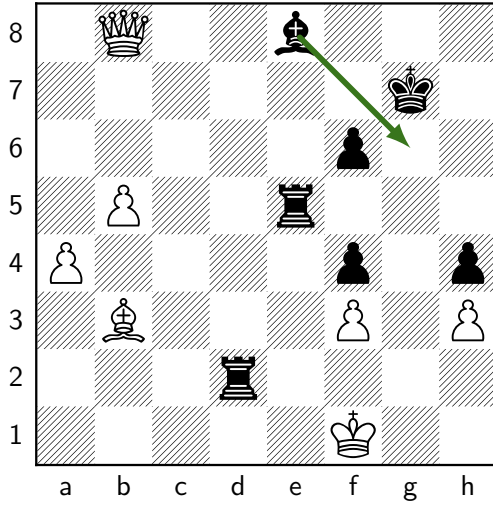
Figure 32: **Move 38, White: f4.** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#3074 (ply 76) [structural distance $d=32$, QAP obj=22]: the master played Rd7; under μ our Pf2 plays the role of that game's Ra7 (a **cross-type** correspondence); cross-type roles: our qb1 as their Rc3, our Bd5 as their qh5; transferred move f2f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3264 (ply 72) [structural distance $d=31$, QAP obj=12]: the master played h3; under μ our Pf2 plays the role of that game's Ph2; cross-type roles: our Bd5 as their Rf1, our Re5 as their Pg2; transferred move f2f4 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3707 (ply 75) [structural distance $d=30$, QAP obj=11]: the master played Rb1; under μ our Pf2 plays the role of that game's Ra8 (a **cross-type** correspondence); cross-type roles: our Bd5 as their Rb8, our qb1 as their Bf8; transferred move f2f4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f4 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



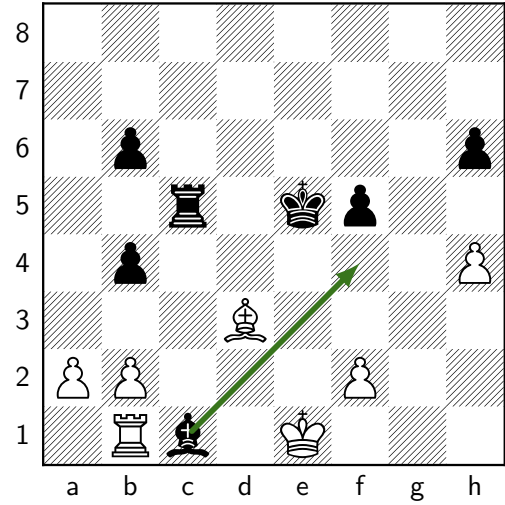
(a)



(b)

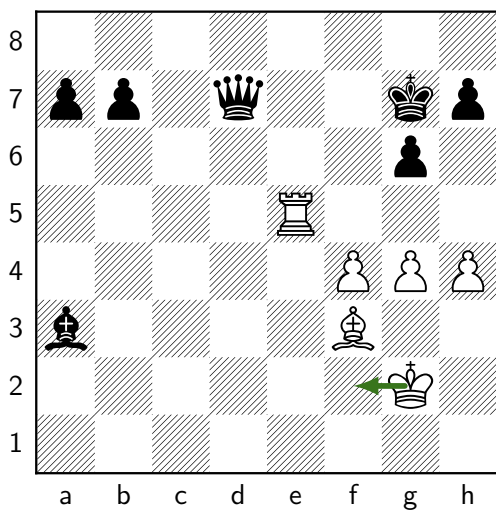


(c)

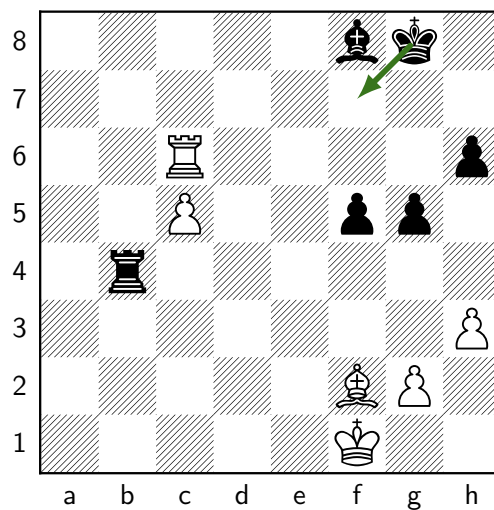


(d)

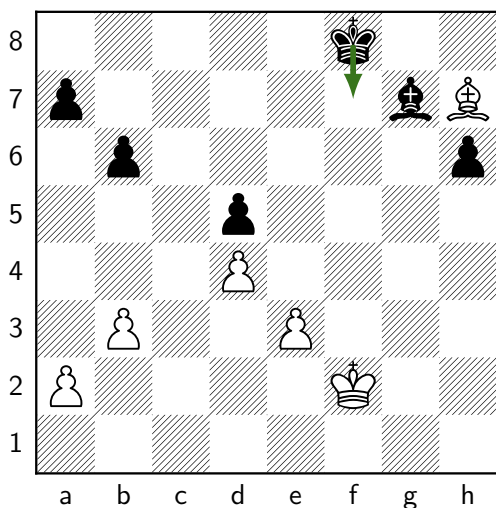
Figure 33: **Move 39, White: Bf3**. (a) Our position; Melanie plays Bf3 (d5f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1314` (ply 61) [structural distance $d=33$, QAP obj=41]: the master played Bc7; under μ our Bd5 plays the role of that game’s Bd3; cross-type roles: our qd1 as their rf6, our ba3 as their pa4; transferred move d5f3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2353` (ply 83) [structural distance $d=33$, QAP obj=23]: the master played Bg6; under μ our Bd5 plays the role of that game’s Be1; cross-type roles: our Re5 as their Pf6, our Bd5 as their Re5; transferred move d5f3 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#445` (ply 85) [structural distance $d=33$, QAP obj=6]: the master played Bf4; under μ our Bd5 plays the role of that game’s Bc8; cross-type roles: our ba3 as their Ph6, our pg6 as their bd3; transferred move d5f3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bf3 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



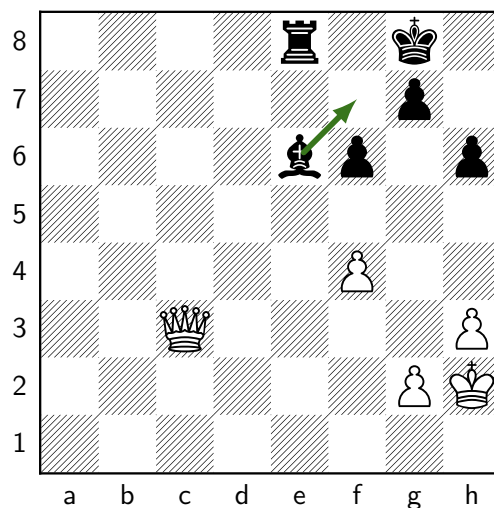
(a)



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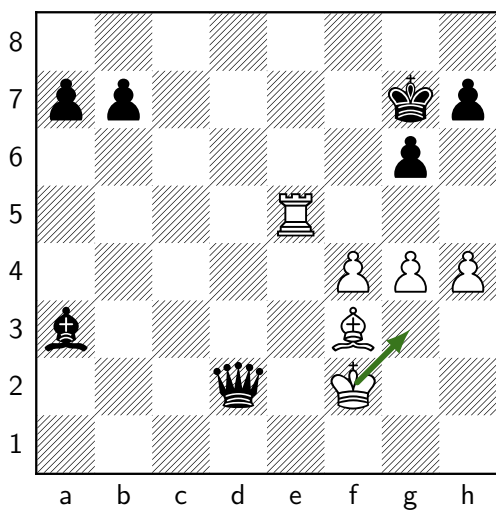


(c)

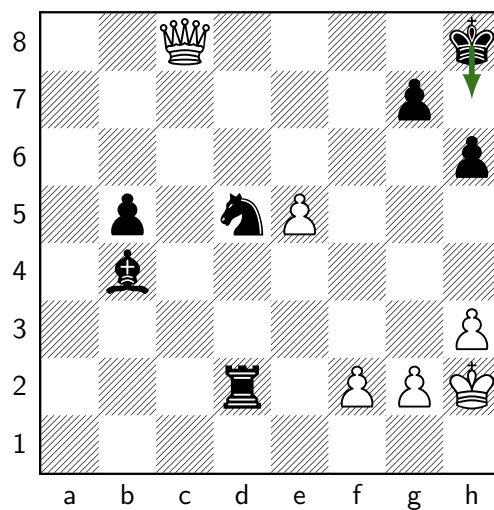


(d)

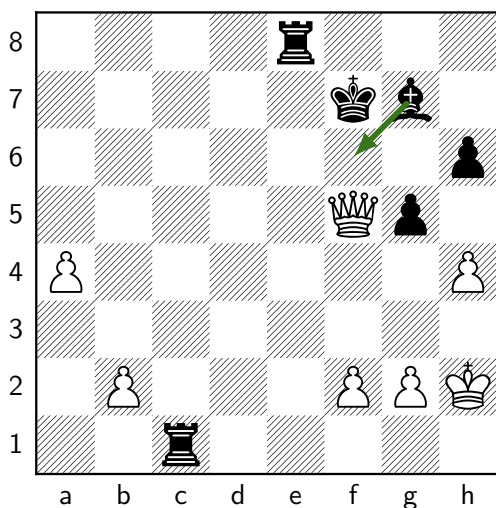
Figure 34: **Move 40, White: Kf2.** (a) Our position; Melanie plays Kf2 (g2f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3425` (ply 85) [structural distance $d=37$, QAP obj=33]: the master played Kf7; under μ our Kg2 plays the role of that game's Kg1; cross-type roles: our qd7 as their Bf8, our Bf3 as their rc6; transferred move g2f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2103` (ply 83) [structural distance $d=35$, QAP obj=20]: the master played Kf7; under μ our Kg2 plays the role of that game's Kf1; cross-type roles: our ba3 as their Pa7, our qd7 as their Bg7; transferred move g2f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1694` (ply 75) [structural distance $d=36$, QAP obj=37]: the master played Bf7; under μ our Bf3 plays the role of that game's Be3; transferred move f3e2. Synthesis: 2 retrieved analogs transferred to Kf2 (summed QAP objective 54, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



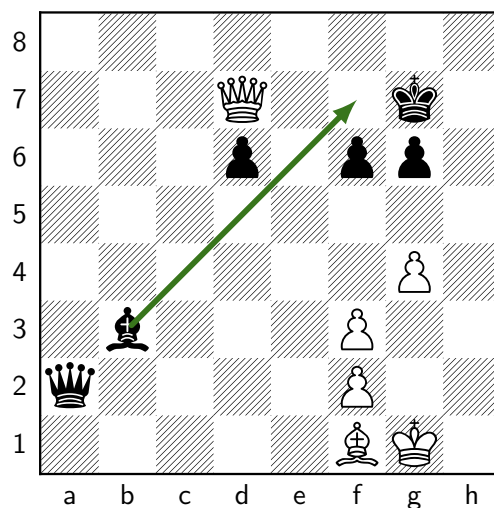
(a)



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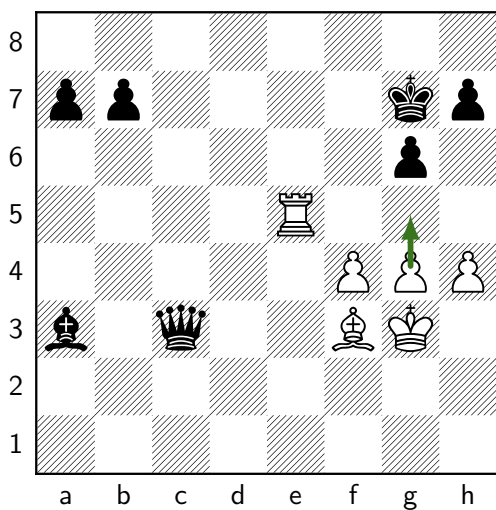


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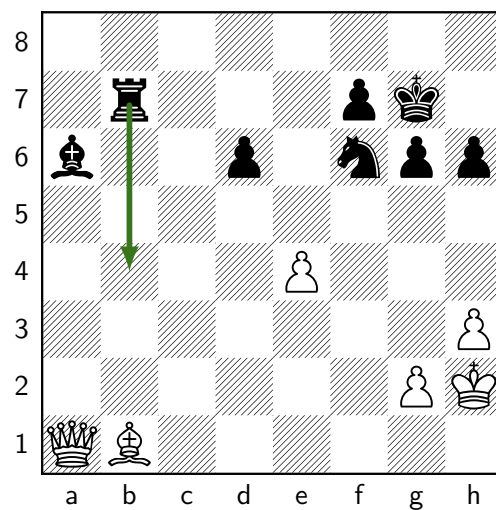


(d)

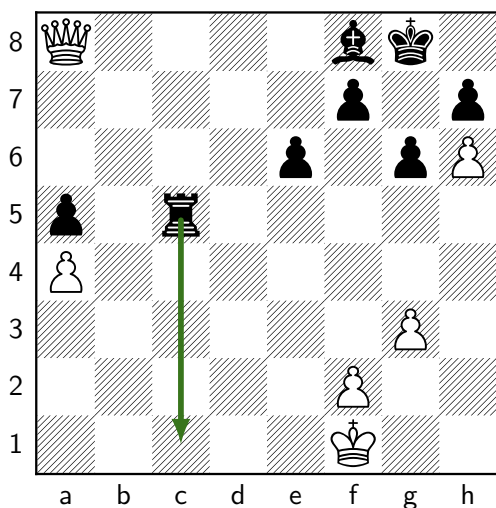
Figure 35: **Move 41, White: Kg3.** (a) Our position; Melanie plays Kg3 (f2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#2002` (ply 81) [structural distance $d=29$, QAP obj=29]: the master played Kh7; under μ our Kf2 plays the role of that game's Kh1; cross-type roles: our Bf3 as their Pg7, our ba3 as their Ph6; transferred move f2g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#375` (ply 73) [structural distance $d=28$, QAP obj=71]: the master played Bf6; under μ our Re5 plays the role of that game's Bg2 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Re8, our ba3 as their pa4; transferred move e5e2. (d) Retrieved from `ccrl_top_p0002#3704` (ply 65) [structural distance $d=30$, QAP obj=46]: the master played Bf7; under μ our Re5 plays the role of that game's Bb6 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Pf6, our ba3 as their pf3; transferred move e5e2. Synthesis: 2 retrieved analogs transferred to Kg3 (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



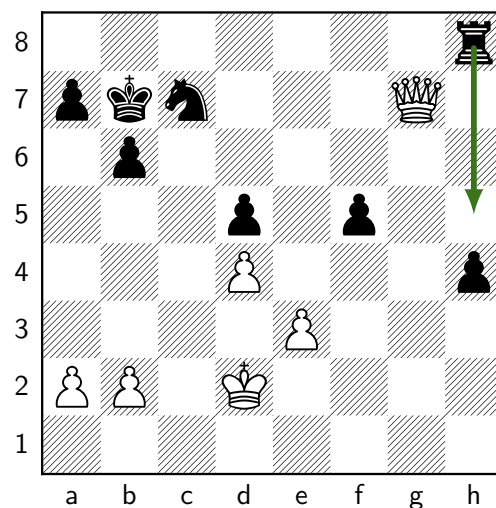
(a)



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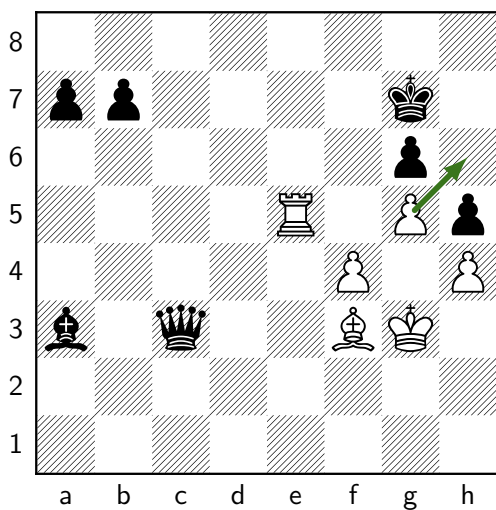


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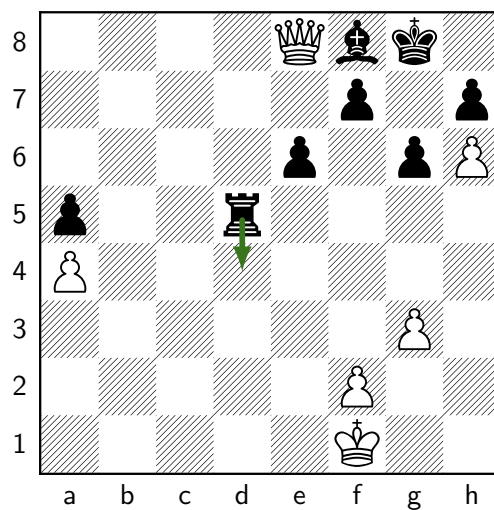


(d)

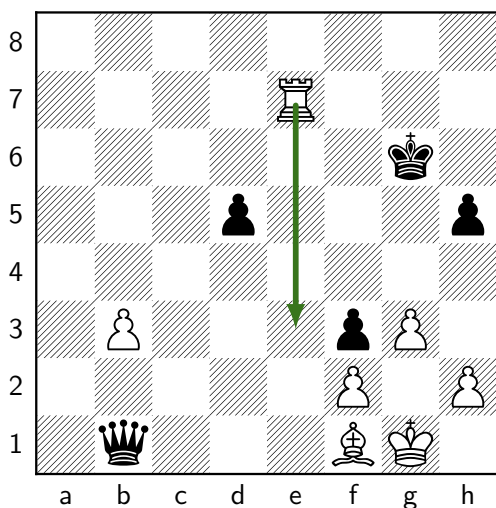
Figure 36: **Move 42, White: g5.** (a) Our position; Melanie plays g5 (g4g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1479` (ply 79) [structural distance $d=34$, QAP obj=63]: the master played Rb4; under μ our Pg4 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our Re5 as their Ba6, our Bf3 as their Nf6; transferred move g4g5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1194` (ply 95) [structural distance $d=33$, QAP obj=42]: the master played Rc1+; under μ our Pg4 plays the role of that game’s Rc4 (a **cross-type** correspondence); cross-type roles: our Re5 as their Pe6, our ba3 as their Pg6; transferred move g4g5 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3903` (ply 81) [structural distance $d=38$, QAP obj=37]: the master played Rh5; under μ our Re5 plays the role of that game’s Rh1; cross-type roles: our Bf3 as their Nc7, our ba3 as their pd4; transferred move e5e8. Synthesis: 2 retrieved analogs transferred to g5 (summed QAP objective 105, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



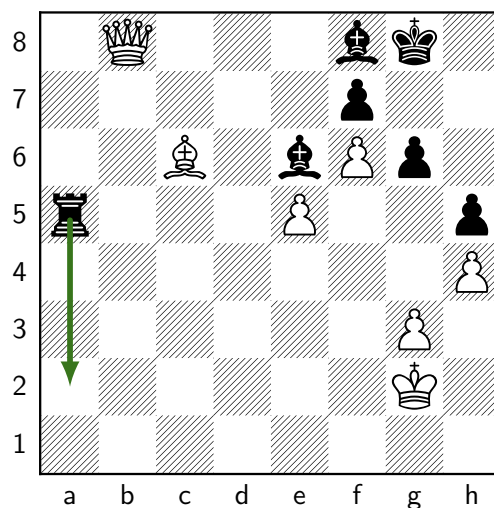
(a)



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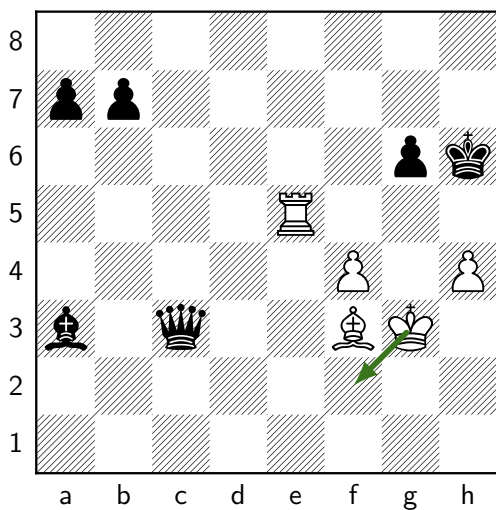


(c)

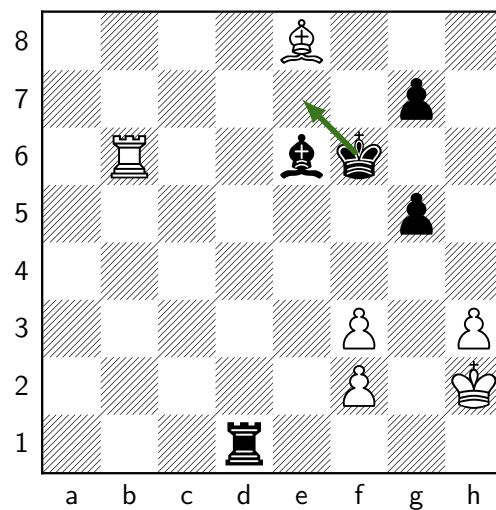


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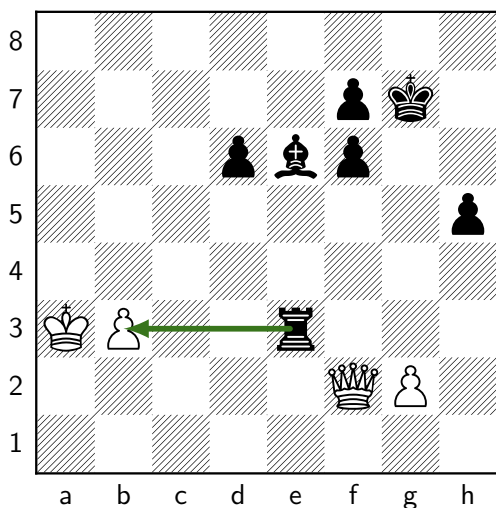
Figure 37: **Move 43, White: gxh6+.** (a) Our position; Melanie plays gxh6+ (g5h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1194` (ply 89) [structural distance $d=27$, QAP obj=64]: the master played Rd4; under μ our Pg5 plays the role of that game's Rd4 (a **cross-type** correspondence); cross-type roles: our Re5 as their Pf7, our Re5 as their Pe6; transferred move g5h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3278` (ply 72) [structural distance $d=39$, QAP obj=27]: the master played Re3; under μ our Pg5 plays the role of that game's Re7 (a **cross-type** correspondence); cross-type roles: our ba3 as their Ph2, our Re5 as their Pb3; transferred move g5h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3660` (ply 91) [structural distance $d=44$, QAP obj=26]: the master played Ra2+; under μ our Pg5 plays the role of that game's Ra4 (a **cross-type** correspondence); cross-type roles: our Ph4 as their bc6, our Re5 as their Be6; transferred move g5h6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to gxh6+ (summed QAP objective 221, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



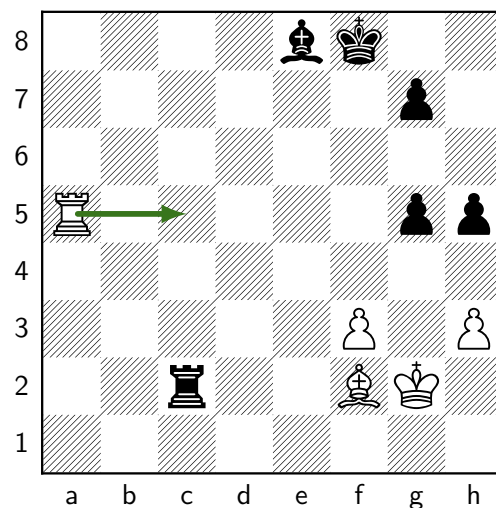
(a)



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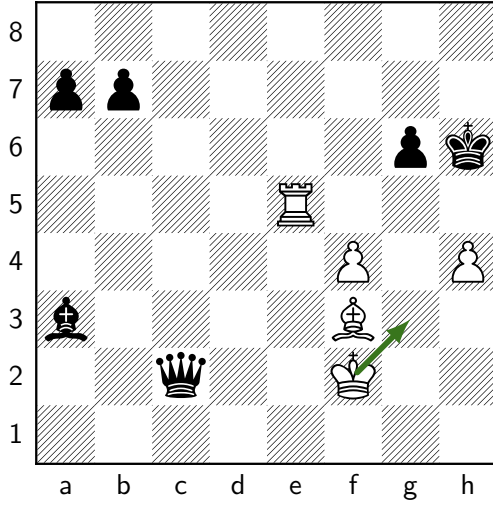


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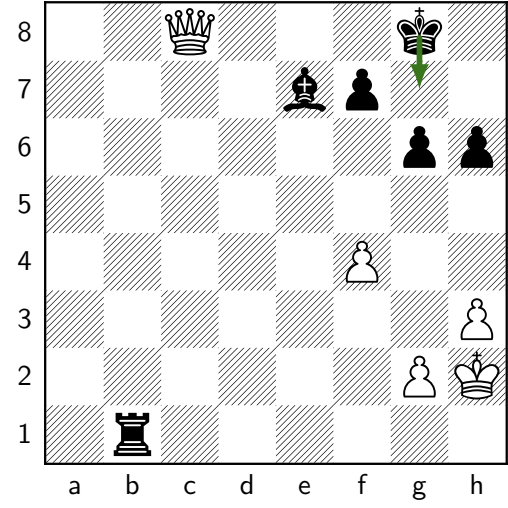


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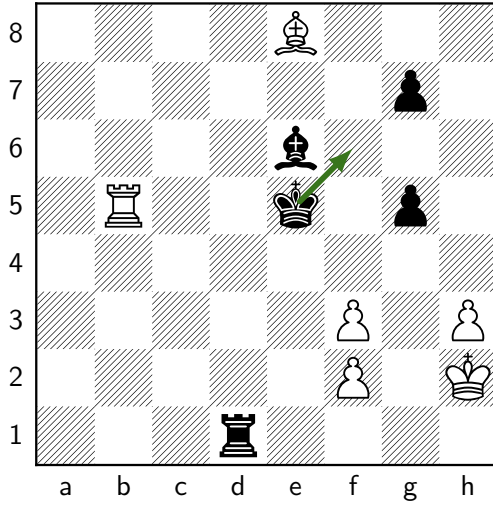
Figure 38: **Move 44, White: Kf2**. (a) Our position; Melanie plays Kf2 (g3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2468` (ply 75) [structural distance $d=31$, QAP obj=55]: the master played Ke7; under μ our Kg3 plays the role of that game's Kf3; cross-type roles: our qc3 as their rb6; transferred move g3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2293` (ply 97) [structural distance $d=37$, QAP obj=32]: the master played Rxb3+; under μ our Re5 plays the role of that game's Re6; cross-type roles: our pg6 as their Be6, our Bf3 as their Pf6; transferred move e5g5. (d) Retrieved from `ccr1_top_p0002#2208` (ply 84) [structural distance $d=37$, QAP obj=32]: the master played Rc5; under μ our Re5 plays the role of that game's Ra5; cross-type roles: our qc3 as their rc2; transferred move e5d5. Synthesis: 1 retrieved analog transferred to Kf2 (summed QAP objective 55, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



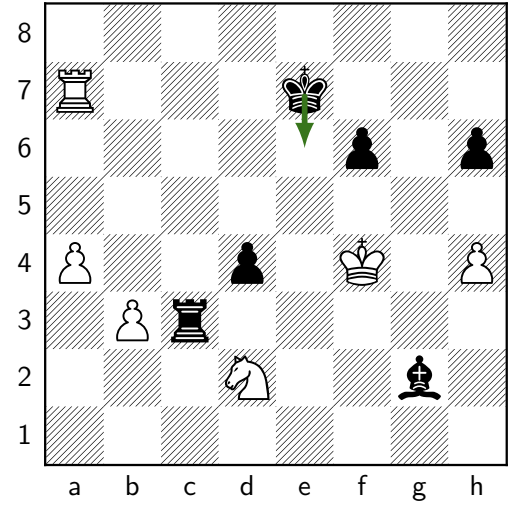
(a)



(b)



(c)



(d)

Figure 39: **Move 45, White: Kg3.** (a) Our position; Melanie plays Kg3 (f2g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3707` (ply 81) [structural distance $d=28$, QAP obj=52]: the master played Kg7; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Re5 as their Be7, our Bf3 as their Pf7; transferred move f2g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2468` (ply 73) [structural distance $d=29$, QAP obj=72]: the master played Kf6; under μ our Kf2 plays the role of that game’s Ke4; cross-type roles: our pb7 as their be8, our ba3 as their Pg7; transferred move f2e3. (d) Retrieved from `ccr1_top_p0000#622` (ply 95) [structural distance $d=29$, QAP obj=53]: the master played Ke6; under μ our Kf2 plays the role of that game’s Ke2; cross-type roles: our qc2 as their ra7, our Bf3 as their Pf6; transferred move f2e3. Synthesis: 1 retrieved analog transferred to Kg3 (summed QAP objective 52, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

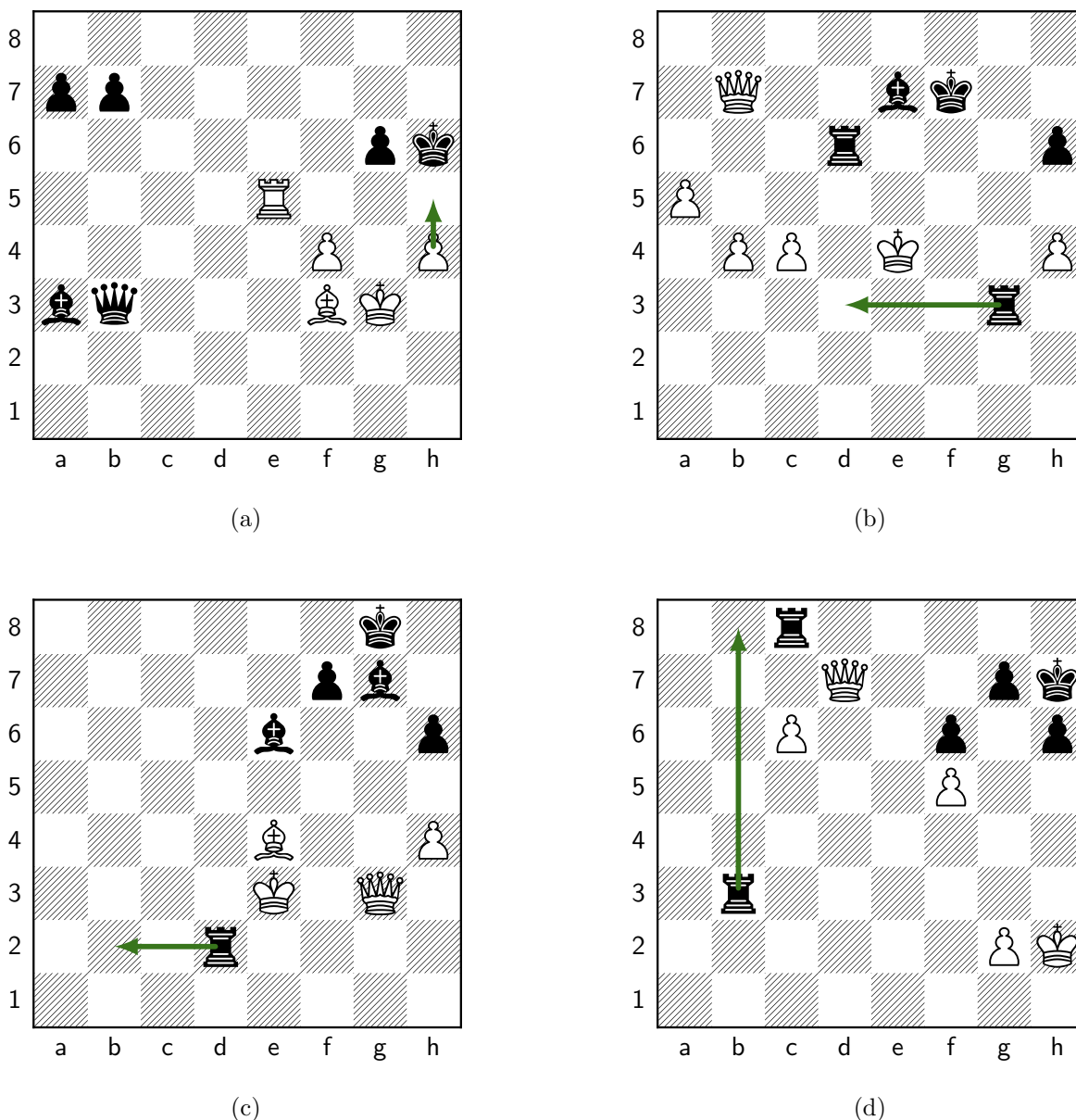
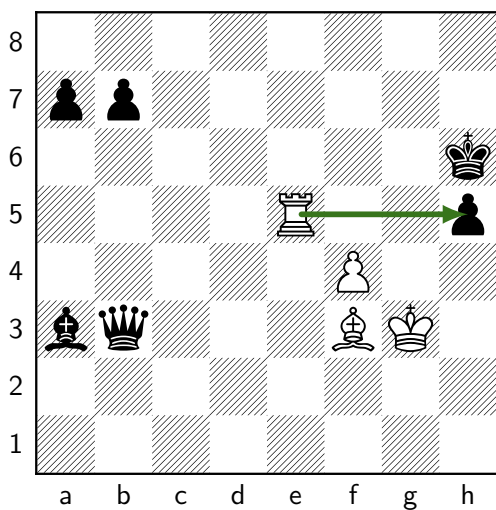
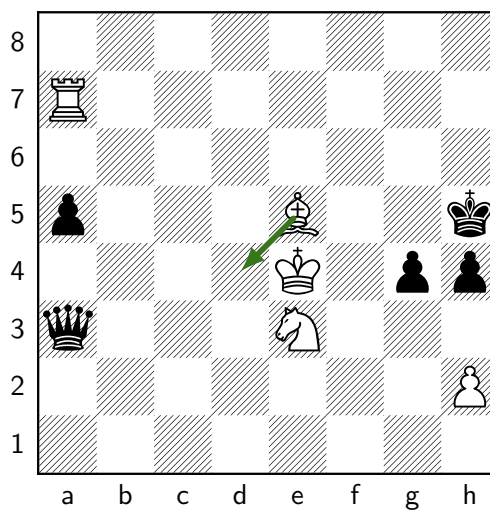


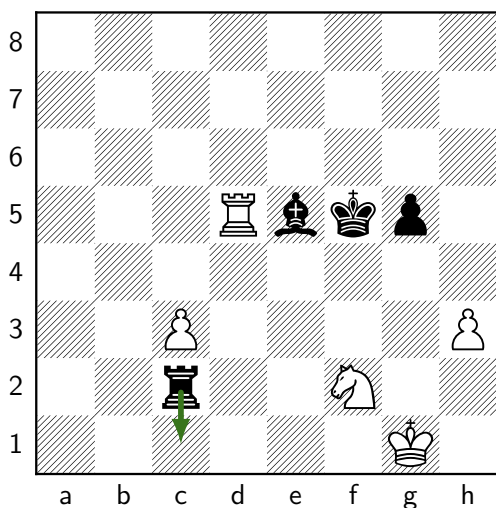
Figure 40: **Move 46, White: h5.** (a) Our position; Melanie plays h5 (h4h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1085` (ply 81) [structural distance $d=36$, QAP obj=61]: the master played Rgd3; under μ our $Ph4$ plays the role of that game's Rg6 (a **cross-type** correspondence); cross-type roles: our ba3 as their pb4; transferred move h4h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#198` (ply 105) [structural distance $d=37$, QAP obj=24]: the master played Rb2; under μ our $Ph4$ plays the role of that game's Rd7 (a **cross-type** correspondence); cross-type roles: our ba3 as their Pf7, our Pf4 as their Be6; transferred move h4h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1496` (ply 91) [structural distance $d=35$, QAP obj=19]: the master played Rbb8; under μ our $Ph4$ plays the role of that game's Rb6 (a **cross-type** correspondence); cross-type roles: our pa7 as their Rc8, our Bf3 as their Pg7; transferred move h4h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to h5 (summed QAP objective 104, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



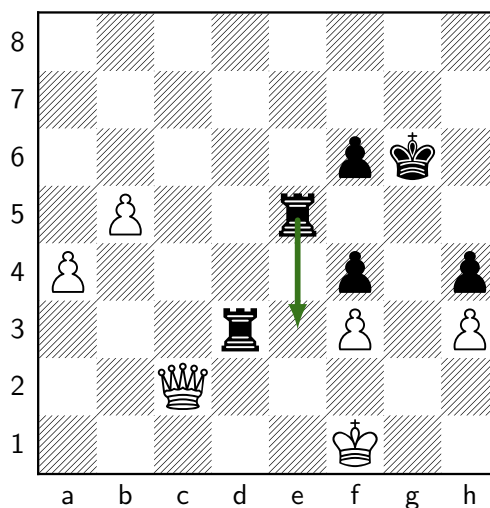
(a)



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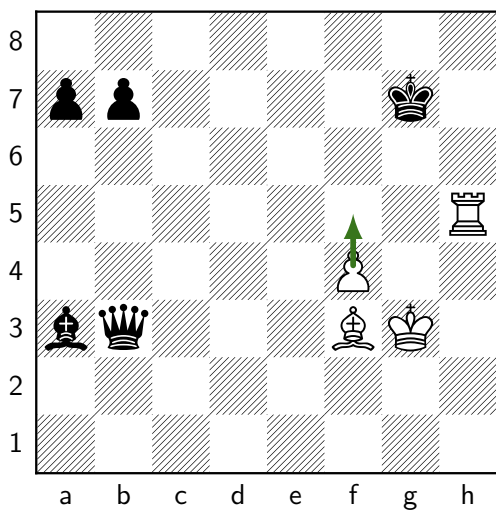


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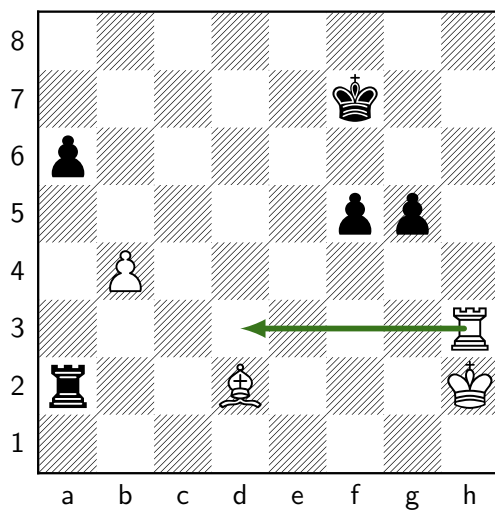


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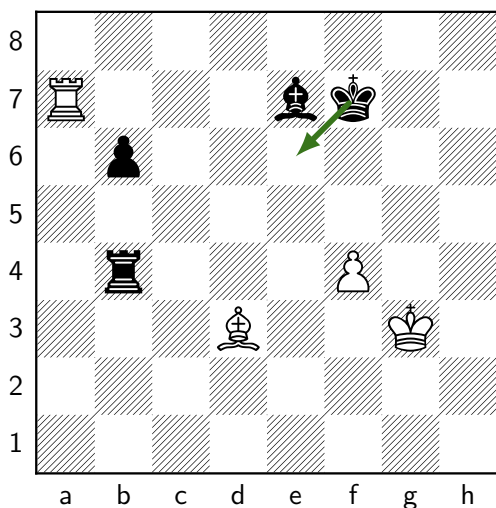
Figure 41: **Move 47, White: Rxh5+.** (a) Our position; Melanie plays Rxh5+ (e5h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2093` (ply 94) [structural distance $d=33$, QAP obj=61]: the master played Bd4; under μ our Re5 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our ba3 as their Ph2, our Bf3 as their Ne3; transferred move e5d5. (c) Retrieved from `ccr1_top_p0003#375` (ply 101) [structural distance $d=34$, QAP obj=31]: the master played Rc1+; under μ our Re5 plays the role of that game’s Rc7; cross-type roles: our qb3 as their rd5, our ba3 as their ph3; transferred move e5e8. (d) Retrieved from `ccr1_top_p0002#2353` (ply 91) [structural distance $d=37$, QAP obj=27]: the master played Re3; under μ our Re5 plays the role of that game’s Re4; cross-type roles: our ba3 as their pb5, our Bf3 as their Rd3; transferred move e5e4. Synthesis: no retrieved analog transferred to Rxh5+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



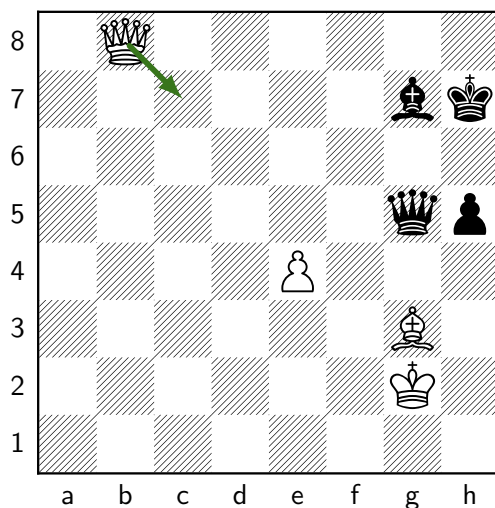
(a)



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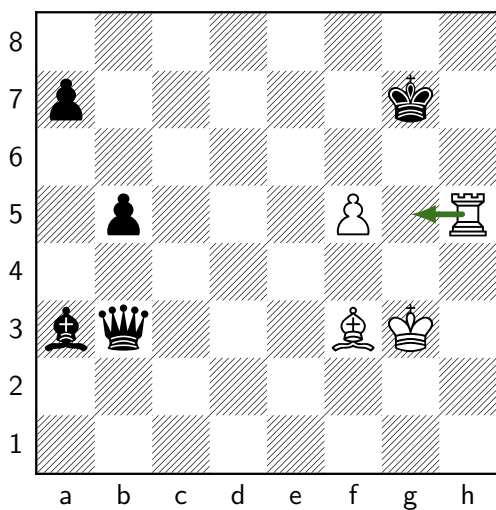


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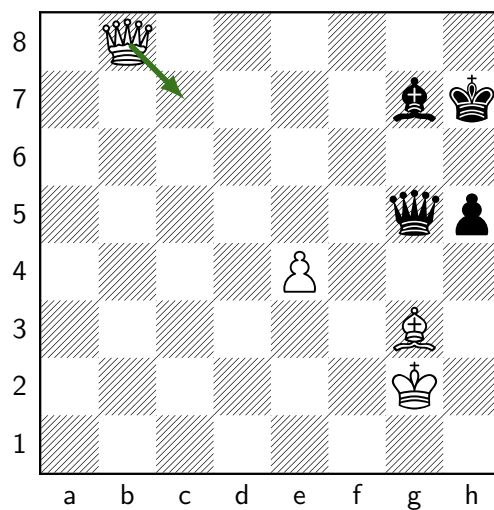


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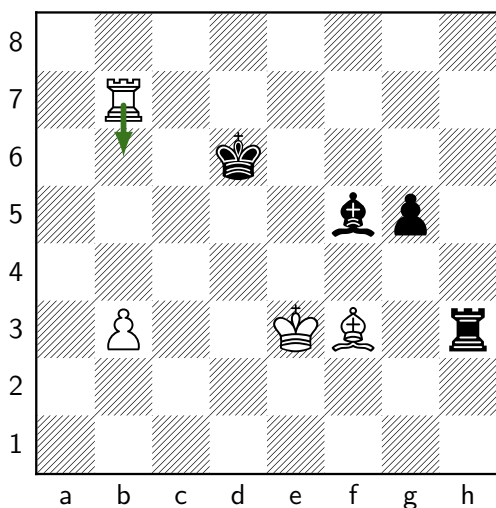
Figure 42: **Move 48, White: f5.** (a) Our position; Melanie plays f5 (f4f5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2448` (ply 90) [structural distance $d=30$, QAP obj=61]: the master played Rd3; under μ our Pf4 plays the role of that game’s Rh3 (a **cross-type** correspondence); cross-type roles: our qb3 as their ra2, our Rh5 as their Pb4; transferred move f4f5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1972` (ply 109) [structural distance $d=28$, QAP obj=46]: the master played Ke6; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our qb3 as their ra7; transferred move g3f2. (d) Retrieved from `ccr1_top_p0003#1158` (ply 84) [structural distance $d=30$, QAP obj=29]: the master played Qc7; under μ our Rh5 plays the role of that game’s Qb8 (a **cross-type** correspondence); transferred move h5g5. Synthesis: 1 retrieved analog transferred to f5 (summed QAP objective 61, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



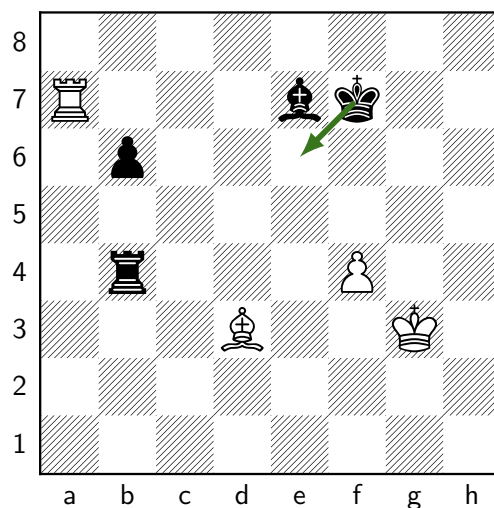
(a)



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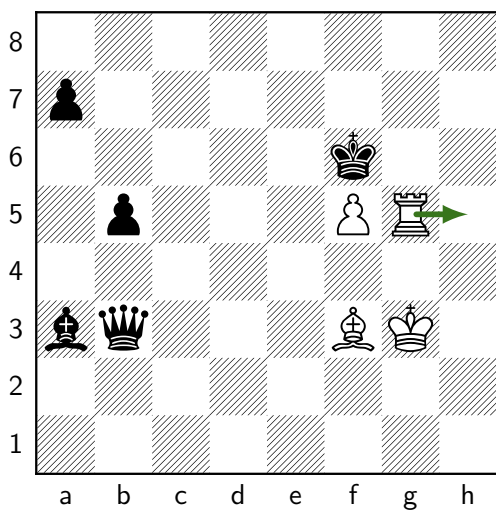


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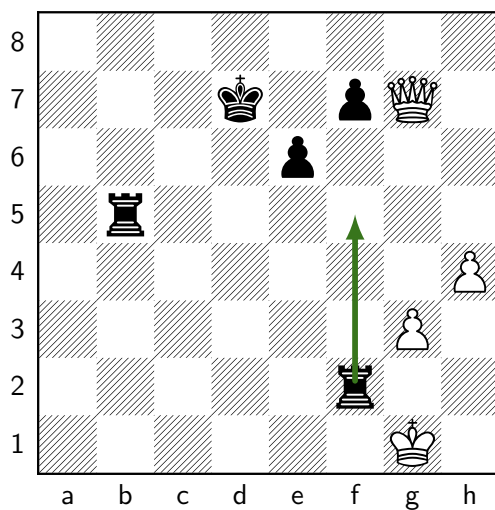


(d)

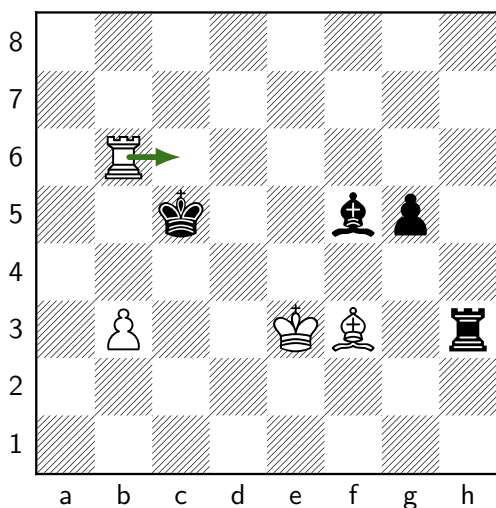
Figure 43: **Move 49, White: Rg5+.** (a) Our position; Melanie plays Rg5+ (h5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1158` (ply 84) [structural distance $d=26$, QAP obj=30]: the master played Qc7; under μ our Rh5 plays the role of that game’s Qb8 (a **cross-type** correspondence); transferred move h5g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3296` (ply 108) [structural distance $d=29$, QAP obj=27]: the master played Rb6+; under μ our Rh5 plays the role of that game’s Rb7; cross-type roles: our ba3 as their Pb3, our qb3 as their rh3; transferred move h5g5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1972` (ply 109) [structural distance $d=26$, QAP obj=45]: the master played Ke6; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our qb3 as their ra7, our ba3 as their pf4; transferred move g3f4. Synthesis: 2 retrieved analogs transferred to Rg5+ (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



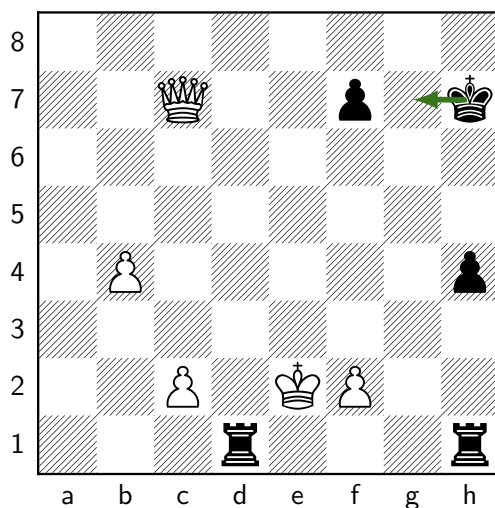
(a)



(b)

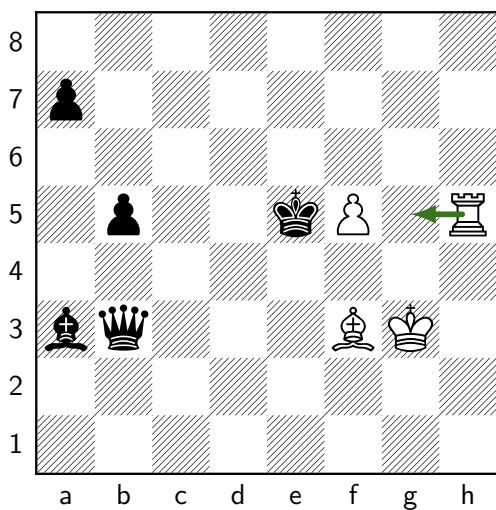


(c)

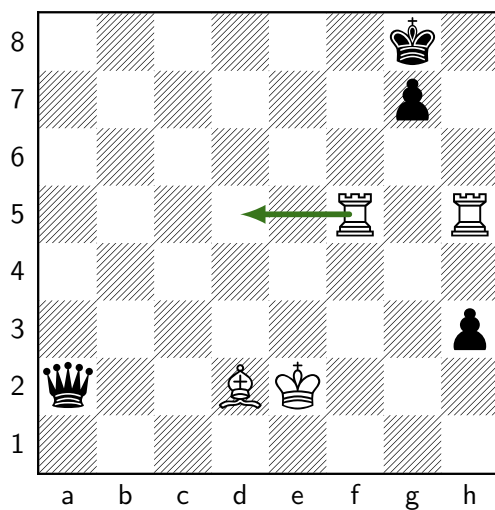


(d)

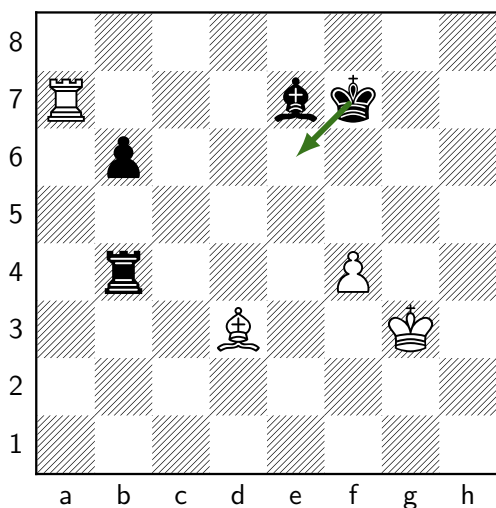
Figure 44: **Move 50, White: Rh5.** (a) Our position; Melanie plays Rh5 (g5h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2590` (ply 115) [structural distance $d=35$, QAP obj=35]: the master played Rff5; under μ our Rg5 plays the role of that game’s Rf7; cross-type roles: our Bf3 as their Pf7, our pb5 as their Rb5; transferred move g5h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3296` (ply 110) [structural distance $d=36$, QAP obj=30]: the master played Rc6+; under μ our Rg5 plays the role of that game’s Rb6; cross-type roles: our qb3 as their rh3, our pb5 as their bf5; transferred move g5g6. (d) Retrieved from `ccr1_top_p0002#365` (ply 97) [structural distance $d=36$, QAP obj=28]: the master played Kg7; under μ our Kg3 plays the role of that game’s Kh2; cross-type roles: our Bf3 as their Pf7, our ba3 as their pb4; transferred move g3h4. Synthesis: 2 retrieved analogs transferred to Rh5 (summed QAP objective 53, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



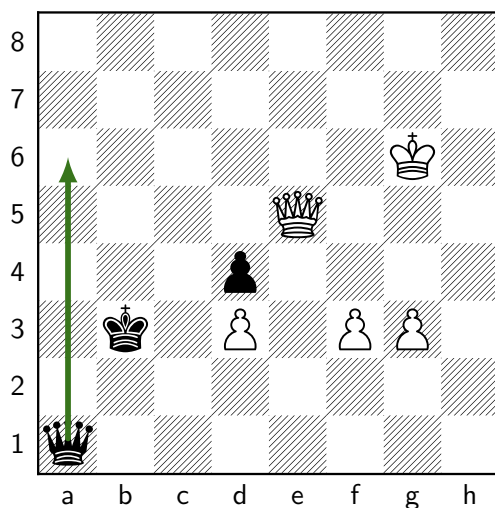
(a)



(b)



(c)



(d)

Figure 45: **Move 51, White: Rg5.** (a) Our position; Melanie plays Rg5 (h5g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3074` (ply 114) [structural distance $d=30$, QAP obj=35]: the master played Rd5; under μ our Rh5 plays the role of that game’s Rf5; cross-type roles: our ke5 as their Rh5, our ba3 as their pg7; transferred move h5g5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1972` (ply 109) [structural distance $d=28$, QAP obj=40]: the master played Ke6; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our qb3 as their ra7, our ba3 as their pf4; transferred move g3f2. (d) Retrieved from `ccr1_top_p0003#1161` (ply 101) [structural distance $d=30$, QAP obj=30]: the master played Qa6+; under μ our Pf5 plays the role of that game’s Qa8 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Pd4, our Rh5 as their pf3; transferred move f5f6. Synthesis: 2 retrieved analogs transferred to Rg5 (summed QAP objective 71, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

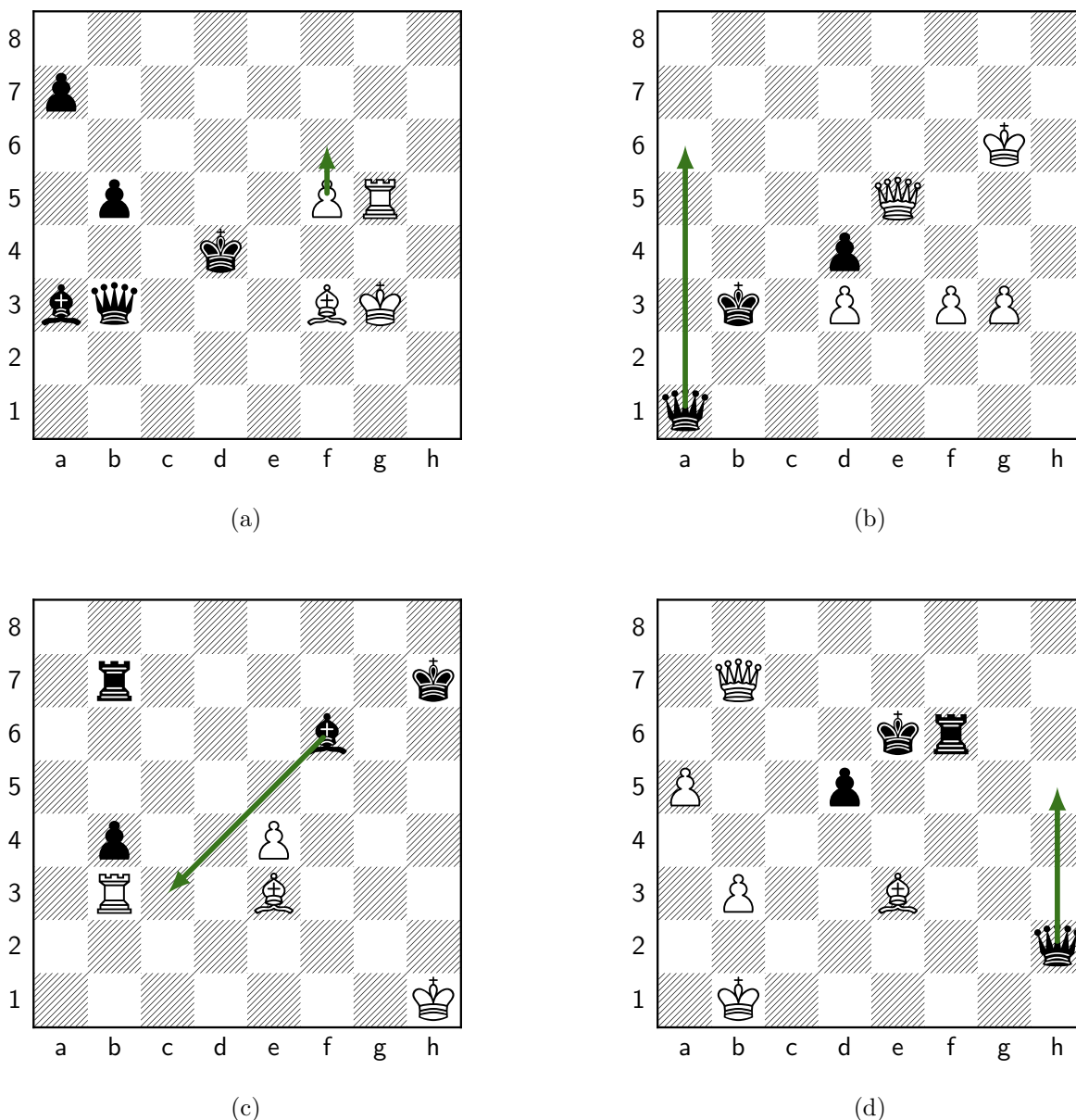


Figure 46: **Move 52, White: f6.** (a) Our position; Melanie plays f6 (f5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1161` (ply 101) [structural distance $d=31$, QAP obj=31]: the master played Qa6+; under μ our Pf5 plays the role of that game’s Qa8 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Pd4, our Rg5 as their pd3; transferred move f5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3728` (ply 119) [structural distance $d=31$, QAP obj=31]: the master played Bc3; under μ our Pf5 plays the role of that game’s Bf3 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Pb4, our qb3 as their rb3; transferred move f5f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2870` (ply 99) [structural distance $d=23$, QAP obj=16]: the master played Qh5; under μ our Pf5 plays the role of that game’s Qh7 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Pd5, our ba3 as their pb3; transferred move f5f6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to f6 (summed QAP objective 78, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

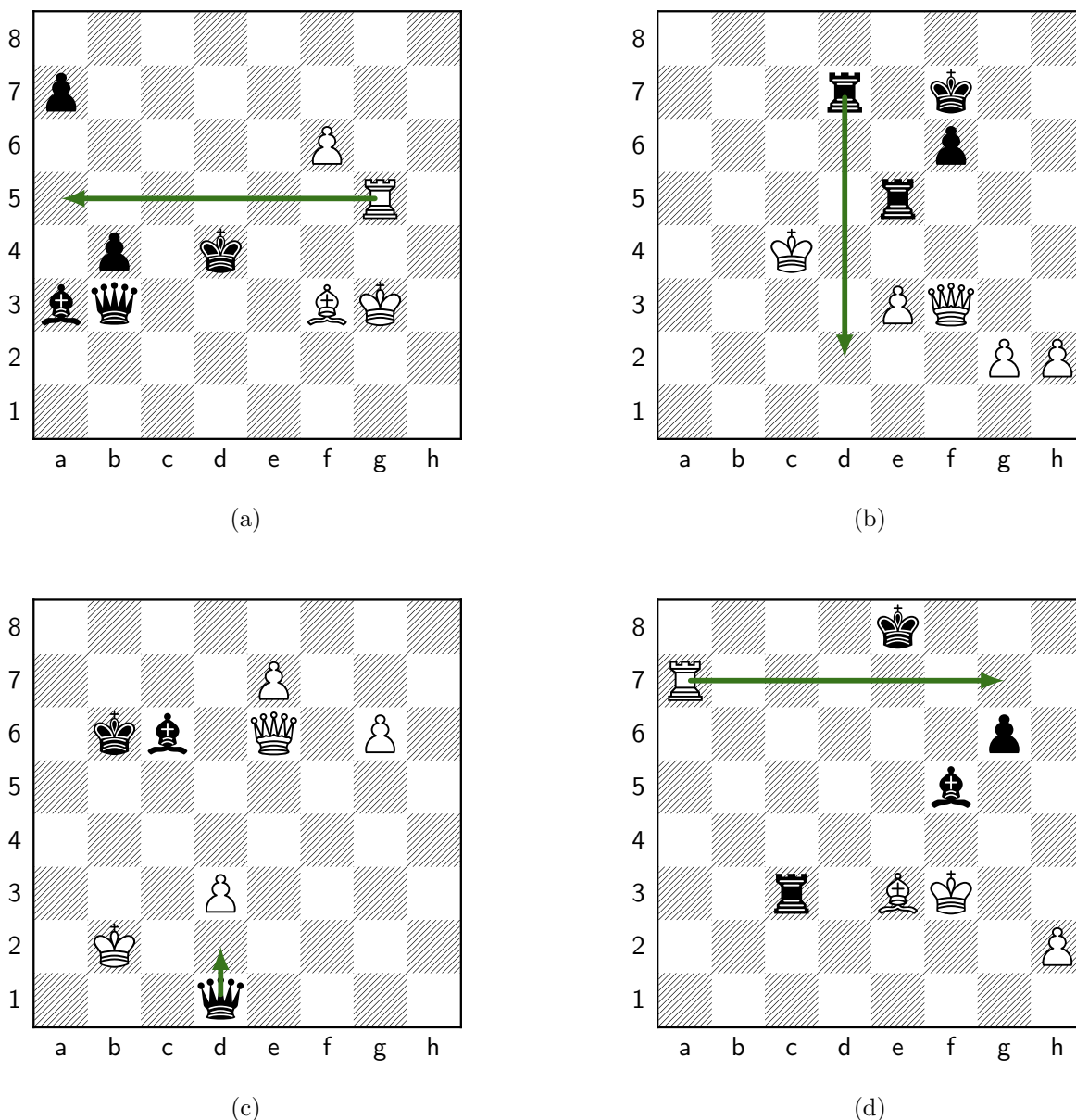
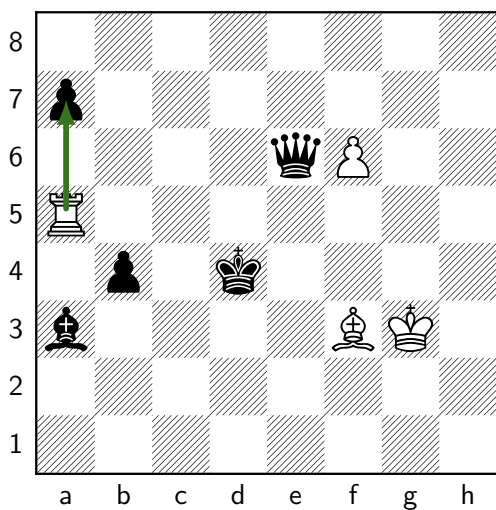
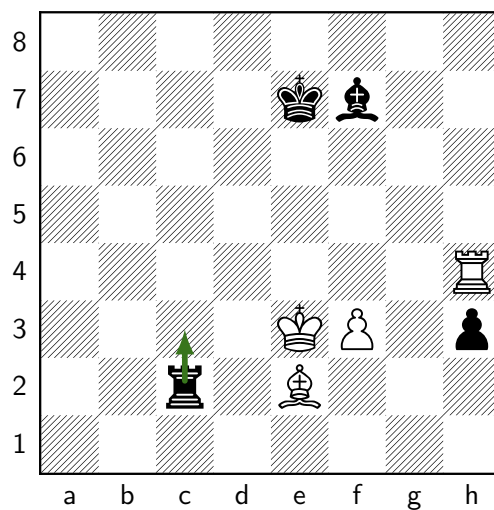


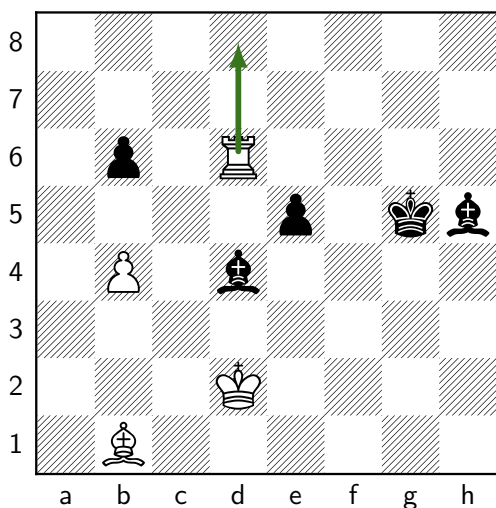
Figure 47: **Move 53, White: Ra5.** (a) Our position; Melanie plays Ra5 (g5a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#1754 (ply 105) [structural distance $d=29$, QAP obj=27]: the master played Rd2; under μ our Rg5 plays the role of that game's Rd2; cross-type roles: our Bf3 as their Pf6, our qb3 as their Re5; transferred move g5a5 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1651 (ply 121) [structural distance $d=26$, QAP obj=26]: the master played Qd2+; under μ our Rg5 plays the role of that game's Qd8 (a **cross-type** correspondence); cross-type roles: our ba3 as their pe7; transferred move g5a5 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1219 (ply 120) [structural distance $d=33$, QAP obj=23]: the master played Rg7; under μ our Rg5 plays the role of that game's Ra7; cross-type roles: our qb3 as their rc3; transferred move g5a5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ra5 (summed QAP objective 75, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



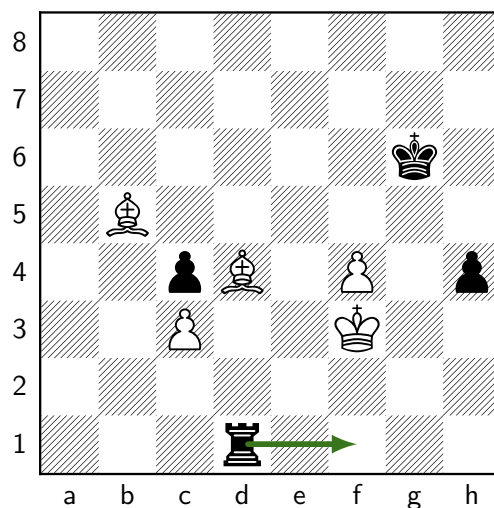
(a)



(b)

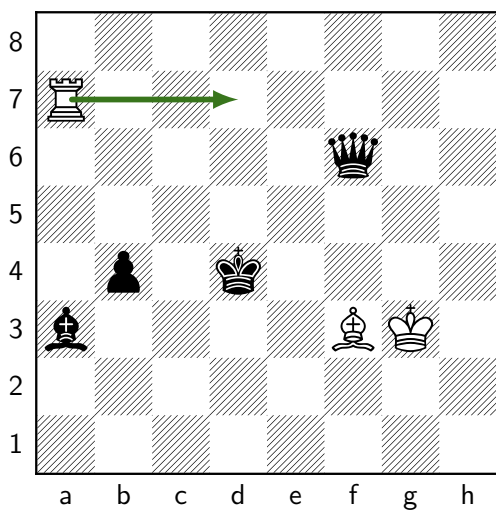


(c)

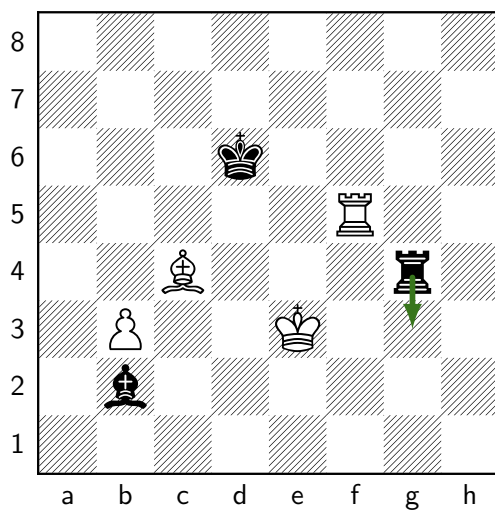


(d)

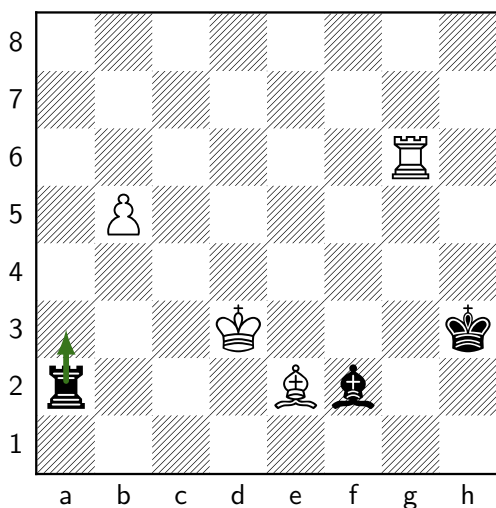
Figure 48: **Move 54, White: Rxa7**. (a) Our position; Melanie plays Rxa7 (a5a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2718` (ply 121) [structural distance $d=24$, QAP obj=34]: the master played Rc3+; under μ our Ra5 plays the role of that game’s Rc7; cross-type roles: our qe6 as their rh4; transferred move a5a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2356` (ply 94) [structural distance $d=24$, QAP obj=33]: the master played Rd8; under μ our Ra5 plays the role of that game’s Rd6; cross-type roles: our qe6 as their bh5; transferred move a5a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3717` (ply 123) [structural distance $d=24$, QAP obj=28]: the master played Rf1+; under μ our Ra5 plays the role of that game’s Rd8; cross-type roles: our qe6 as their bb5, our Bf3 as their pf4; transferred move a5a7 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Rxa7 (summed QAP objective 142, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



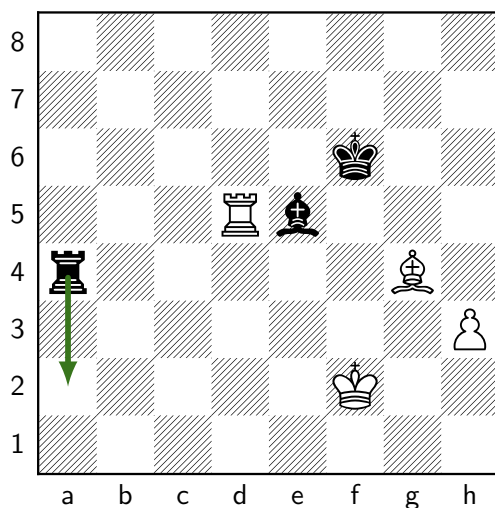
(a)



(b)

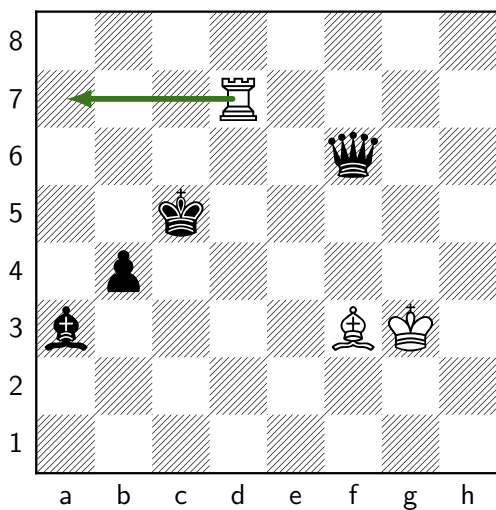


(c)

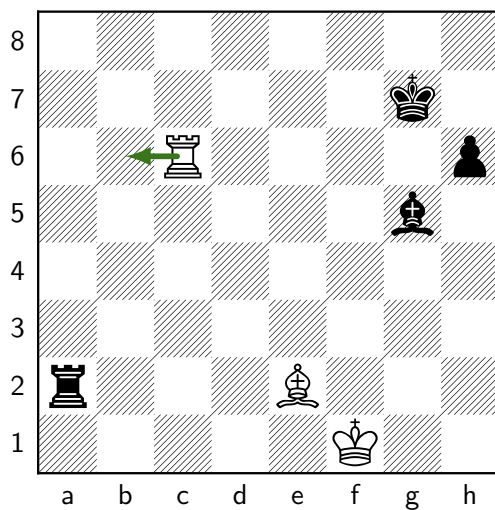


(d)

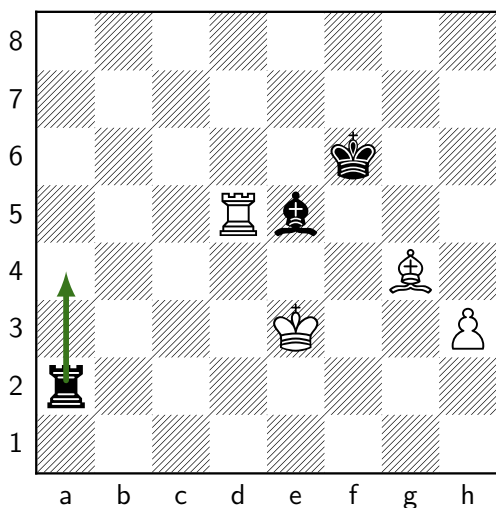
Figure 49: **Move 55, White: Rd7+.** (a) Our position; Melanie plays Rd7+ (a7d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#164` (ply 93) [structural distance $d=13$, QAP obj=18]: the master played Rg3+; under μ our Ra7 plays the role of that game's Rg5; cross-type roles: our qf6 as their rf5; transferred move a7d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#274` (ply 121) [structural distance $d=16$, QAP obj=15]: the master played Ra3+; under μ our Ra7 plays the role of that game's Ra7; cross-type roles: our qf6 as their rg6; transferred move a7d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1081` (ply 107) [structural distance $d=14$, QAP obj=9]: the master played Ra2+; under μ our Ra7 plays the role of that game's Ra5; cross-type roles: our qf6 as their rd5; transferred move a7d7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rd7+ (summed QAP objective 42, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



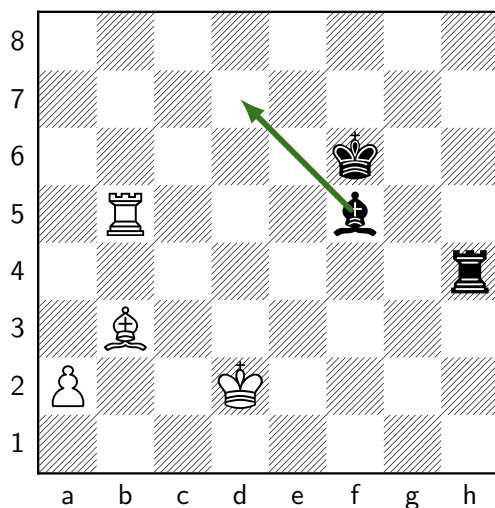
(a)



(b)

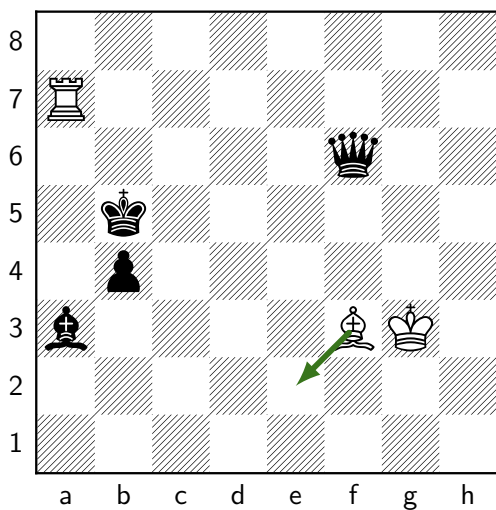


(c)

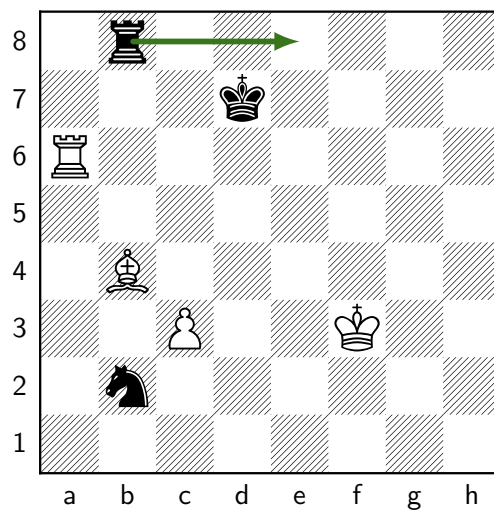


(d)

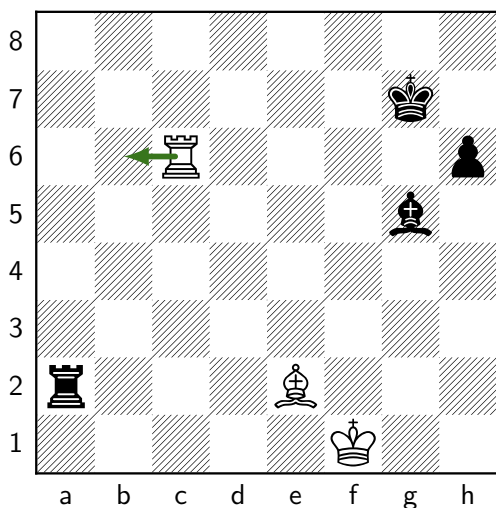
Figure 50: **Move 56, White: Ra7.** (a) Our position; Melanie plays Ra7 (d7a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0003#2365` (ply 114) [structural distance $d=15$, QAP obj=17]: the master played Rb6; under μ our Rd7 plays the role of that game's Rc6; cross-type roles: our qf6 as their ra2; transferred move d7a7 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#1081` (ply 109) [structural distance $d=14$, QAP obj=10]: the master played Ra4; under μ our Rd7 plays the role of that game's Ra7; cross-type roles: our qf6 as their rd5; transferred move d7a7 (**endorses this move**). (d) Retrieved from `ccr1_top.p0003#1750` (ply 119) [structural distance $d=15$, QAP obj=20]: the master played Bd7; under μ our Bf3 plays the role of that game's Bf4; cross-type roles: our qf6 as their rb5; transferred move f3e2. Synthesis: 2 retrieved analogs transferred to Ra7 (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



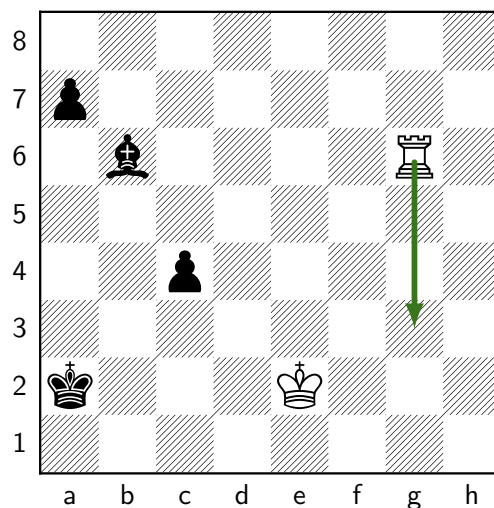
(a)



(b)

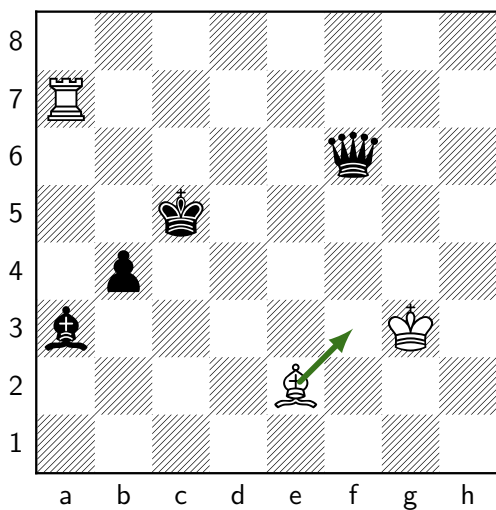


(c)

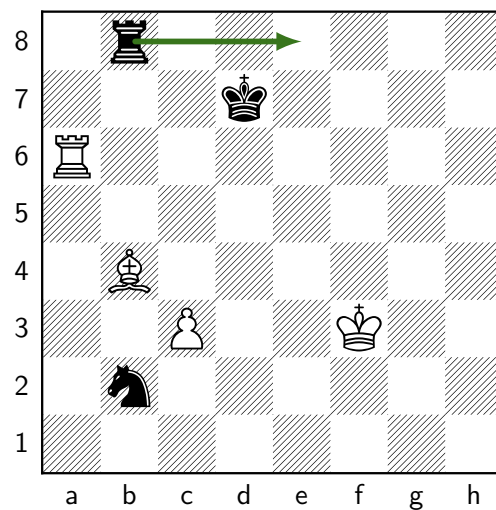


(d)

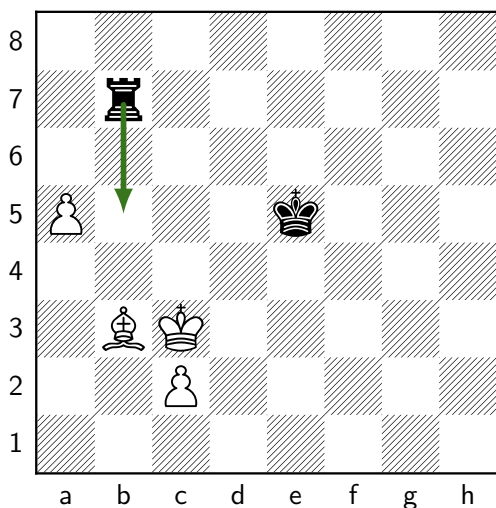
Figure 51: **Move 57, White: Be2+ (f3e2)**. (a) Our position; Melanie plays **Be2+** (f3e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3149` (ply 117) [structural distance $d=16$, QAP obj=20]: the master played **Re8**; under μ our *Ra7* plays the role of that game’s *Rb1*; cross-type roles: our *qf6* as their *ra6*, our *Bf3* as their *Nb2*; transferred move **a7c7**. (c) Retrieved from `ccr1_top_p0003#2365` (ply 114) [structural distance $d=14$, QAP obj=17]: the master played **Rb6**; under μ our *Ra7* plays the role of that game’s *Rc6*; cross-type roles: our *qf6* as their *ra2*, our *pb4* as their *bg5*; transferred move **a7c7**. (d) Retrieved from `ccr1_top_p0001#1167` (ply 104) [structural distance $d=16$, QAP obj=15]: the master played **Rg3**; under μ our *Ra7* plays the role of that game’s *Rg6*; cross-type roles: our *qf6* as their *pc4*; transferred move **a7c7**. Synthesis: no retrieved analog transferred to **Be2+** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



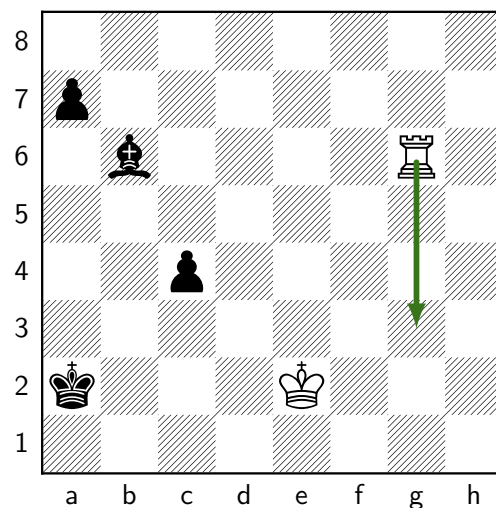
(a)



(b)

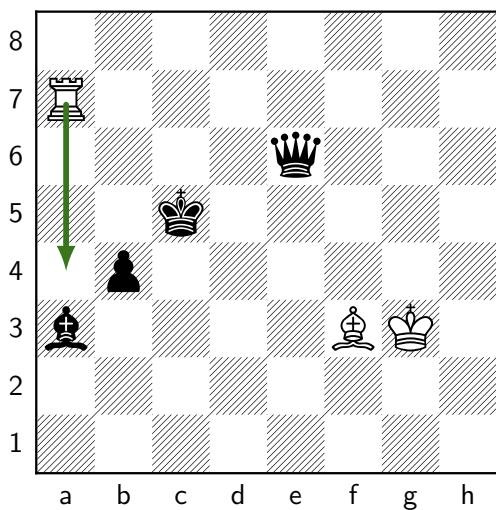


(c)

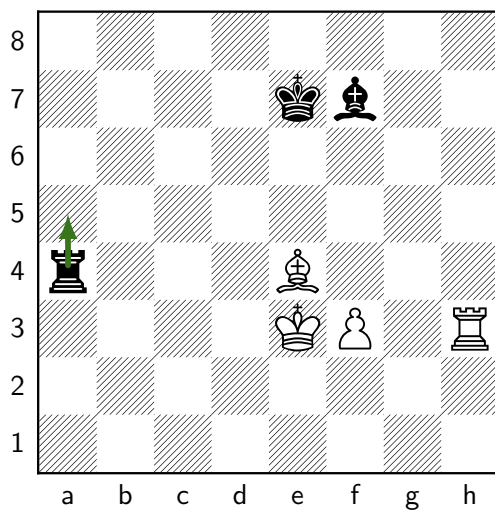


(d)

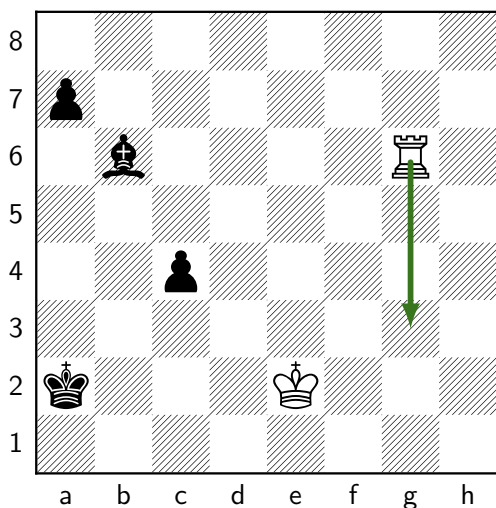
Figure 52: **Move 58, White: Bf3**. (a) Our position; Melanie plays Bf3 (e2f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3149` (ply 117) [structural distance $d=12$, QAP obj=24]: the master played Re8; under μ our Ra7 plays the role of that game’s Rb1; cross-type roles: our qf6 as their ra6, our Be2 as their Nb2; transferred move a7a4. (c) Retrieved from `ccr1_top_p0000#204` (ply 109) [structural distance $d=11$, QAP obj=22]: the master played Rb5; under μ our Ra7 plays the role of that game’s Rb2; cross-type roles: our qf6 as their pa5; transferred move a7a4. (d) Retrieved from `ccr1_top_p0001#1167` (ply 104) [structural distance $d=11$, QAP obj=19]: the master played Rg3; under μ our Ra7 plays the role of that game’s Rg6; cross-type roles: our qf6 as their pc4; transferred move a7a4. Synthesis: no retrieved analog transferred to Bf3 – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



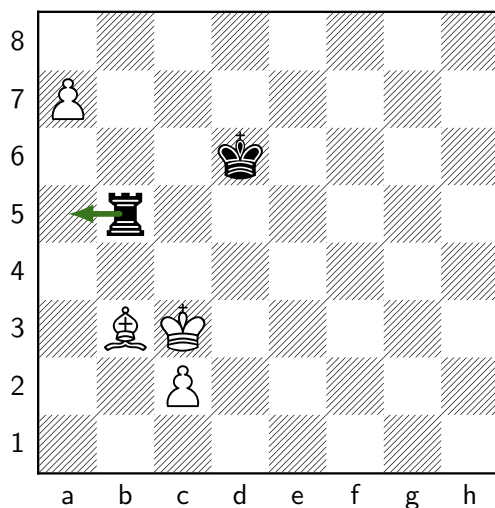
(a)



(b)



(c)



(d)

Figure 53: **Move 59, White: Ra4.** (a) Our position; Melanie plays **Ra4** (a7a4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2718` (ply 131) [structural distance $d=10$, QAP obj=25]: the master played **Ra5**; under μ our *Ra7* plays the role of that game's *Ra5*; cross-type roles: our *qe6* as their *rh3*; transferred move **a7a4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1167` (ply 104) [structural distance $d=15$, QAP obj=18]: the master played **Rg3**; under μ our *Ra7* plays the role of that game's *Rg6*; cross-type roles: our *Bf3* as their *pc4*; transferred move **a7a4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#204` (ply 113) [structural distance $d=15$, QAP obj=14]: the master played **Ra5**; under μ our *Ra7* plays the role of that game's *Rb4*; cross-type roles: our *qe6* as their *pa7*; transferred move **a7a4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Ra4** (summed QAP objective 58, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

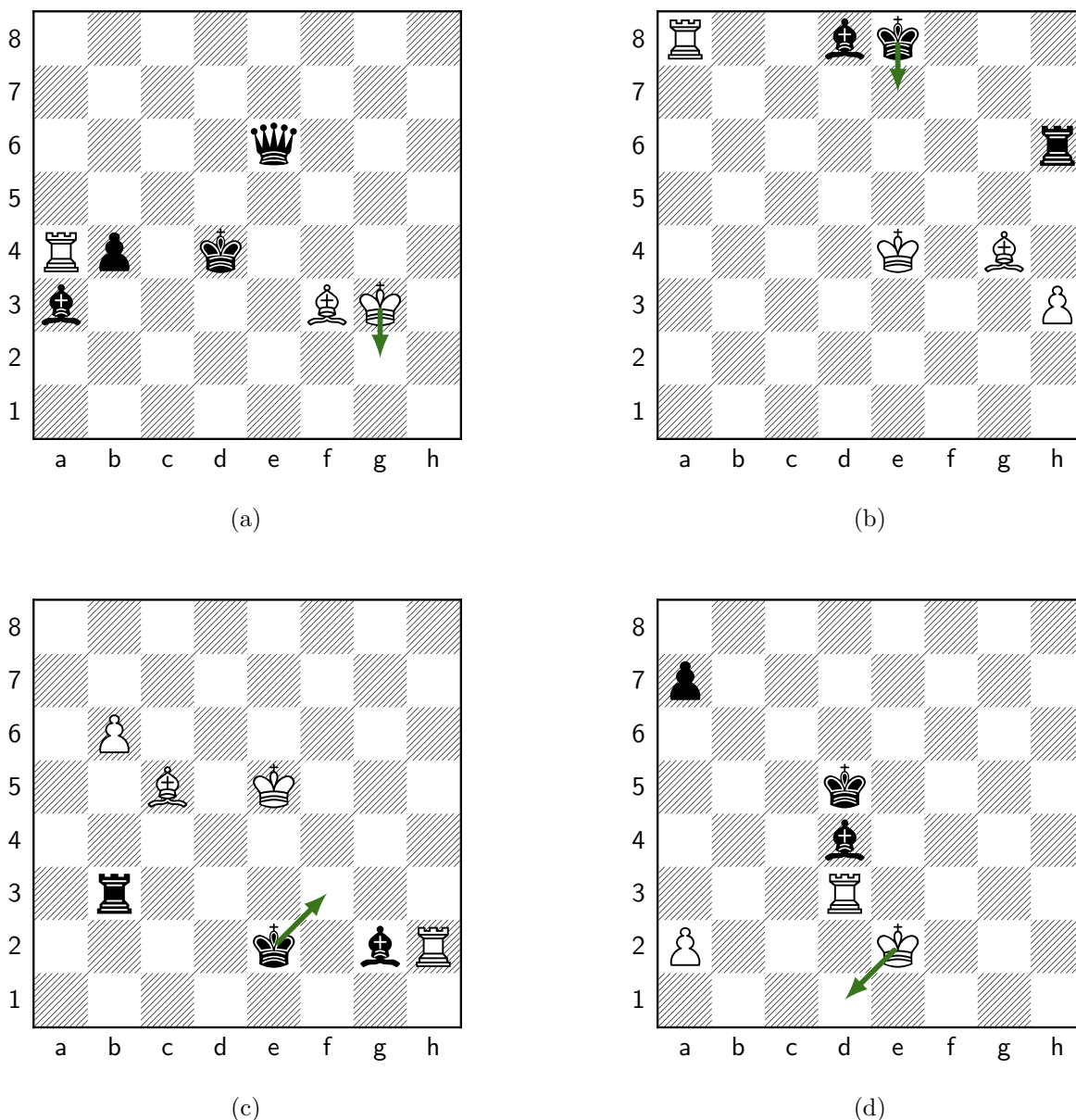
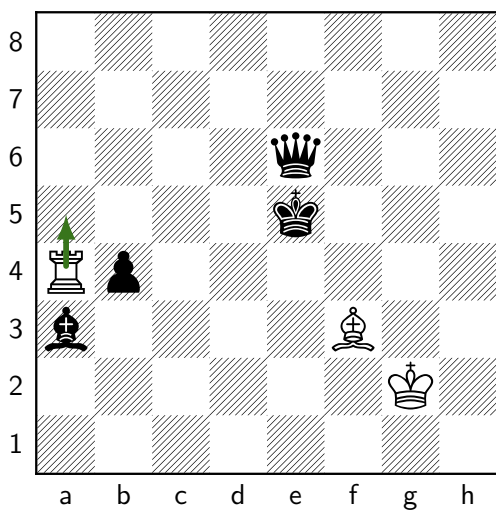
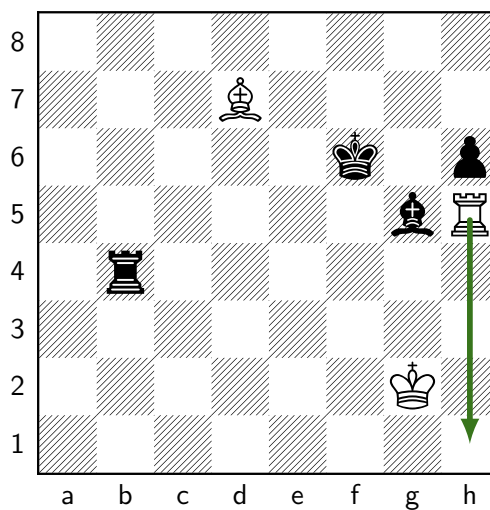


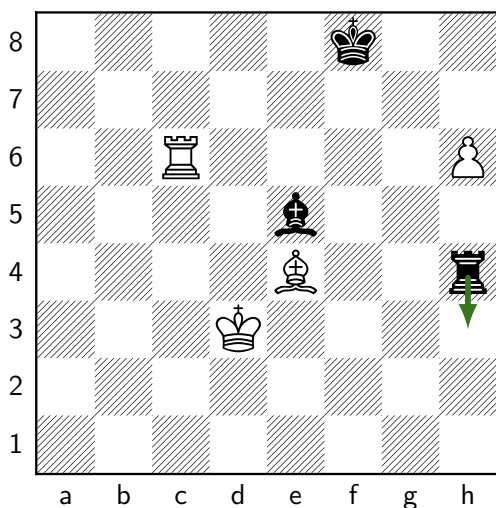
Figure 54: **Move 60, White: Kg2.** (a) Our position; Melanie plays Kg2 (g3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#479` (ply 125) [structural distance $d=16$, QAP obj=27]: the master played Ke7; under μ our Kg3 plays the role of that game’s Ke1; cross-type roles: our pb4 as their Bd8, our qe6 as their Rh6; transferred move g3g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2991` (ply 109) [structural distance $d=18$, QAP obj=19]: the master played Kf3; under μ our Kg3 plays the role of that game’s Ke7; cross-type roles: our qe6 as their Rb3; transferred move g3g2 (**endorses this move**). (d) Retrieved from `ccrl_top_p0003#1213` (ply 118) [structural distance $d=18$, QAP obj=19]: the master played Kd1; under μ our Kg3 plays the role of that game’s Ke2; cross-type roles: our Bf3 as their Pa2; transferred move g3g2 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg2 (summed QAP objective 92, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



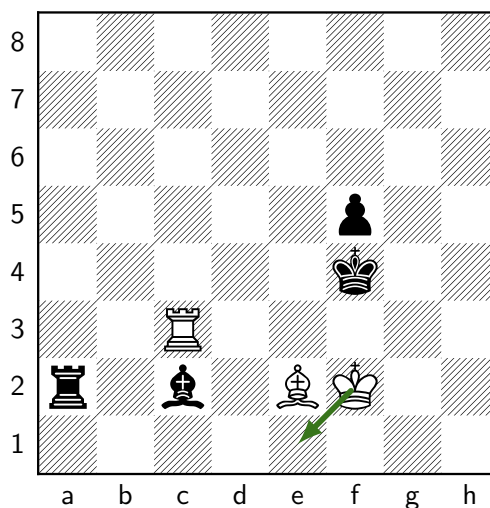
(a)



(b)

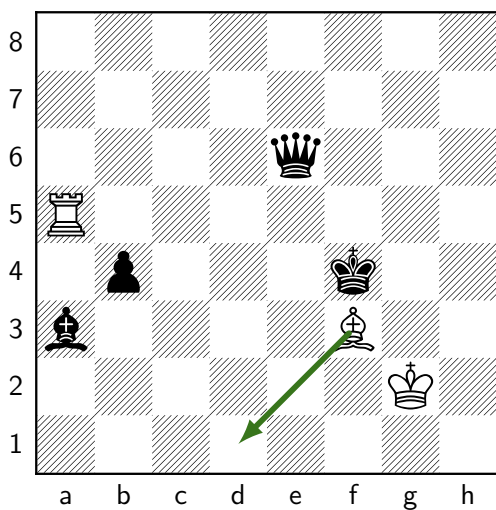


(c)

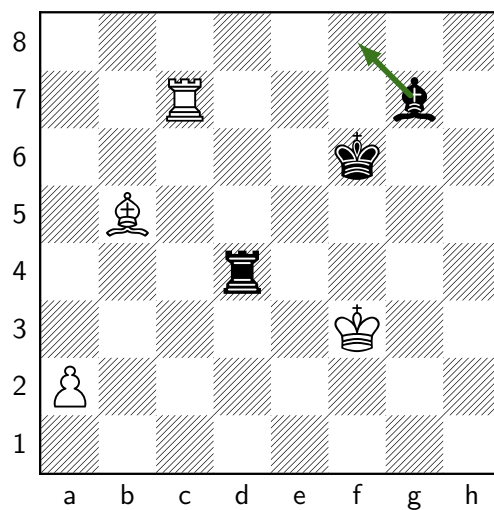


(d)

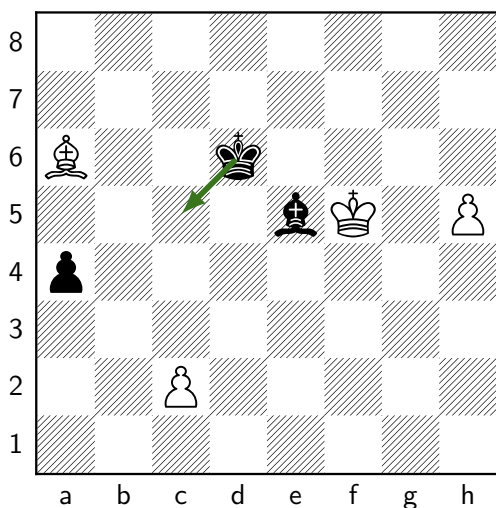
Figure 55: **Move 61, White: Ra5+ (a4a5).** (a) Our position; Melanie plays Ra5+ (a4a5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#370` (ply 112) [structural distance $d=21$, QAP obj=38]: the master played Rh1; under μ our Ra4 plays the role of that game's Rh5; cross-type roles: our qe6 as their rb4; transferred move a4a5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#76` (ply 113) [structural distance $d=24$, QAP obj=10]: the master played Rh3+; under μ our Ra4 plays the role of that game's Rh5; cross-type roles: our qe6 as their rc6; transferred move a4a5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3840` (ply 126) [structural distance $d=24$, QAP obj=22]: the master played Ke1; under μ our Kg2 plays the role of that game's Kf2; cross-type roles: our pb4 as their ra2, our qe6 as their pf5; transferred move g2f2. Synthesis: 2 retrieved analogs transferred to Ra5+ (summed QAP objective 48, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



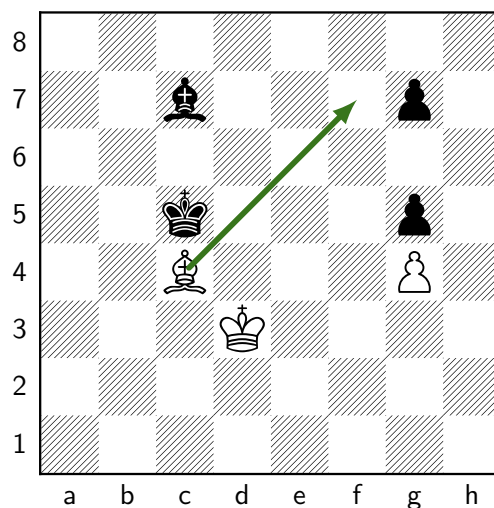
(a)



(b)

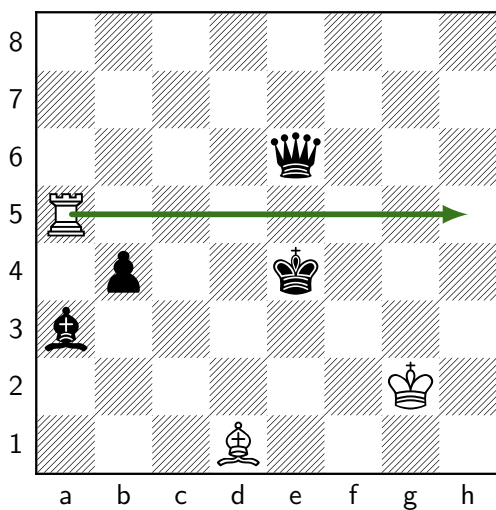


(c)

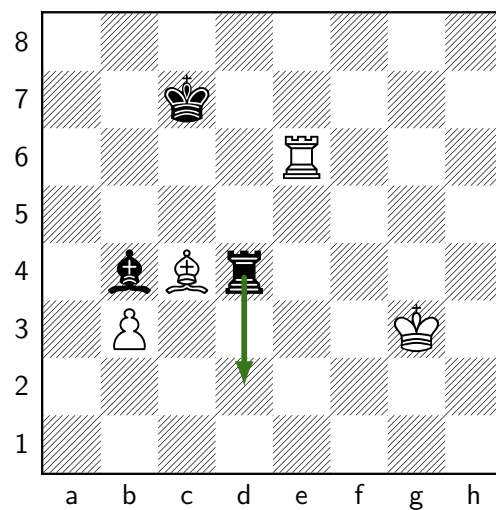


(d)

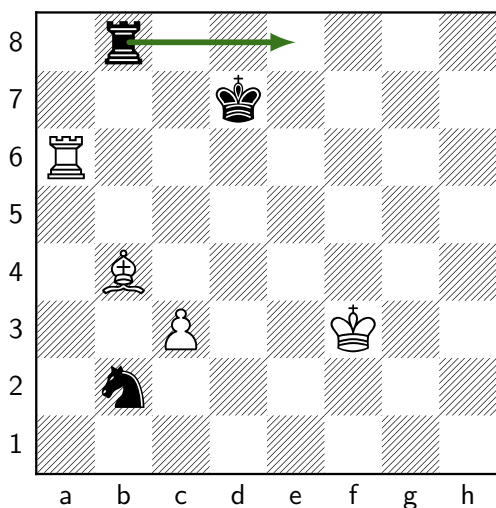
Figure 56: **Move 62, White: Bd1.** (a) Our position; Melanie plays Bd1 (f3d1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3495` (ply 129) [structural distance $d=16$, QAP obj=6]: the master played Bf8; under μ our Bf3 plays the role of that game’s Bg2; cross-type roles: our kf4 as their rc7, our qe6 as their bb5; transferred move f3d1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2547` (ply 109) [structural distance $d=17$, QAP obj=25]: the master played Kc5; under μ our Kg2 plays the role of that game’s Kd3; cross-type roles: our Ra5 as their Pa4, our qe6 as their pc2; transferred move g2h2. (d) Retrieved from `ccr1_top_p0002#2271` (ply 120) [structural distance $d=17$, QAP obj=25]: the master played Bf7; under μ our Bf3 plays the role of that game’s Bc4; cross-type roles: our qe6 as their Pg4, our Ra5 as their pg5; transferred move f3b7. Synthesis: 1 retrieved analog transferred to Bd1 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



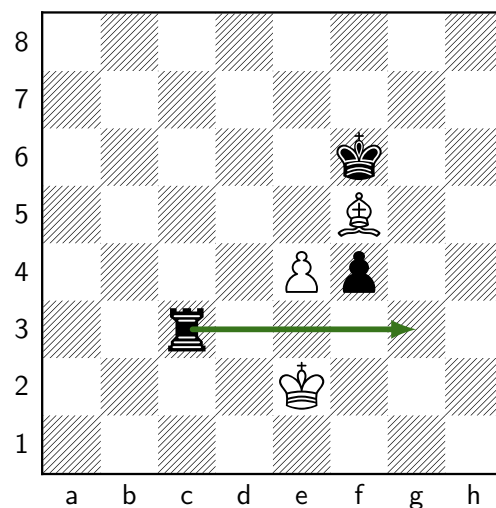
(a)



(b)

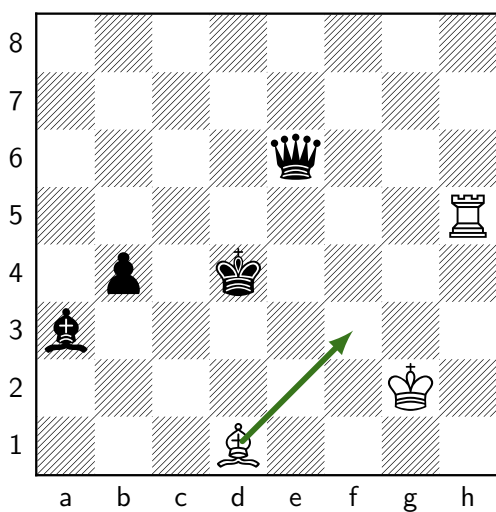


(c)

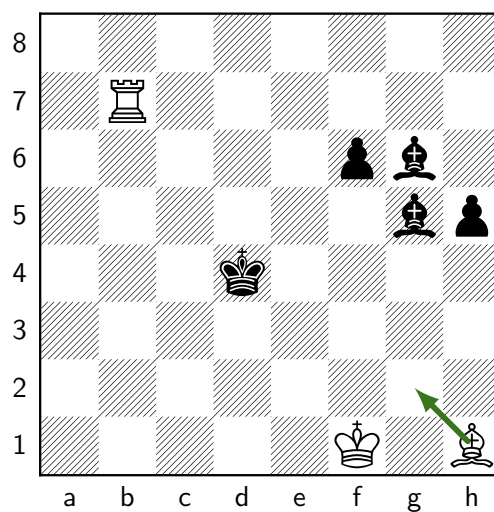


(d)

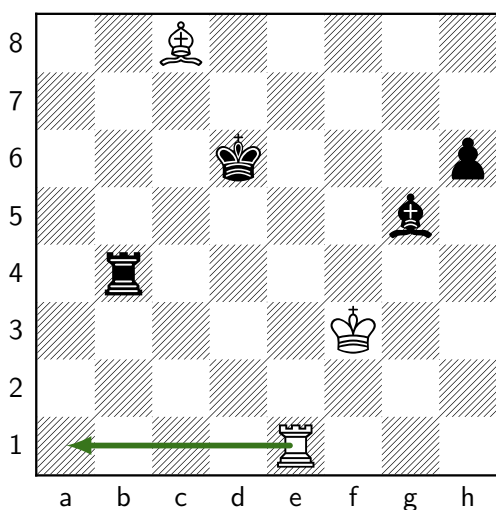
Figure 57: **Move 63, White: Rh5**. (a) Our position; Melanie plays Rh5 (a5h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1643` (ply 117) [structural distance $d=12$, QAP obj=14]: the master played Rd2; under μ our Ra5 plays the role of that game’s Rd5; cross-type roles: our qe6 as their re6; transferred move a5h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3149` (ply 117) [structural distance $d=12$, QAP obj=12]: the master played Re8; under μ our Ra5 plays the role of that game’s Rb1; cross-type roles: our qe6 as their ra6, our Bd1 as their Nb2; transferred move a5h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1559` (ply 117) [structural distance $d=13$, QAP obj=1]: the master played Rg3; under μ our Ra5 plays the role of that game’s Rc6; cross-type roles: our Bd1 as their Pf4; transferred move a5h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rh5 (summed QAP objective 27, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



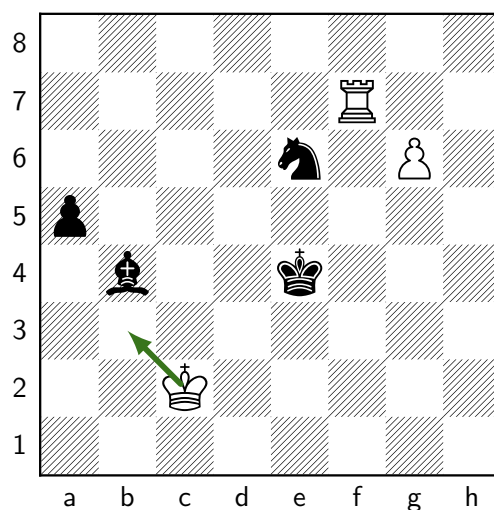
(a)



(b)



(c)



(d)

Figure 58: **Move 64, White: Bf3**. (a) Our position; Melanie plays Bf3 (d1f3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2863` (ply 114) [structural distance $d=10$, QAP obj=4]: the master played Bg2; under μ our Bd1 plays the role of that game’s Bh1; cross-type roles: our qe6 as their bg6; transferred move d1f3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#370` (ply 120) [structural distance $d=9$, QAP obj=4]: the master played Ra1; under μ our Rh5 plays the role of that game’s Re1; cross-type roles: our qe6 as their rb4; transferred move h5h4. (d) Retrieved from `ccr1_top_p0001#3983` (ply 140) [structural distance $d=12$, QAP obj=4]: the master played Kb3; under μ our Kg2 plays the role of that game’s Kc2; cross-type roles: our qe6 as their ne6, our Bd1 as their Pg6; transferred move g2g3. Synthesis: 1 retrieved analog transferred to Bf3 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

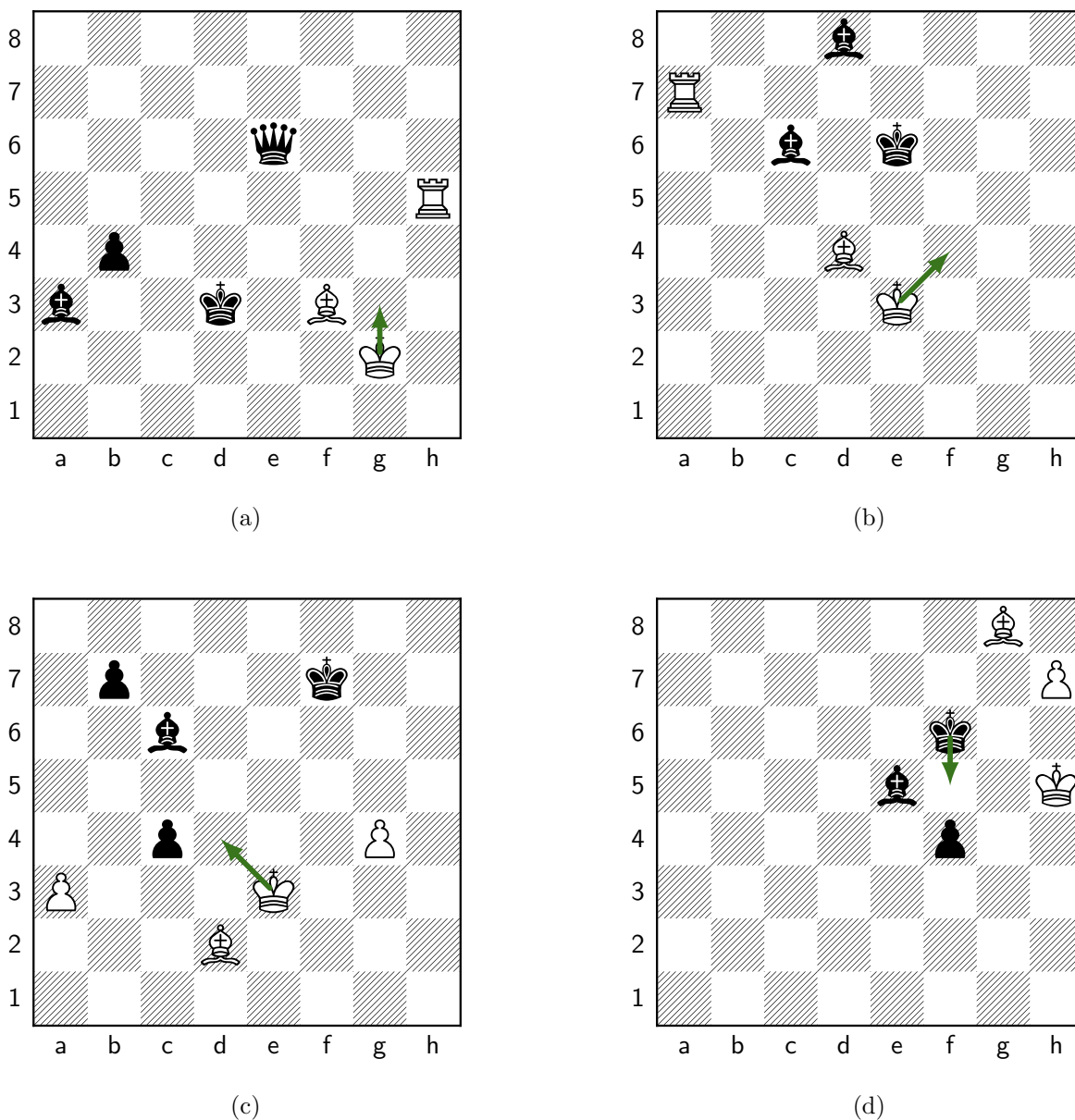
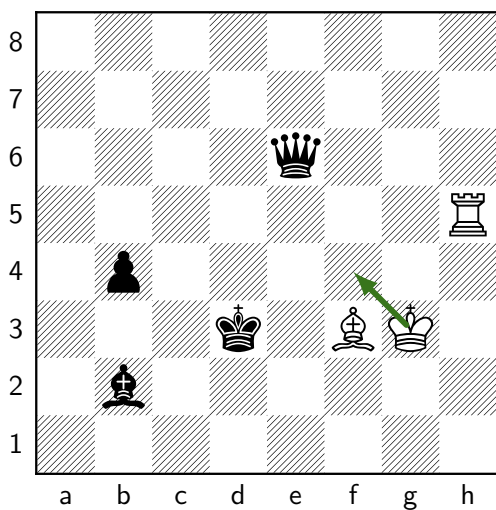
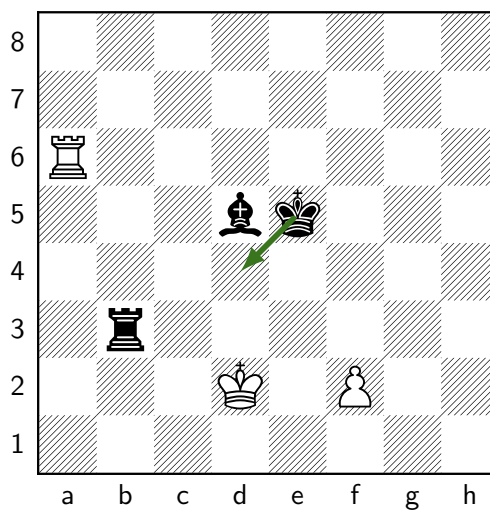


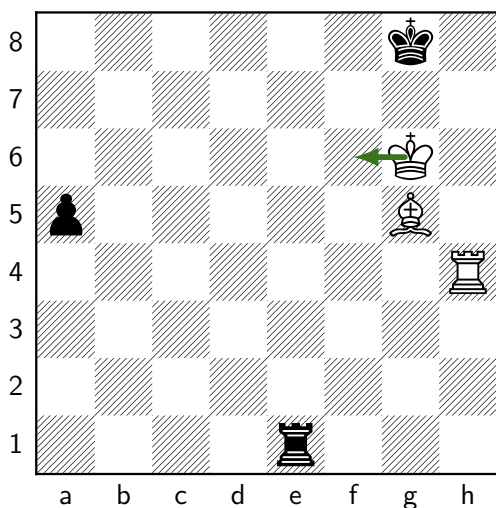
Figure 59: **Move 65, White: Kg3.** (a) Our position; Melanie plays Kg3 (**g2g3**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2456` (ply 144) [structural distance $d=14$, QAP obj=4]: the master played Kf4; under μ our Kg2 plays the role of that game’s Ke3; cross-type roles: our qe6 as their bd8; transferred move g2g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#16` (ply 132) [structural distance $d=16$, QAP obj=3]: the master played Kd4; under μ our Kg2 plays the role of that game’s Ke3; cross-type roles: our Rh5 as their Pa3, our qe6 as their pc4; transferred move g2g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2362` (ply 143) [structural distance $d=11$, QAP obj=3]: the master played Kf5; under μ our Kg2 plays the role of that game’s Kf3; cross-type roles: our qe6 as their bg8, our Rh5 as their ph7; transferred move g2g3 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg3 (summed QAP objective 9, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



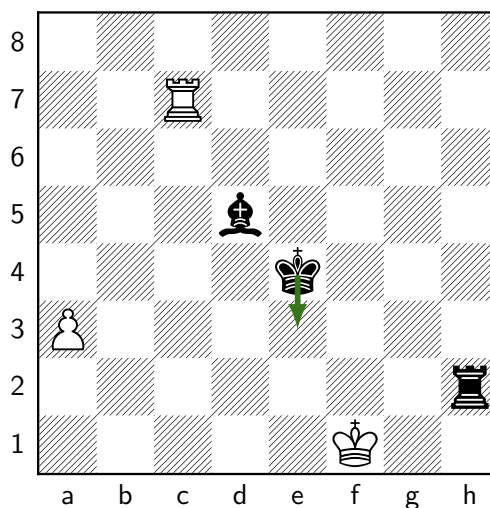
(a)



(b)

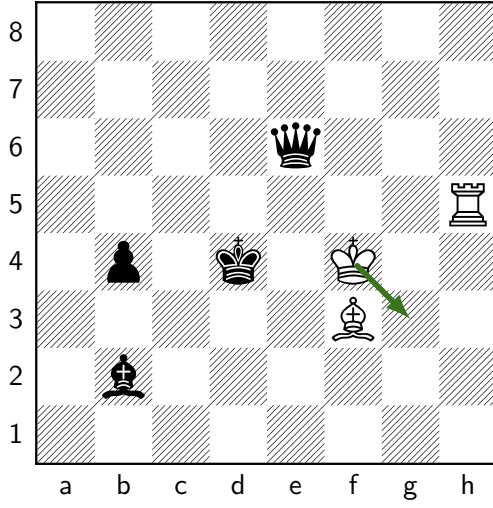


(c)

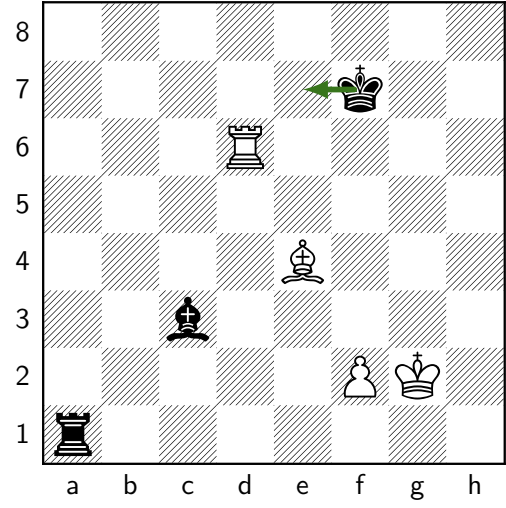


(d)

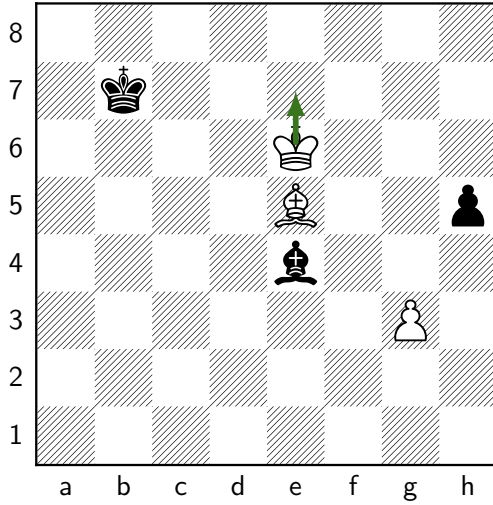
Figure 60: **Move 66, White: Kf4**. (a) Our position; Melanie plays Kf4 (g3f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2447` (ply 141) [structural distance $d=10$, QAP obj=4]: the master played Kd4; under μ our $Kg3$ plays the role of that game’s $Ke4$; cross-type roles: our $qe6$ as their $ra6$, our $bb2$ as their $pf2$; transferred move g3f4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3657` (ply 144) [structural distance $d=9$, QAP obj=3]: the master played Kf6; under μ our $Kg3$ plays the role of that game’s $Kg6$; cross-type roles: our $qe6$ as their $re1$, our $bb2$ as their $pa5$; transferred move g3f4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#94` (ply 127) [structural distance $d=10$, QAP obj=1]: the master played Ke3; under μ our $Kg3$ plays the role of that game’s $Ke5$; cross-type roles: our $qe6$ as their $rc7$, our $Rh5$ as their $Bd5$; transferred move g3f4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kf4 (summed QAP objective 12, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



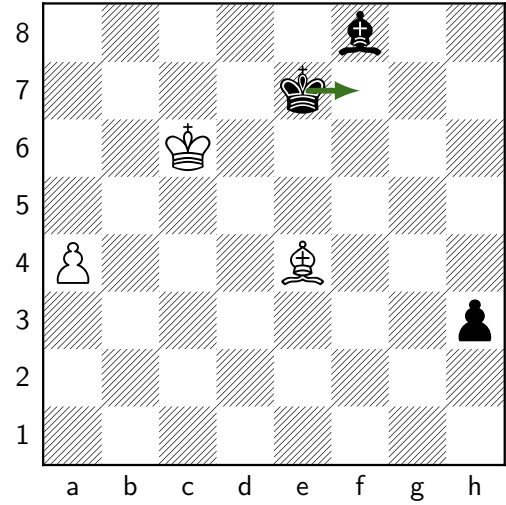
(a)



(b)



(c)



(d)

Figure 61: **Move 67, White: Kg3**. (a) Our position; Melanie plays Kg3 (f4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1799` (ply 125) [structural distance $d=15$, QAP obj=4]: the master played Ke7; under μ our Kf4 plays the role of that game’s Kf2; cross-type roles: our qe6 as their rd6; transferred move f4g3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3678` (ply 136) [structural distance $d=15$, QAP obj=2]: the master played Ke7; under μ our Kf4 plays the role of that game’s Ke6; cross-type roles: our qe6 as their Pg3, our Rh5 as their Be5; transferred move f4g3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2626` (ply 121) [structural distance $d=14$, QAP obj=2]: the master played Kf7; under μ our Kf4 plays the role of that game’s Ke2; cross-type roles: our Rh5 as their Ph3; transferred move f4g3 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg3 (summed QAP objective 11, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

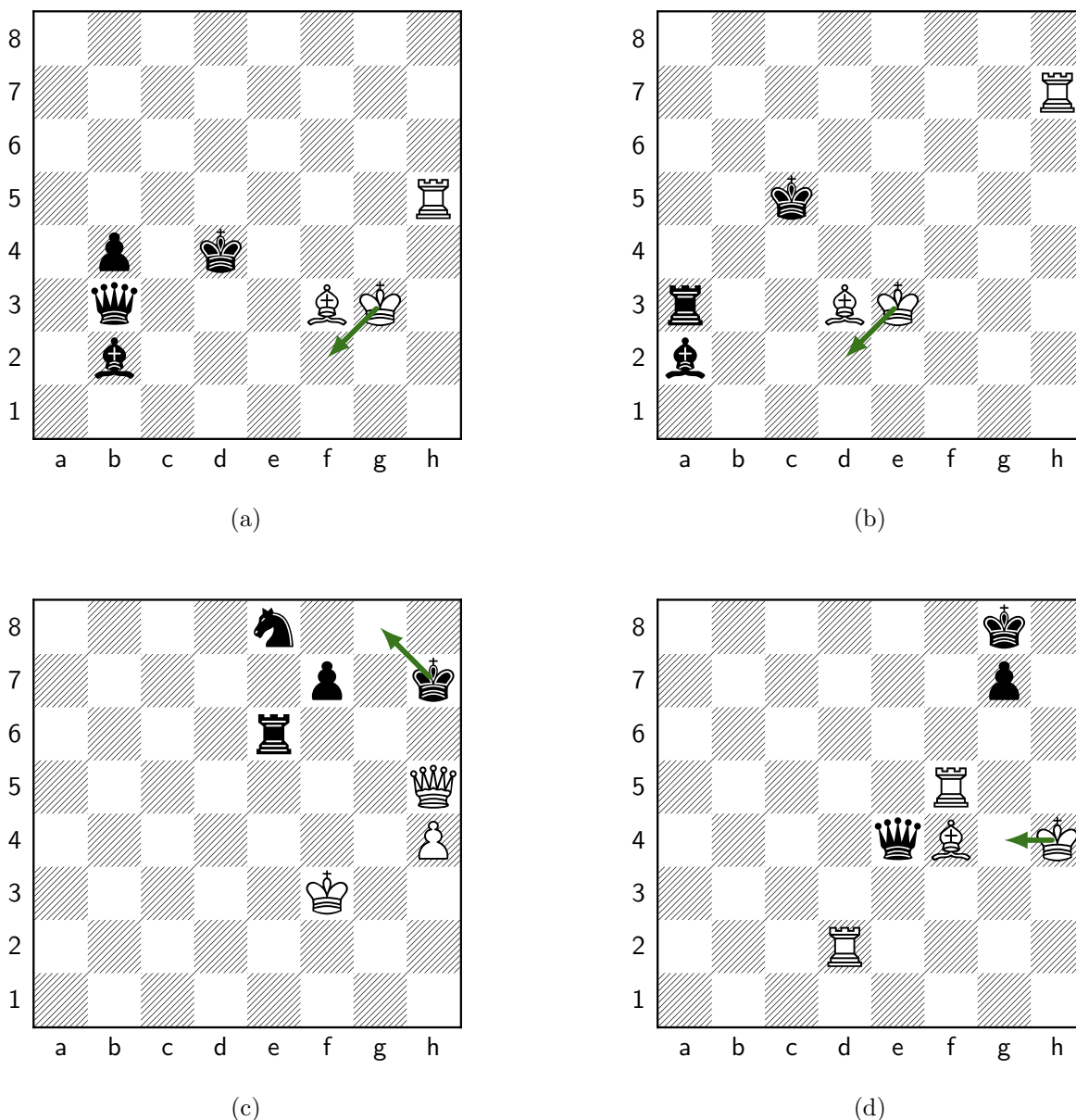
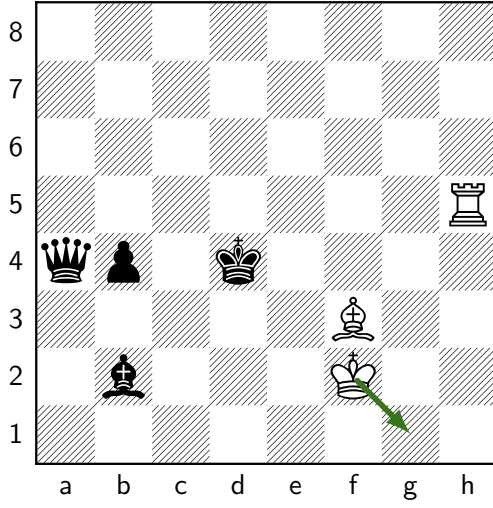
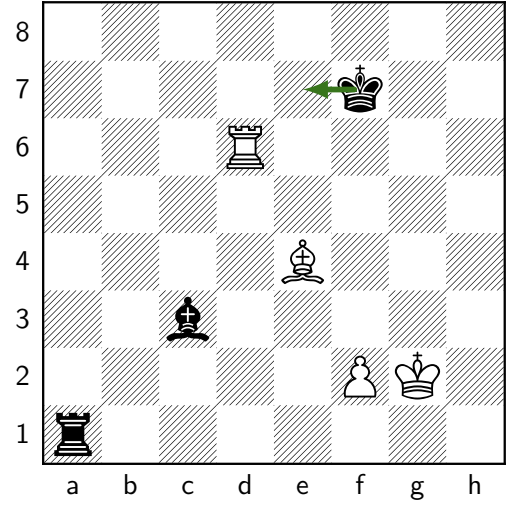


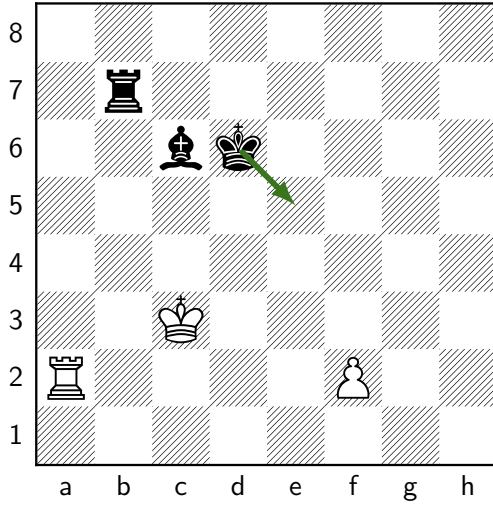
Figure 62: **Move 68, White: Kf2.** (a) Our position; Melanie plays Kf2 (g3f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2781` (ply 120) [structural distance $d=28$, QAP obj=39]: the master played Kd2; under μ our Kg3 plays the role of that game’s Ke3; cross-type roles: our qb3 as their ra3; transferred move g3f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2397` (ply 119) [structural distance $d=27$, QAP obj=16]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kh2; cross-type roles: our bb2 as their Ne8, our Bf3 as their Pf7; transferred move g3f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3074` (ply 150) [structural distance $d=28$, QAP obj=8]: the master played Kg4; under μ our Kg3 plays the role of that game’s Kh4; cross-type roles: our pb4 as their Rf5, our bb2 as their pg7; transferred move g3f2 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf2 (summed QAP objective 63, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



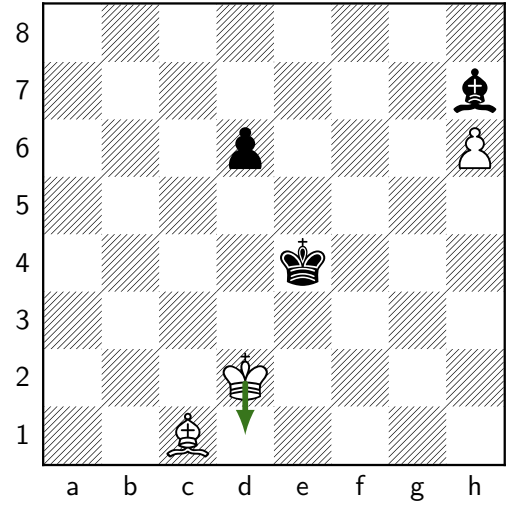
(a)



(b)



(c)



(d)

Figure 63: **Move 69, White: Kg1 (f2g1).** (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1799` (ply 125) [structural distance $d=12$, QAP obj=4]: the master played Ke7; under μ our Kf2 plays the role of that game’s Kf2; cross-type roles: our qa4 as their rd6; transferred move f2g1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2447` (ply 133) [structural distance $d=13$, QAP obj=2]: the master played Ke5; under μ our Kf2 plays the role of that game’s Kd3; cross-type roles: our qa4 as their ra2, our bb2 as their pf2; transferred move f2g1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#486` (ply 130) [structural distance $d=15$, QAP obj=1]: the master played Kd1; under μ our Kf2 plays the role of that game’s Kd2; cross-type roles: our qa4 as their Ph6; transferred move f2g1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg1 (summed QAP objective 8, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

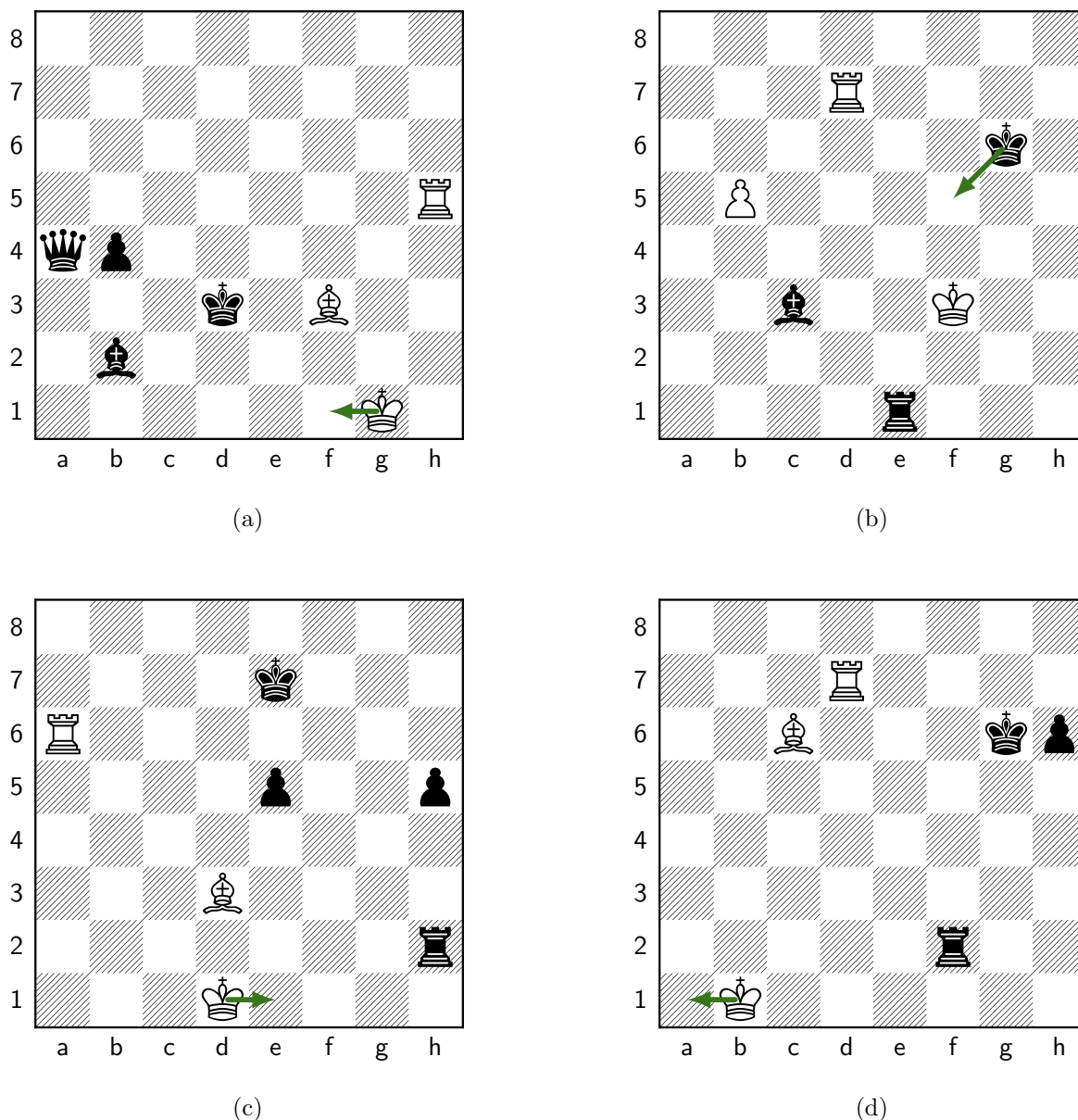
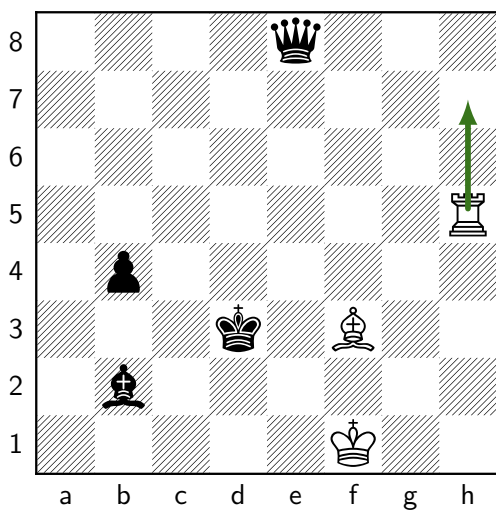
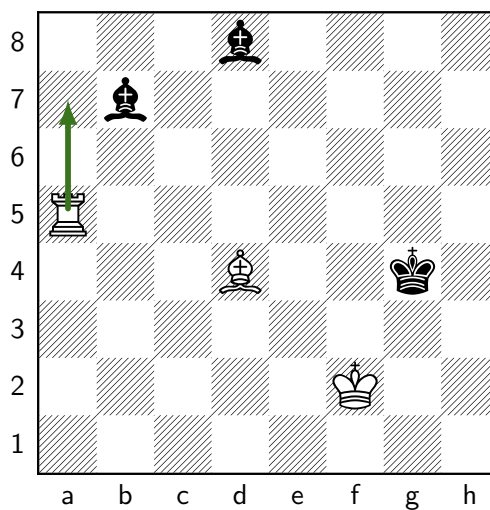


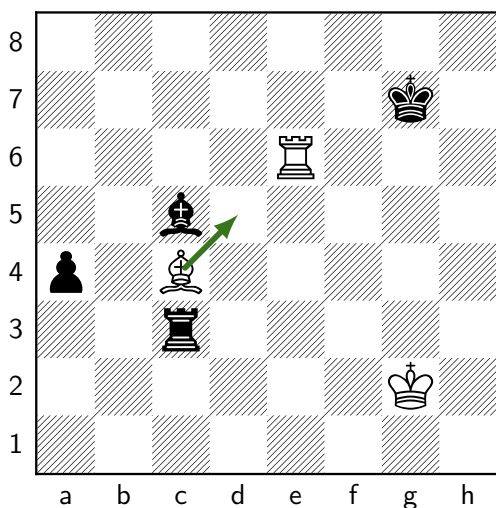
Figure 64: **Move 70, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2255` (ply 155) [structural distance $d=10$, QAP obj=3]: the master played Kf5; under μ our *Kg1* plays the role of that game’s *Kg3*; cross-type roles: our *qa4* as their *rd7*, our *bb2* as their *pb5*; transferred move g1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3941` (ply 122) [structural distance $d=10$, QAP obj=2]: the master played Ke1; under μ our *Kg1* plays the role of that game’s *Kd1*; cross-type roles: our *qa4* as their *rh2*, our *bb2* as their *ph5*; transferred move g1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3510` (ply 122) [structural distance $d=10$, QAP obj=1]: the master played Ka1; under μ our *Kg1* plays the role of that game’s *Kb1*; cross-type roles: our *qa4* as their *rf2*, our *bb2* as their *ph6*; transferred move g1f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kf1 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



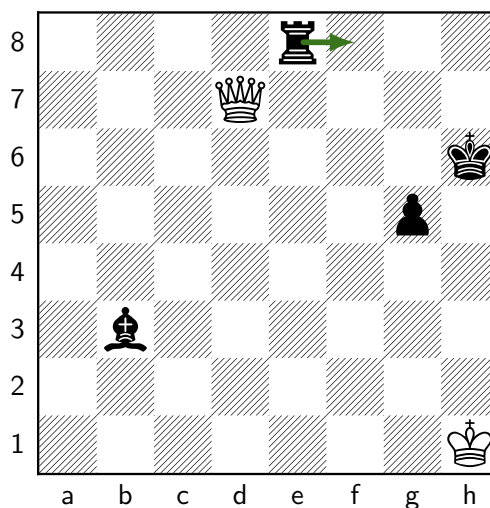
(a)



(b)

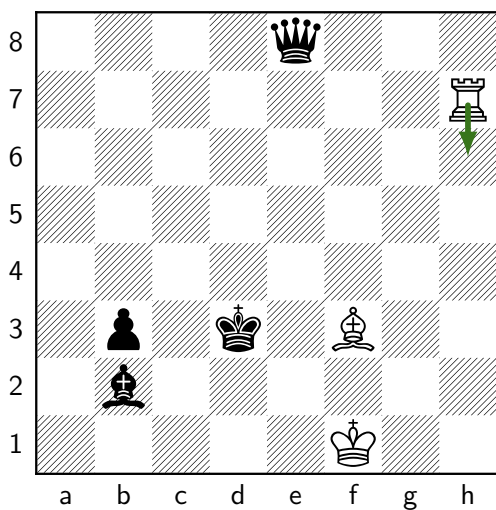


(c)

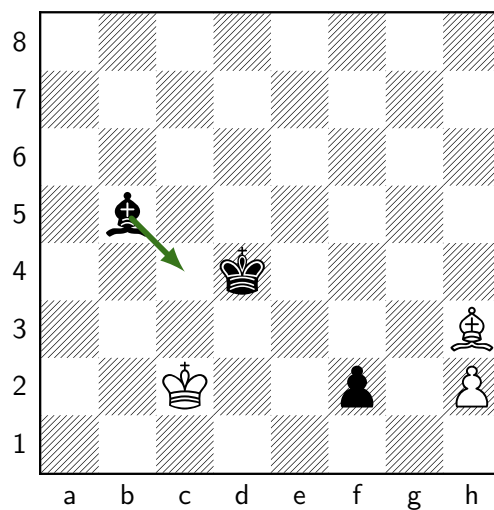


(d)

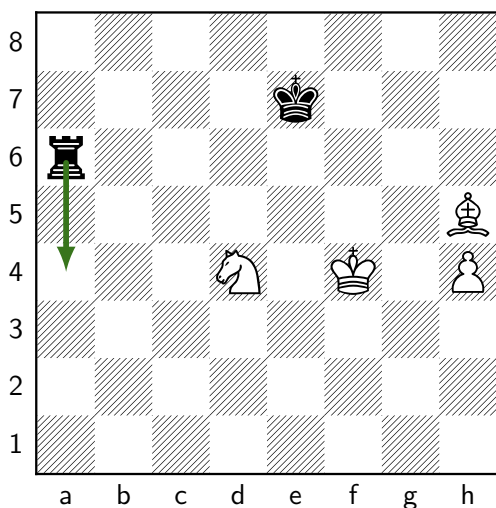
Figure 65: **Move 71, White: Rh7.** (a) Our position; Melanie plays Rh7 (**h5h7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2456` (ply 132) [structural distance $d=13$, QAP obj=20]: the master played Ra7; under μ our Rh5 plays the role of that game’s Ra5; cross-type roles: our qe8 as their bd8; transferred move h5h7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#749` (ply 154) [structural distance $d=9$, QAP obj=12]: the master played Bd5; under μ our Rh5 plays the role of that game’s Bc4 (a **cross-type** correspondence); cross-type roles: our qe8 as their rc3, our bb2 as their pa4; transferred move h5c5. (d) Retrieved from `ccr1_top_p0001#1694` (ply 125) [structural distance $d=13$, QAP obj=8]: the master played Rf8; under μ our Rh5 plays the role of that game’s Re1; cross-type roles: our bb2 as their Pg5; transferred move h5h2. Synthesis: 1 retrieved analog transferred to Rh7 (summed QAP objective 20, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



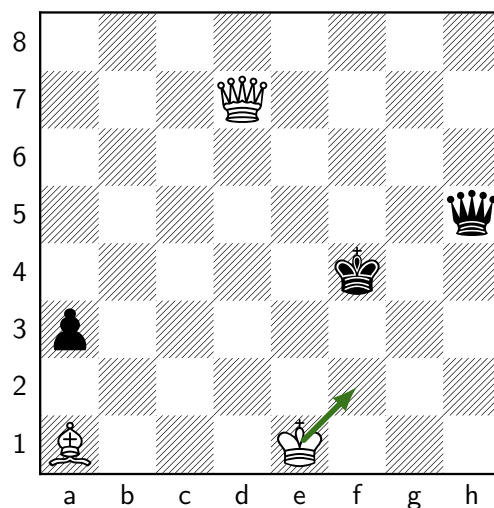
(a)



(b)

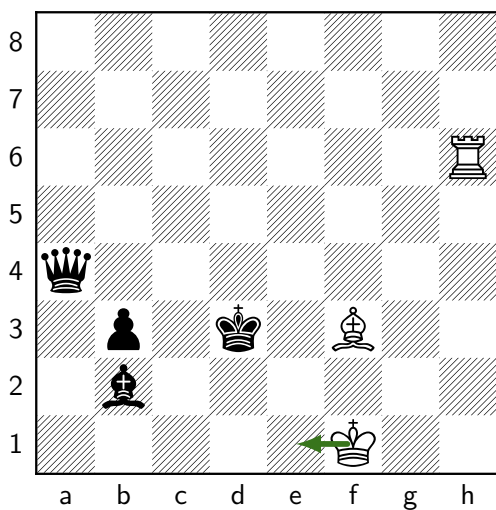


(c)

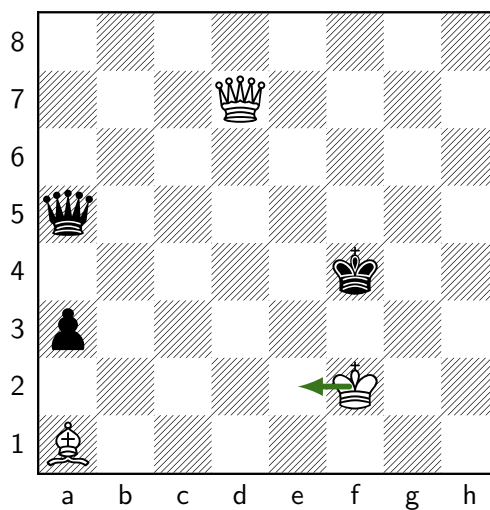


(d)

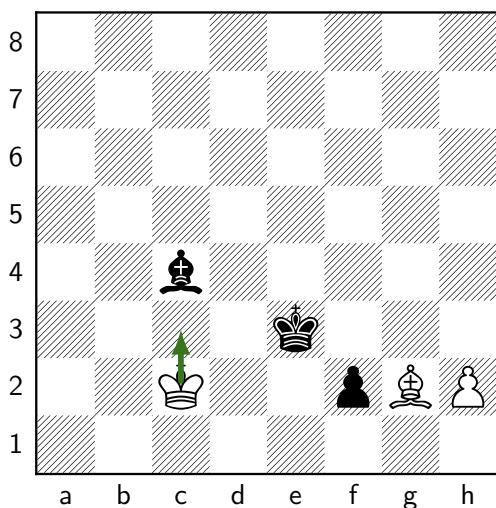
Figure 66: **Move 72, White: Rh6.** (a) Our position; Melanie plays Rh6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3988` (ply 127) [structural distance $d=7$, QAP obj=2]: the master played Bc4; under μ our Rh7 plays the role of that game’s Bb4 (a **cross-type** correspondence); cross-type roles: our Bf3 as their Pf2; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2943` (ply 139) [structural distance $d=7$, QAP obj=2]: the master played Ra4; under μ our Rh7 plays the role of that game’s Ra3; cross-type roles: our qe8 as their nd4; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1019` (ply 156) [structural distance $d=8$, QAP obj=2]: the master played Kf2; under μ our Kf1 plays the role of that game’s Ke1; cross-type roles: our Rh7 as their Ba1, our bb2 as their pa3; transferred move f1f2. Synthesis: 2 retrieved analogs transferred to Rh6 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



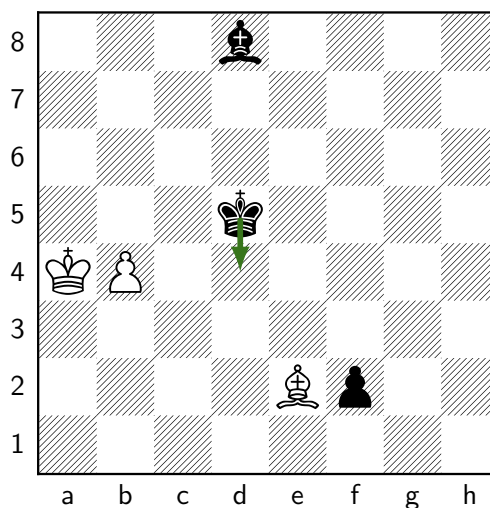
(a)



(b)

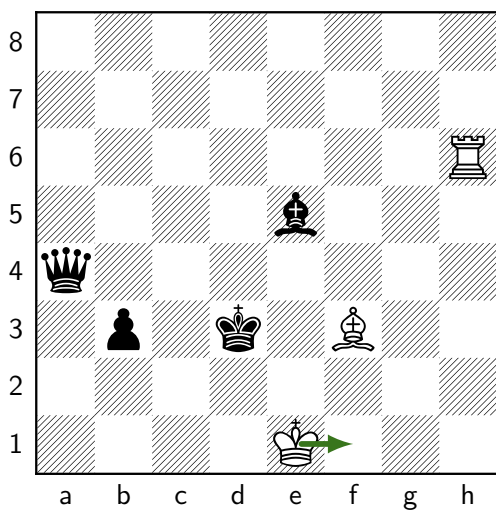


(c)

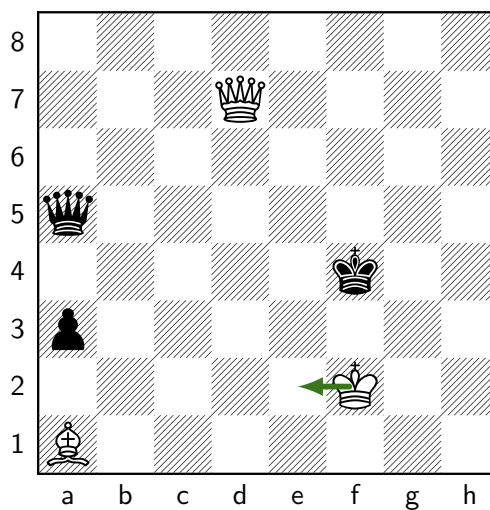


(d)

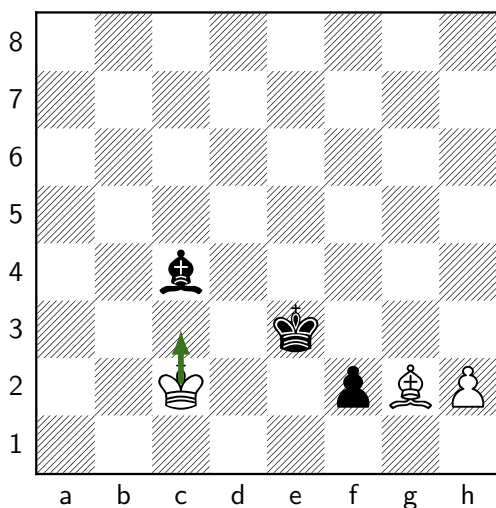
Figure 67: **Move 73, White: Ke1**. (a) Our position; Melanie plays Ke1 (f1e1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1019` (ply 158) [structural distance $d=8$, QAP obj=2]: the master played Ke2; under μ our *Kf1* plays the role of that game’s *Kf2*; cross-type roles: our *Rh6* as their *Ba1*, our *Bf3* as their *Qd7*; transferred move f1e1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3988` (ply 130) [structural distance $d=11$, QAP obj=2]: the master played Kc3; under μ our *Kf1* plays the role of that game’s *Kc2*; cross-type roles: our *bb2* as their *Ph2*, our *qa4* as their *bc4*; transferred move f1e1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2397` (ply 139) [structural distance $d=11$, QAP obj=1]: the master played Kd4; under μ our *Kf1* plays the role of that game’s *Kd4*; cross-type roles: our *qa4* as their *pb4*; transferred move f1e1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ke1 (summed QAP objective 5, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



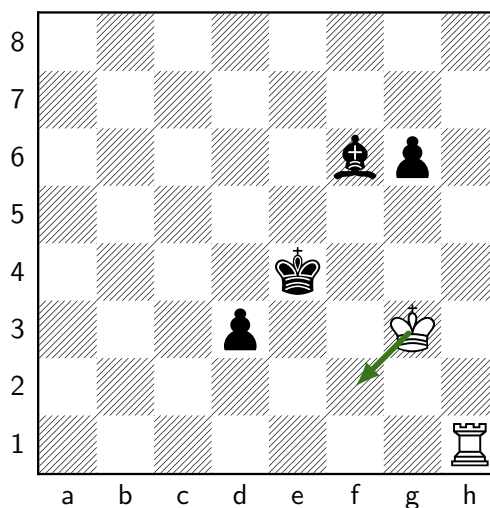
(a)



(b)



(c)



(d)

Figure 68: **Move 74, White: Kf1.** (a) Our position; Melanie plays Kf1 (e1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1019` (ply 158) [structural distance $d=6$, QAP obj=3]: the master played Ke2; under μ our Ke1 plays the role of that game’s Kf2; cross-type roles: our Rh6 as their Ba1, our Bf3 as their Qd7; transferred move e1f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3988` (ply 130) [structural distance $d=9$, QAP obj=2]: the master played Kc3; under μ our Ke1 plays the role of that game’s Kc2; cross-type roles: our be5 as their Ph2, our qa4 as their bc4; transferred move e1f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1253` (ply 148) [structural distance $d=9$, QAP obj=2]: the master played Kf2; under μ our Ke1 plays the role of that game’s Kg3; cross-type roles: our Bf3 as their pd3, our qa4 as their bf6; transferred move e1f1 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kf1 (summed QAP objective 7, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

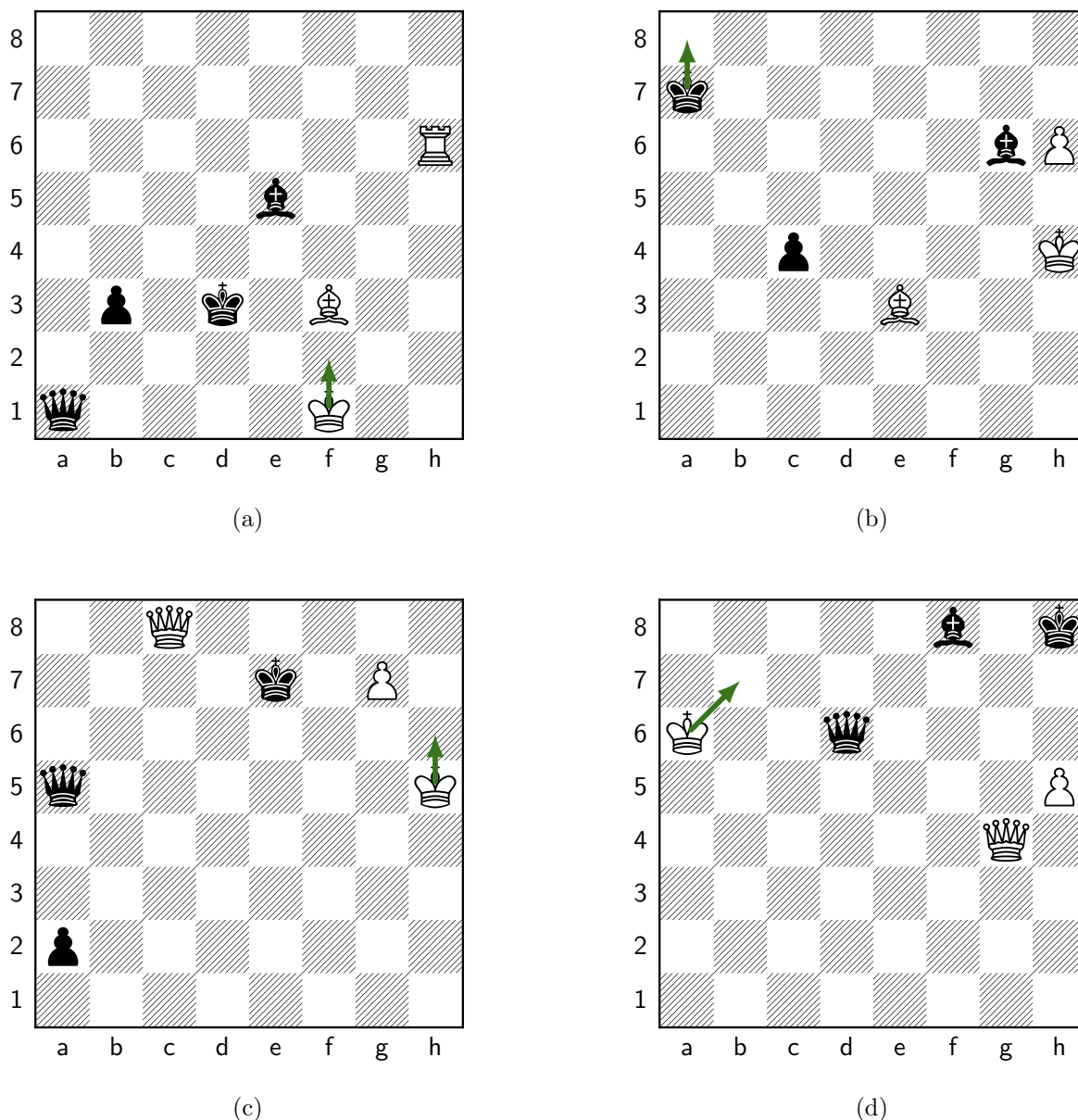
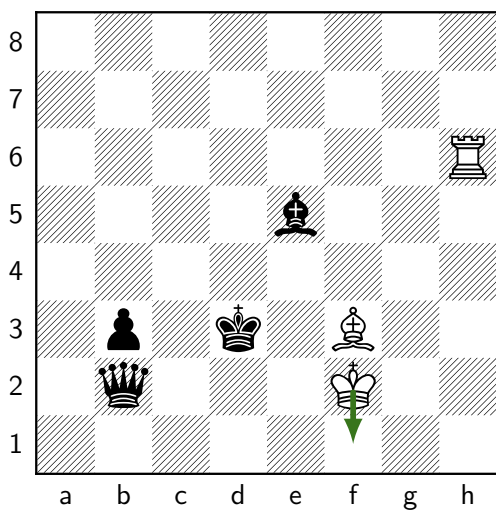
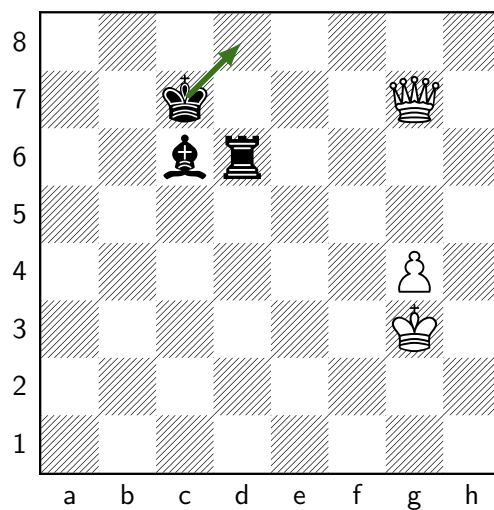


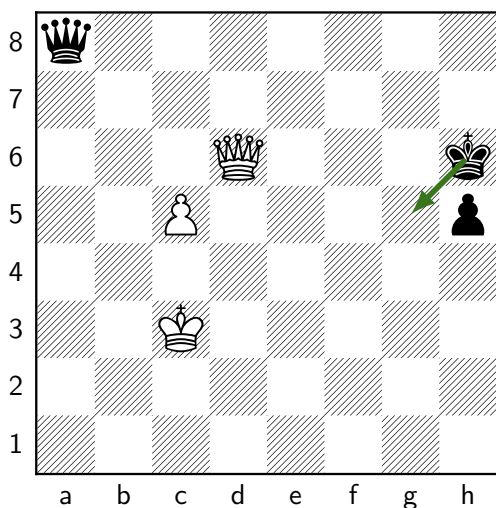
Figure 69: **Move 75, White: Kf2.** (a) Our position; Melanie plays Kf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2807 (ply 155) [structural distance $d=16$, QAP obj=53]: the master played Ka8; under μ our Kf1 plays the role of that game's Ka2; cross-type roles: our be5 as their ph6, our ga1 as their be3; transferred move f1f2 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2429 (ply 136) [structural distance $d=17$, QAP obj=47]: the master played Kh6; under μ our Kf1 plays the role of that game's Kh5; cross-type roles: our be5 as their pa2, our Bf3 as their Qc8; transferred move f1f2 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#705 (ply 142) [structural distance $d=16$, QAP obj=41]: the master played Kb7; under μ our Kf1 plays the role of that game's Ka6; cross-type roles: our Bf3 as their Qg4, our Rh6 as their Ph5; transferred move f1f2 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kf2 (summed QAP objective 313, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



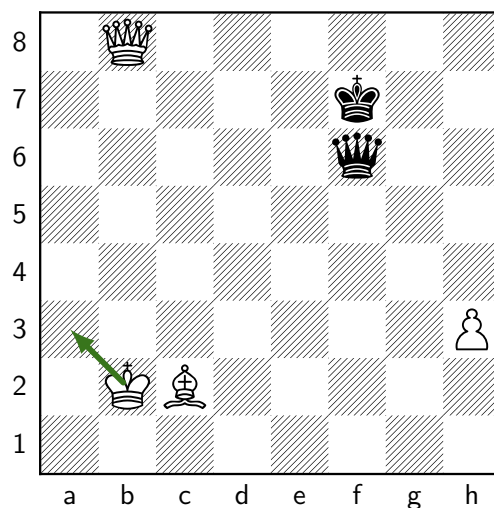
(a)



(b)

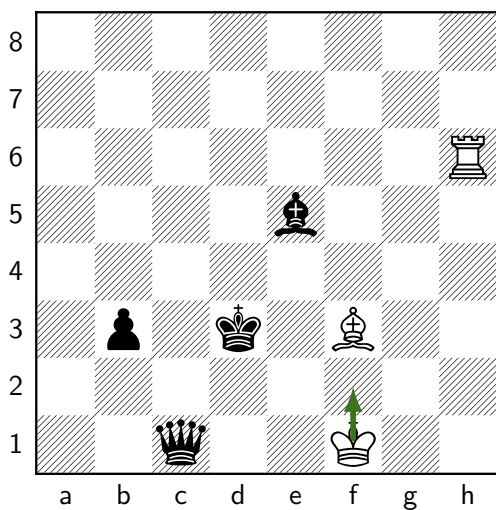


(c)

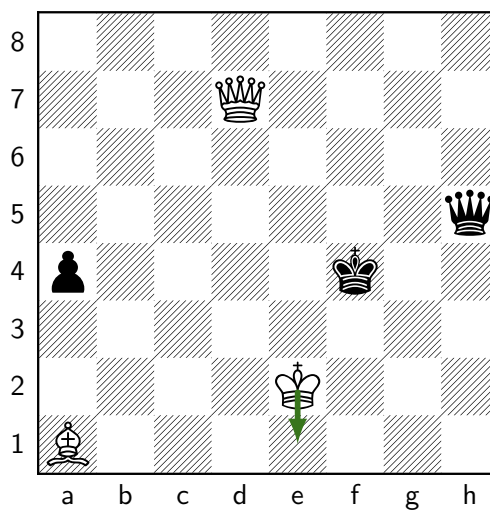


(d)

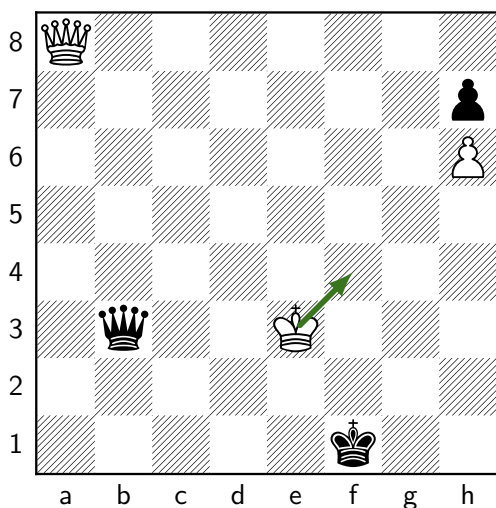
Figure 70: **Move 76, White: Kf1**. (a) Our position; Melanie plays Kf1 (f2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2792` (ply 139) [structural distance $d=27$, QAP obj=70]: the master played Kd8; under μ our Kf2 plays the role of that game's Kc2; cross-type roles: our be5 as their Rd6; transferred move f2f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2312` (ply 167) [structural distance $d=24$, QAP obj=68]: the master played Kg5; under μ our Kf2 plays the role of that game's Kh3; cross-type roles: our Rh6 as their Qa8, our be5 as their pc5; transferred move f2f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#430` (ply 144) [structural distance $d=27$, QAP obj=68]: the master played Ka3; under μ our Kf2 plays the role of that game's Kb2; cross-type roles: our Rh6 as their Qb8; transferred move f2f1 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf1 (summed QAP objective 563, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



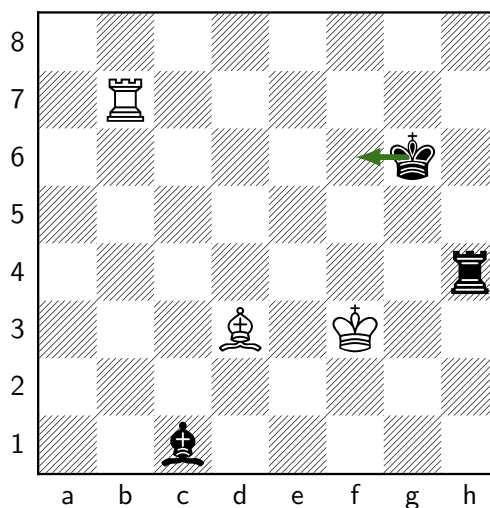
(a)



(b)

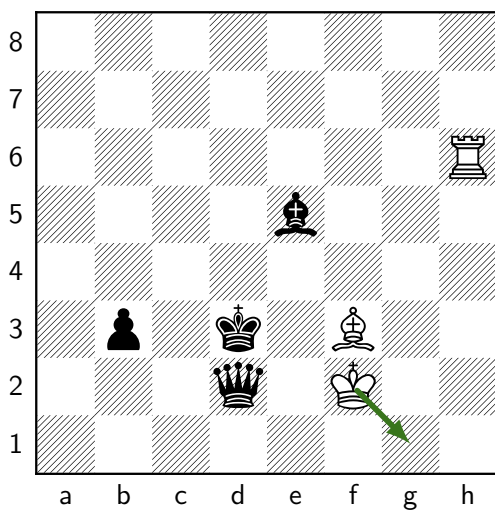


(c)

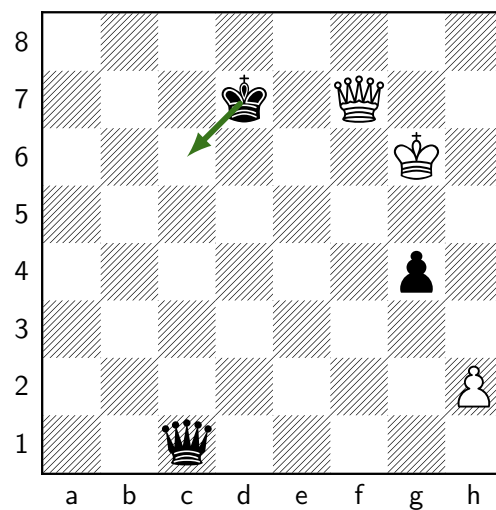


(d)

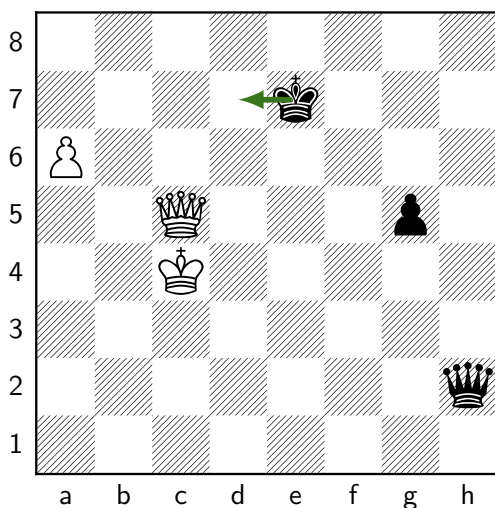
Figure 71: **Move 77, White: Kf2**. (a) Our position; Melanie plays Kf2 (f1f2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1019` (ply 154) [structural distance $d=9$, QAP obj=68]: the master played Ke1; under μ our Kf1 plays the role of that game's Ke2; cross-type roles: our Rh6 as their Qd7; transferred move f1f2 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2313` (ply 154) [structural distance $d=11$, QAP obj=68]: the master played Kf4; under μ our Kf1 plays the role of that game's Ke3; cross-type roles: our Rh6 as their Ph6, our be5 as their ph7; transferred move f1f2 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#201` (ply 149) [structural distance $d=12$, QAP obj=67]: the master played Kf6; under μ our Kf1 plays the role of that game's Kg3; cross-type roles: our be5 as their rb7, our qc1 as their bd3; transferred move f1f2 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kf2 (summed QAP objective 546, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



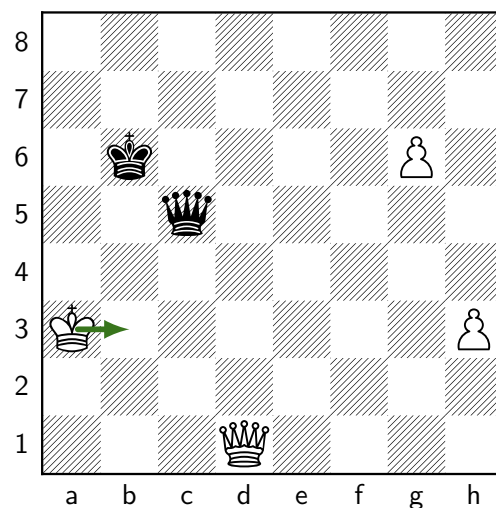
(a)



(b)



(c)



(d)

Figure 72: **Move 78, White: Kg1**. (a) Our position; Melanie plays Kg1 (f2g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccl1_top_p0000#998` (ply 153) [structural distance $d=16$, QAP obj=70]: the master played Kc6; under μ our Kf2 plays the role of that game’s Kd2; cross-type roles: our Rh6 as their Pg4, our be5 as their ph2; transferred move f2g1 (**endorses this move**). (c) Retrieved from `ccl1_top_p0000#88` (ply 163) [structural distance $d=14$, QAP obj=68]: the master played Kd7; under μ our Kf2 plays the role of that game’s Ke2; cross-type roles: our Rh6 as their Pg5, our Bf3 as their Qh2; transferred move f2g1 (**endorses this move**). (d) Retrieved from `ccl1_top_p0001#1772` (ply 150) [structural distance $d=19$, QAP obj=68]: the master played Kb3; under μ our Kf2 plays the role of that game’s Ka3; cross-type roles: our Bf3 as their Qd1, our be5 as their Ph3; transferred move f2g1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg1 (summed QAP objective 341, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

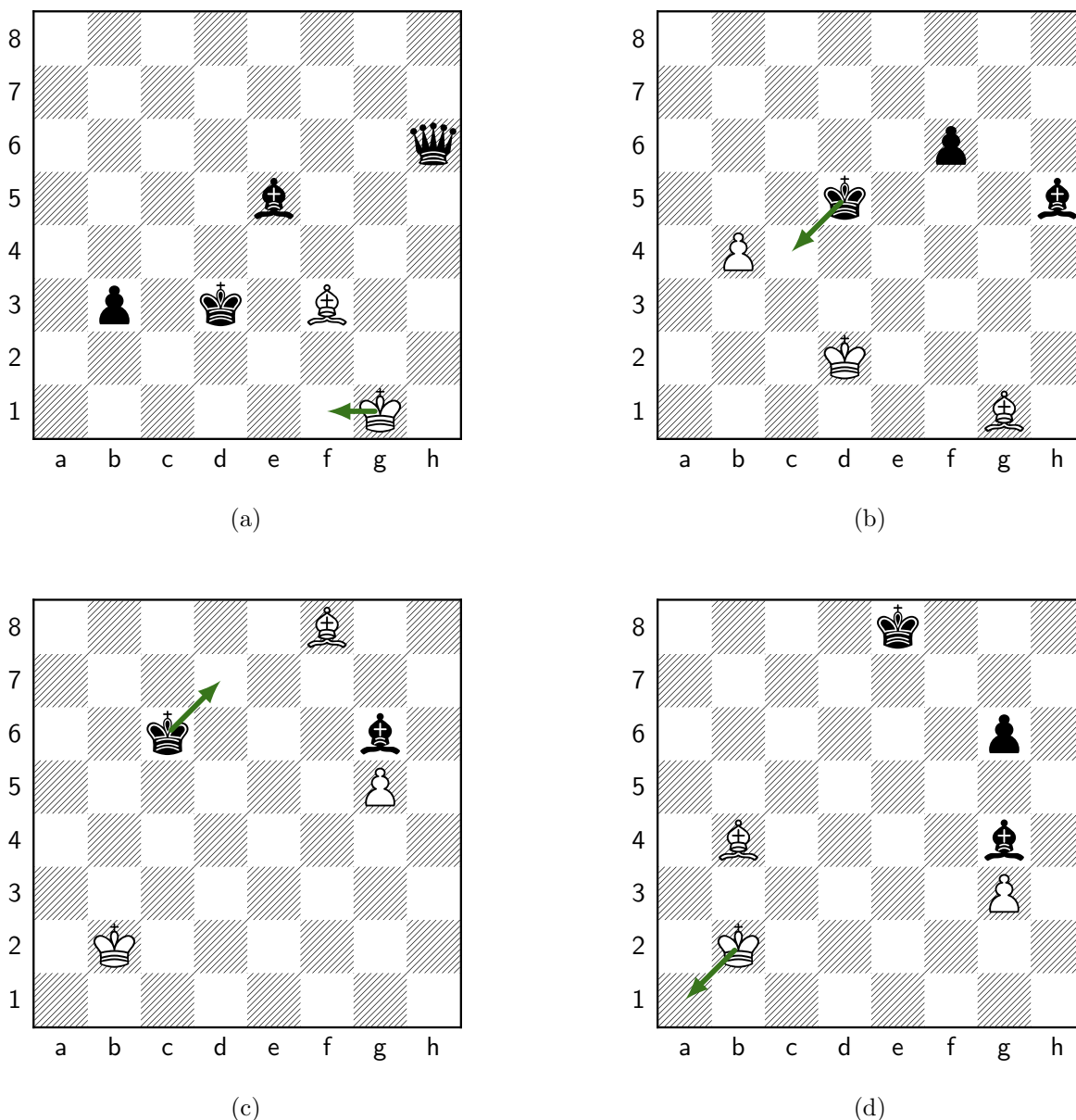


Figure 73: **Move 79, White: Kf1.** (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0001#1488 (ply 153) [structural distance $d=3$, QAP obj=1]: the master played Kc4; under μ our Kg1 plays the role of that game’s Kd4; cross-type roles: our be5 as their pb4, our qh6 as their bg1; transferred move g1f1 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1609 (ply 169) [structural distance $d=3$, QAP obj=1]: the master played Kd7; under μ our Kg1 plays the role of that game’s Kc3; cross-type roles: our qh6 as their bf8; transferred move g1f1 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#3122 (ply 144) [structural distance $d=3$, QAP obj=1]: the master played Ka1; under μ our Kg1 plays the role of that game’s Kb2; cross-type roles: our qh6 as their pg6; transferred move g1f1 (**endorses this move**). Synthesis: 7 retrieved analogs transferred to Kf1 (summed QAP objective 6, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

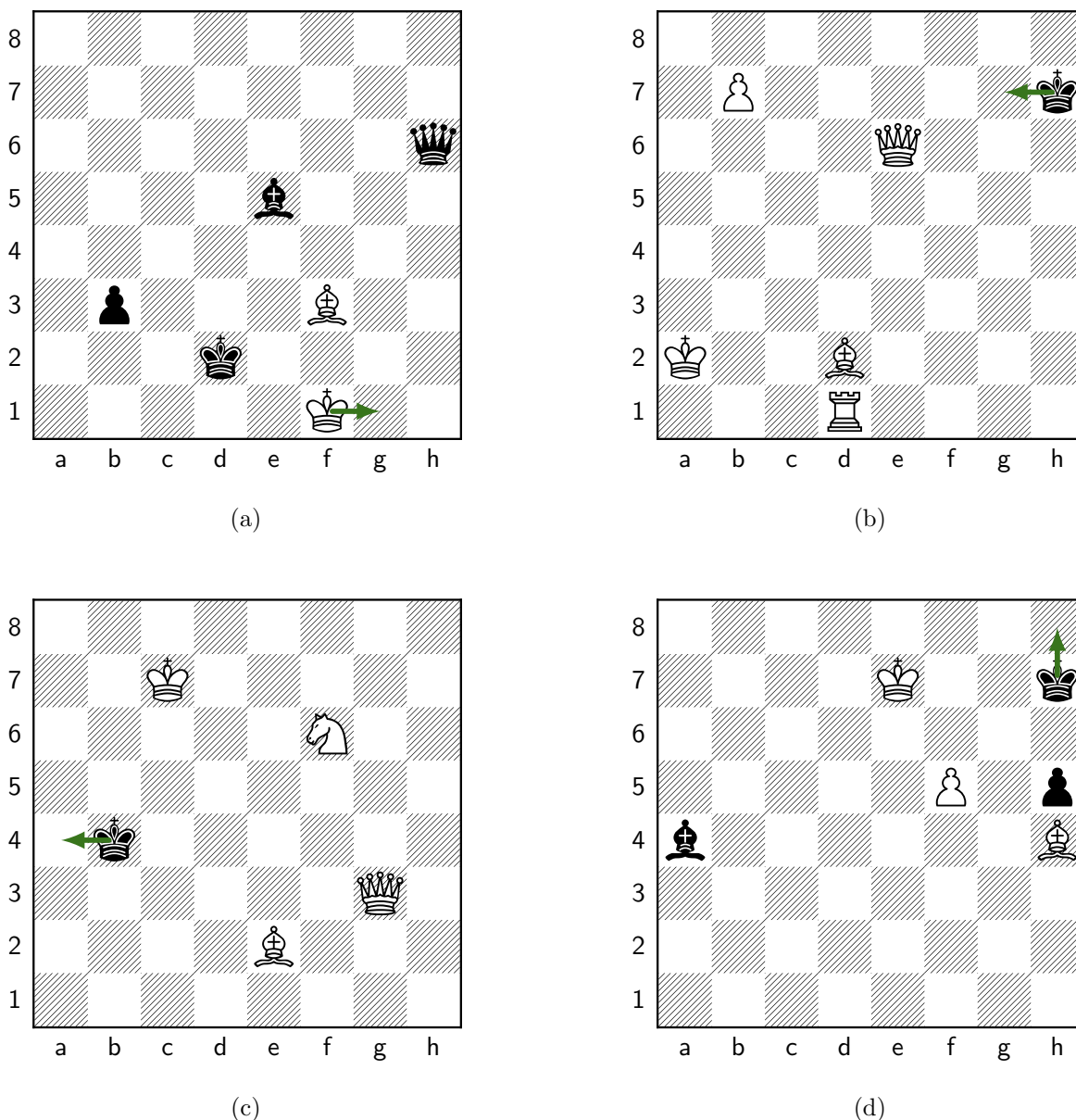


Figure 74: **Move 80, White: Kg1.** (a) Our position; Melanie plays Kg1 (f1g1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#41 (ply 151) [structural distance $d=7$, QAP obj=2]: the master played Kg7; under μ our *Kf1* plays the role of that game’s *Kh2*; cross-type roles: our *be5* as their *pb7*, our *pb3* as their *bd2*; transferred move f1g1 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#1344 (ply 169) [structural distance $d=5$, QAP obj=2]: the master played Ka4; under μ our *Kf1* plays the role of that game’s *Kb5*; cross-type roles: our *Bf3* as their *nf6*, our *pb3* as their *be2*; transferred move f1g1 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1199 (ply 145) [structural distance $d=7$, QAP obj=2]: the master played Kh8; under μ our *Kf1* plays the role of that game’s *Kh2*; cross-type roles: our *be5* as their *Ph5*, our *qh6* as their *bh4*; transferred move f1g1 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Kg1 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

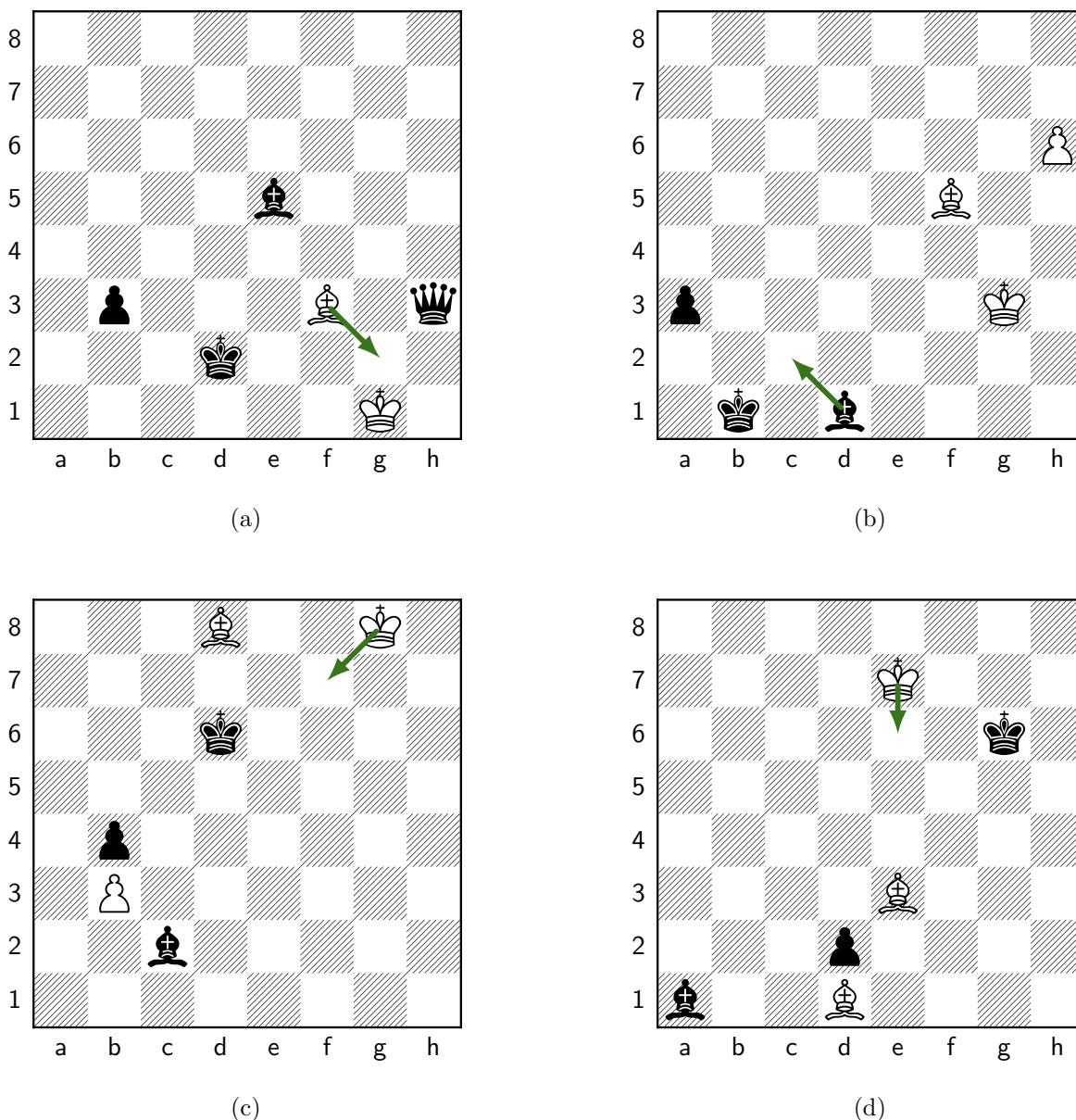
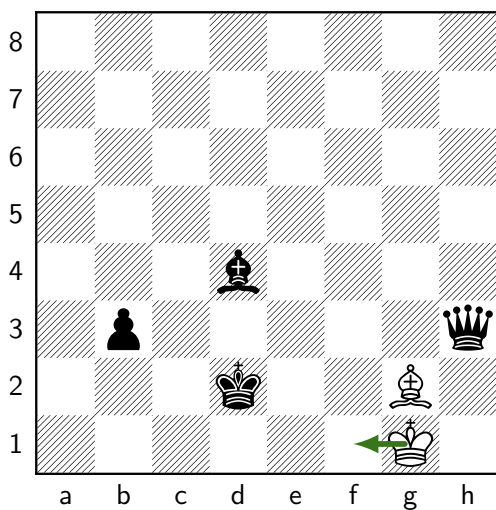
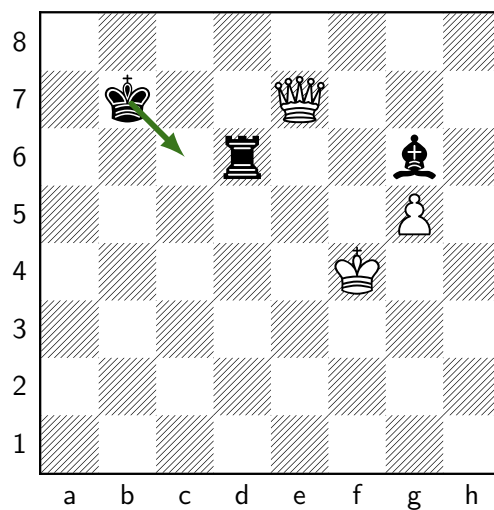


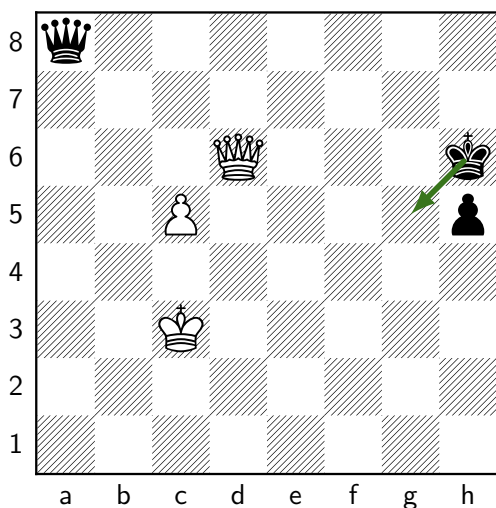
Figure 75: **Move 81, White: Bg2.** (a) Our position; Melanie plays Bg2 (f3g2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#369` (ply 147) [structural distance $d=8$, QAP obj=1]: the master played Bc2; under μ our Bf3 plays the role of that game’s Bd8; cross-type roles: our qh3 as their bf5, our be5 as their Pa3; transferred move f3g2 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2838` (ply 148) [structural distance $d=9$, QAP obj=12]: the master played Kf7; under μ our Kg1 plays the role of that game’s Kg8; cross-type roles: our qh3 as their bc2, our Bf3 as their Pb3; transferred move g1f2. (d) Retrieved from `ccrl_top_p0000#2923` (ply 156) [structural distance $d=9$, QAP obj=12]: the master played Ke6; under μ our Kg1 plays the role of that game’s Ke7; cross-type roles: our pb3 as their ba1, our Bf3 as their pd2; transferred move g1f2. Synthesis: 1 retrieved analog transferred to Bg2 (summed QAP objective 1, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



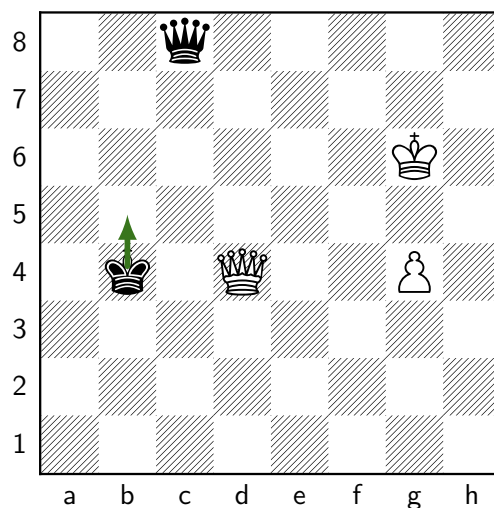
(a)



(b)



(c)



(d)

Figure 76: **Move 82, White: Kf1**. (a) Our position; Melanie plays Kf1 (g1f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#2792 (ply 157) [structural distance $d=20$, QAP obj=66]: the master played Kc6; under μ our Kg1 plays the role of that game’s Kb2; cross-type roles: our bd4 as their qe7, our pb3 as their Rd6; transferred move g1f1 (**endorses this move**). (c) Retrieved from ccr1_top_p0003#2312 (ply 167) [structural distance $d=20$, QAP obj=50]: the master played Kg5; under μ our Kg1 plays the role of that game’s Kh3; cross-type roles: our bd4 as their qd6, our Bg2 as their Ph5; transferred move g1f1 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#1434 (ply 153) [structural distance $d=15$, QAP obj=44]: the master played Kb5; under μ our Kg1 plays the role of that game’s Kb5; cross-type roles: our bd4 as their qd4, our Bg2 as their pg4; transferred move g1f1 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Kf1 (summed QAP objective 337, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.